

Development of a Web-Based Quality Management System to Organize and Manage Policies at Asia Pacific College

Quality Management System

A Software Design Project Proposal Submitted to the School of Engineering
in Partial Fulfilment of the Requirements for the Degree
Bachelor of Science in Computer Engineering

Asia Pacific College

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Chapter 1

PROJECT OVERVIEW

1.1 Introduction

Asia Pacific College (APC), a private, non-profit college in the Philippines, is well known for its commitment to providing high-quality education and programs that meet industry standards. APC, which specializes in subjects such as information technology, engineering, and business administration, is always adapting to meet the changing needs of both learners and professionals. This commitment to excellence has helped to maintain its position as a top higher education institution.

Every successful academic institution is built around good policy management. Policies are fundamental guidelines that govern how an institution functions, assuring compliance with standards, establishing accountability, and promoting consistency in decision-making processes. Policies have a significant effect on an educational institution's academic standards, organizational processes, and overall strategic orientation. They enable institutions to respond effectively to technological, societal, and industry changes, ensuring institutional relevance and competitiveness.

A Quality Management System (QMS) is a systematic technique for improving the creation, implementation, and maintenance of institutional policies. A QMS provides a structured framework, that enables organizations to standardize processes, achieve operational efficiency, and maintain a high degree of quality in all operational areas (Wong, 2024). In the education sector, a well-implemented QMS ensures that institutional policies are not only accessible and understandable but also regularly updated to reflect the institution's aims and regulatory needs.

For schools like APC, implementing a QMS is more than just a matter of compliance; it is an essential element of institutional growth and sustainability. A QMS promotes transparency by giving stakeholders clear, structured information about policies and procedures. It also develops collaboration by providing a single platform for maintaining and communicating policies, ensuring that all the staff—from administrators and academics to students—understand and follow institutional directions.

A quality management system is important for more than just an institution's internal processes. It has a significant impact on the institution's capacity to achieve external standards, such as accreditation requirements, which are essential for sustaining credibility and reputation. A QMS can use modern technologies to improve administrative processes, increase accountability, and support continual improvement in various areas, such as academic quality, operational effectiveness, and stakeholder satisfaction.

In the 21st century, technology is altering the educational landscape, digital tools and systems are no longer optional but required. According to Santos (2021), digitalization is advancing globally, particularly in industrial and societal infrastructures. Hence, digital transformation has become a pillar of industrial policy. A modernized QMS symbolizes this trend by utilizing technology to improve policy administration and support the institution's aim of offering innovative and high-quality education. APC believes that implementing such a system not only corresponds with its commitment to excellence but also equips the institution to meet the challenges and opportunities of a fast-changing educational environment.

1.2 Problem Statement

The existing quality management system (QMS) at Asia Pacific College (APC) has multiple challenges that hinder its function and efficiency. Based on our interview with Engr. Einstein Yong, one of the faculty for the school of engineering. He mentioned that the current system is outdated and not user-friendly, making it difficult for him to navigate the QMS. This problem results in delays in the implementation of policies.

In another interview with Engr. Leonardo Samaniego, Executive Director of the School of Engineering, indicated that one of the key problems with the present QMS is the absence of a function to track the progress of policy implementation. He stated that without a proper tracking system, it is difficult to determine whether policies are being verified, approved, or uploaded to the system. The lack of a tracking process can result in inefficiencies, miscommunication, and policy execution delays, all of which impact the overall quality assurance process.

Similarly, Lorena Rabago, the current Quality Assurance Director, discussed her views on the current system. According to her, there are no major issues with the overall functionality. However, she raised a concern about the policy revision process. Currently, policy revisions are carried out manually, which not only takes time but also causes delays in implementation.

In a separate interview, Jo Anne De Lacuesta, a former Quality Assurance Director, stated that Asia Pacific College's existing quality management system is inefficient. She explained that the system is missing essential functions like a search and filter tool, making it difficult to find policies quickly. She also pointed out the necessity for a new system with an enhanced user interface and experience (UI/UX) design. Furthermore, the current submission process demands that documents be delivered by email, making it a tedious process. She also stated that proposing changes in policies requires obtaining a form from the Central Document Controller (CDC) and waiting for the response. This procedure frequently results in delays, making policy revisions less efficient.

Furthermore, in an interview with Normalyn Bachar, the Current Lead Internal Auditor. She mentioned that the current use of control codes in policy documents lacks clarity, making it confusing for the user to identify where the policy document can be found. Another problem she mentioned is the lack of storage for the uploaded policies. This results in them using online repositories like OneDrive or SharePoint making it vulnerable to hacking.

In an interview with Maylene Mallari, the current Chief Librarian, she stated that all departments meet once a year to evaluate and revise their policies. These conversations seek to identify which policies need to be improved or changed to better fulfill the needs of the institution. When asked about using a Change Request Form, a document that must be filed to the Quality Assurance office when making policy changes, she said that she had never used one. Furthermore, she was unaware that such a form existed or that it was required for documenting policy revisions. When asked about her thoughts on the existing system, she stated that she is satisfied with the overall process of reviewing policies. However, she stated that she was never fully taught or trained in how to document and update policies. This shows a lack of communication or formal training on the proper procedures for managing policy updates.

From these interviews, the researchers were able to assess that the current QMS lacks functions that address the needs of its stakeholders. Addressing the problems regarding the outdated and difficult-to-use system, tracking policy implementation, manual policy implementation process, and communication is essential for APC to increase operational effectiveness while keeping its dedication to quality management. A well-functioning quality management system is significant for institutional performance because it guarantees that policies are organized, accessible, and compliant with standards. By simplifying the policy management process, the institution can increase its ability to respond to change, maintain compliance, and develop an environment of continuous improvement.

According to Rais et. al. (2021), Critical to the successful deployment of a quality culture is the institutionalization of an integrated Quality Management System in which formally documented processes work according to the Vision and Mission of an institute. The researcher's project focuses on creating a web-based quality management system that corresponds to the specific issues raised at APC. The suggested solutions will include the use of automation to reduce manual processes like email-based approvals and version control.

As per Rais et. al. (2021), Commitment to Continuous Quality Improvement (CQI) to close the loop through effective feedback, would ensure that the planned outcomes are attained to the satisfaction of all the stakeholders and that the process overall is improving consistently and continuously. This will encourage a community of constant improvement and inclusivity throughout the organization.

With that, our proposed system aims to offer centralized policy management and automated processes, reducing the likelihood of errors. The redesigned UI/UX will improve the user experience, ensuring the system is more intuitive and accessible.

1.3 Target Audience

The proposed Policy Management System is designed as a tool that addresses the needs of these various stakeholders within the institution:

1.3.1 Administrator

The administrator is responsible for assigning roles to users and ensuring that each has the proper permissions to carry out their task. Possible individuals who can hold the Administrator role include the Quality Assurance Director, ITRO, or other authorized personnel who have been granted this responsibility within the system.

1.3.2 Quality Assurance Director

The role of the Quality Assurance (QA) Director is to review and analyze reports, supervise the implementation of processes, and make strategic decisions to improve the effectiveness and efficiency of the quality management system. The proposed system includes modules that would automatically generate reports for the QA director to analyze, it may be useful to the QA director in making key decisions for the improvement of the system.

1.3.3 Quality Assurance Personnel

The QA personnel are responsible for the management of policies within the institution. They ensure that all policies align with institutional standards and regulatory requirements. With the modules that the system provides, it may be used by the QA personnel to aid their task in managing policies.

1.3.4 Staff

The role of the staff, teaching and non-teaching is to comply with the institution's policies, procedures, and quality standards. While actively contributing to the achievement of organizational goals and the continuous improvement of services. With the proposed system being centralized, it would help the staff in the implementation of policies.

1.4 Objectives

1.4.1. General

This project aims to design and develop a web-based quality management system that provides various departments of Asia Pacific College with a centralized platform to manage institutional quality policies, procedures, and processes.

1.4.2. Specific

The specific objectives that this website aims to provide are the following:

- A facility for users to submit new and revised policies.
- A module for the review and approval process of proposed policies.
- A facility for uploading approved policies into a repository.
- A module where users can provide feedback regarding specific policies
- A notification system for policy updates.
- A facility to monitor reports created from the system.

1.5 Scope of work

1.5.1. Scope

The proposed Quality Management System features four user roles. These are the administrator, quality assurance Director, quality assurance personnel, and staff. The role system follows a hierarchical structure, which means that starting from the top role, they can access every module below them. Thus, each role is assigned the following modules:

Administrator

- Deletion or management of user accounts.

Quality assurance director

- Automatic generation of reports such as:
 - Number of approved policies in the repository.
 - List of rejected policy proposals, with reason for rejection.
 - Number of reviewed/verified/approved documents per person.
 - Feedback received and actions taken.
 - Percentage of people who always provide/ feedback.
- Create, delete, and manage organizational structures and policy organization.
- Appointment of persons to verify and approve proposed policies

Quality assurance personnel

- Review of proposed policies.
- Upload, store, and organize approved policies.
- Automated version numbering and editable control code for policy documents

Staff

- Verification and approval of proposed policies.
- Downloading templates for the creation and revision of policies.
- Submission of new and revised policies.

General

- User account creation using an APC email, with verification via email for confirmation.
- A task manager that organizes tasks based on the date submitted.
- The viewing of policy documents will feature watermarks.
- Automatic generation of a digital signature for the signing of documents using reference code.
- Tracking of policy implementation process status.

- Audit trail facility will provide policy revision history with details, including:
 - Person who authored, verified, and approved the policy.
 - Date of publication for each process.
 - Version number of each revision
 - Description of changes made to the policy.
- Location of policies or staff using keywords or filter and sorting options such as by date, department, or policy type.
- Submission of comments, suggestions, and concerns regarding specific policies.
- Automated sending of alerts regarding policy implementation.

1.5.2. Delimitation

This project will be limited to designing and developing a Web-based Quality Management System by mainly focusing on the functionality of each feature. Specifically, it will be delimited to the following:

- The proposed system is designed for desktop computers or laptops.
- File sharing functionality is not included in the feedback module.
- Notifications are limited to policy updates and changes within the system and do not include external communication channels like email or SMS.
- Automatic role assignment is not supported.

1.5.3. Assumptions

- Users are assumed to possess basic knowledge of navigating and operating a web-based platform.
- Users are expected to have a fundamental understanding of the processes involved in creating, submitting, verifying, and approving quality policies.
- The system is expected to have a stable connection to the server to operate properly.
- Potential restrictions on data transfer size, and speed.

1.5.4. Constraints

- The system will be hosted on the AIR Hub server.
- The system will abide by the Republic Act 10173, or the Data Privacy Act (DPA) of 2012.
- The system is required to have a stable internet connection.

1.6 Definition of Terms

Term/Acronym	Definition
Asia Pacific College Information System (APCIS)	The current system of APC is used for different functions, like room reservations and employee management. for the context of this project, we will be mainly focusing on the QMS portion of the system.
Audit Trail Facility	A system feature that records all modifications made to policies within the QMS, providing traceability and accountability for auditing purposes.
AIR Hub Server	The server platform that will be used to host the proposed QMS will provide the necessary infrastructure for the system's operation. It ensures the data security of the system.
Monitoring Mechanism	A feature within the proposed QMS that tracks the status of policy proposals, ensuring that each policy is properly reviewed, updated, and implemented.

Notification System	A feature within the QPMS that sends automated alerts to users about updates and status regarding the policy process, ensuring that stakeholders are informed of important policy developments.
Quality Assurance (QA) Staff	These individuals are responsible for monitoring and evaluating policies, procedures, and practices to ensure they align with established standards and regulatory requirements.
Quality Management System (QMS)	A quality management system is a structured process that defines and documents an organization's processes, procedures, and responsibilities for achieving quality policies, practices, and objectives.
Reports module	A system feature that provides an overview of key performance indicators regarding policy management. This includes the number of approved policies, rejected proposals, and feedback received, allowing the Quality Assurance Director to track the effectiveness of the policy management process.
Search and Filter Functions	Tools within the proposed QMS will allow users to find specific policies, staff, or related information by using keywords, applying filters, or sorting based on categories.
Staff	The academic and administrative personnel, who include the directors and executive directors at APC, will use the proposed QPMS to submit, review, and provide feedback on institutional policies.
Stakeholders	Individuals or groups involved with quality management. This includes the staff, administrators, and regulatory bodies who are affected or involved in the creation, review, and implementation of institutional policies.
Subject Matter Experts	Individuals who possess specialized knowledge, expertise, and experience in a particular area. These individuals are designated to verify and approve the policies based on their expertise.
Version Control	The process of managing and tracking the different versions of a policy document. This ensures that users can identify the most current and relevant version of each policy and prevents the use of outdated information.
Version Labeling	The practice of assigning specific labels or identifiers to each version of a policy allows users to easily track revisions and updates.

Chapter 2

ANALYSIS

2.1. Situation Analysis

2.1.1 Current process

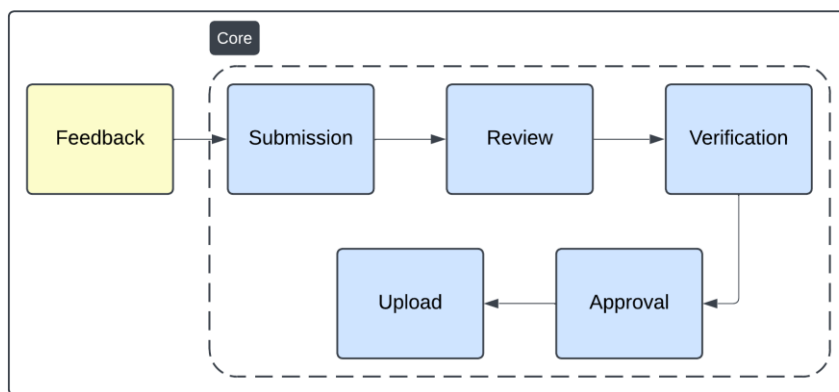


Figure 2.1 Policy Implementation Process

Illustrated in Figure 2.1 is the current policy implementation at Asia Pacific College. Currently, there is an organized sequence involved in the basic process of implementing a policy. In the submission procedure, any policy whether it be new or revised, it must be typed in an editable form like the Change Request Form, which is requested from the Central Document Controller from the Quality Assurance. Once the editable file is filled out with the details of the proposed policy, it is then submitted through email.

Once the policy proposal is submitted, it will be received by the immediate head. Who will then review the proposal, if they find it insufficient or unsatisfactory they will reject it. If they deemed it acceptable and up to standard, they would verify the proposal. After verification, they will then send the verified policy to the executive head.

Once the executive head receives the verified proposal they would check the document. Much like the immediate head, the executive head can also reject the proposal if they are not satisfied with it, and if deemed sufficient, the executive head would give their approval. They would send the approved proposed policy back to the Quality Assurance.

After being verified and approved by the immediate and executive heads the proposed policy is then returned to the Quality Assurance's central document controller. The proposed policy will then pass to the ITRO who will upload the policy to the Asia Pacific College Information System as a new or revised policy.

The feedback process, although a minor process, is still a crucial procedure. Here feedback, reports, and reactions to a particular policy are sent through the Quality Assurance from the policy makers or vice versa. This feedback may range from positive to negative and if it is deemed that a policy needs a revision or update, the whole process begins anew, starting from the submission procedure.

2.1.2 User Persona

Quality Assurance Director



Figure 2.2 *Liz Katherine*

On Figure 2.2, Liz Katherine, a 50-year-old Quality Assurance Director at Asia Pacific College, values her peace and solitude. In her free time, she enjoys immersing herself in books or watching her favorite shows on her TV, phone, or laptop, liking moments of quiet relaxation.

As the Quality Assurance Director, she is responsible for leading the Quality Assurance Team, ensuring that both active and proposed policies meet established standards and align with the organization's goals. She is also tasked with overseeing compliance with these policies and

ensuring they are regularly reviewed for effectiveness. Her role requires clear and consistent communication with QA personnel, and staff, as most reports and feedback originate from them. This information is crucial for making informed decisions to enhance the organization's standards and drive continuous improvement.

Quality Assurance Personnel



Figure 2.3 *Maria Elena Cruz*

Miss Maria Elena Cruz as shown in figure 2.3, is a 33-year-old employee of Asia Pacific College, enjoys watching foreign shows in her free time. She serves as a member of the Quality Assurance Team, where her primary responsibilities include ensuring that the college's processes and services meet quality standards and align with institutional and regulatory expectations. Her work involves compiling documents, reviewing policies to ensure compliance with regulations, and overseeing the verification and approval processes for new or updated policies. While her tasks may appear straightforward, the complexity of the processes and the challenges she encounters make her role more demanding than it seems.

Much of the files Maria handles are change request forms, which are essential for updating policies. During the verification and approval process, she forwards these forms to department heads within Asia Pacific College (APC) for their review and approval of the proposed policies. Once the policies are verified and approved, Maria coordinates with the Information Technology and Resource Office (ITRO) to upload the finalized policies into the Asia Pacific College Information System (APCIS).

Staff



Figure 2.4 *Nathaniel James Carter*

As shown in Figure 2.4, Nathaniel James Carter, a 30-year-old employee of Asia Pacific College, serves as the department head of the registrar. He is a family man and spends much of his free time with his kids and wife and go out hiking with his friends in his spare time.

As the department head of the registrar, his primary responsibilities include overseeing academic records, managing the registration process, and coordinating the graduation process. However, with the increasing number of students each academic year, it has become evident that the current procedure is insufficient. The existing procedure is time-consuming, slow, and prone to errors

Driven to make things easier for him, his staff, and the enrolling students, he wants to develop a new enrollment procedure. He and his team collaborated, deliberated, and sought input from other departments and facilities to finalize the proposed procedure. The final step in the process is to submit the proposed procedure to the Quality Assurance team for verification and approval.

2.1.3 Use Case Scenario

Quality Assurance Director

Liz starts her day with a cup of coffee and a stack of reports from Maria. As she goes through them, she thoroughly studies the details, looking for patterns, inefficiencies, or indications of system problems. Some problems are simple to identify, such as process delays or communication gaps, but others need a deeper look to determine their root causes. Liz understands that every move she makes will have an impact on how efficiently the system functions, so she takes the time to carefully assess the reports. She considers many alternatives and determines which ones will be most effective in improving the system's structure. It's a difficult assignment, but she recognizes the need to make informed decisions to keep procedures operating smoothly.

Liz oversees not only the management of the system, but also the organization of policies. She ensures that each policy is properly categorized, making it easier for staff to access the information they require. Proper organization is important because it enhances clarity, assures accessibility, and maintains policies in line with the company's process. Without proper categorization, policies can become unclear, resulting in misunderstandings and inefficiencies. Liz takes this job very seriously; she understands that a well-structured policy framework leads to a more organized and successful workplace. While the process takes time and careful attention, she is committed to maintaining order and ensuring that all policies serve their intended purposes.

Quality Assurance Personnel

Maria sits at her desk, flipping through a large stack of documents. It was another stressful day at work, and she had a lot to examine. As part of her role as QA personnel, she must ensure that all policies adhered to the organizational standards. Occasionally, she would receive request from staffs for editable files like a change request form to revise a certain policy but before she hands over the documents, a reason must be given for the request. Occasionally she would receive a verified and approved change request form sent to her by an executive head in which she will post it the Asia Pacific College Information System.

Later that afternoon, Maria focused on policy compliance. She analyzed policies from the Registrar's office and discovered a problem, the graduation procedure did not meet the organization's criteria. Without hesitation, she contacted the Registrar's office and requested a correction. It is critical that all policies followed the institutional regulations and ensure that students had a smooth graduation process.

Maria completed her monthly report before the end of the day. She collected information on policy approvals, feedback, and other relevant data. After reviewing it one last time, she submits the reports to the Quality Assurance Director. Her work contributes to monitoring and optimizing the organization's operations, ensuring that everything worked as smoothly as possible.

Staff

Nathaniel works in the office of registrar, as the head of the office he is responsible for ensuring that the office of the registrar's policies are well-structured, accurate, and meet institutional requirements. His duties include developing, verifying, approving, and updating policies, as well as providing feedback as needed. When Nathaniel creates a new policy, he follows the template provided by the quality assurance office. For example, he just developed a guideline for the registration procedure of students. He thoroughly fills out each area of the template, making sure to include every necessary detail, and then sends the document to the quality assurance office for review.

Nathaniel is also in charge of verifying policies submitted to the department. When he receives a policy for verification, he carefully reviews it to ensure that it follows the school's set rules and quality standards. After studying the paper, he signs it as an acknowledgement of verification and immediately sends it to the executive head. Nathaniel is occasionally responsible with approving policies that fulfill the required standards. When tasked with approving a policy, he carefully reviews its contents to ensure that it is consistent with the institution's goals and quality expectations. If everything is in order, he signs the document to give acknowledgement and returns it to the Quality Assurance Office.

While reviewing policies, Nathaniel may identify places for improvement. When he sees a policy that needs to be revised, he contacts the Quality Assurance office so that the responsible individual can re-evaluate and revise the policy as needed. For example, Nathaniel once came across a policy that lacked clarity, so he highlighted it for review and proposed revisions.

One day, Nathaniel received feedback on a graduation policy that required revision. After carefully examining the document, he made the necessary adjustments and completed the change request form provided by the quality assurance office. He then sent the change request form to the quality assurance office for further review and approval. Nathaniel's attention to detail and commitment to quality help keep the school's policies organized, compliant, and up to date, ensuring that administrative operations run smoothly and that the institution's approach to various problems is consistent.

2.1.4 Pain Points

Quality Assurance Director

Liz is constantly monitoring reports submitted by the quality assurance personnel. But due to the high workload of the quality assurance personnel, there is a delay in receiving the reports this results in delays in response to the issues that need to be addressed.

Another issue is the lack of a feedback system to address the concerns of the other stakeholders. Addressing this issue would promote improvement within the quality management system and would increase the quality of service provided by the institution.

Quality Assurance Personnel

Maria is the Quality Assurance Internal Lead Auditor. Every day, she encounters multiple difficulties that make her job challenging. One of the most significant issues is the lack of clear roles in policy review. Without clear roles, deadlines are missed and decisions become inconsistent. This slows down and complicates the review process.

Another concern Maria has, is a lack of version control. Policies are constantly revised, but there is no system to identify the most recent version. As a result, people may continue to utilize outdated policies without realizing it. This results in confusion and noncompliance with current standards.

Maria also recognizes an issue with the dissemination of information. There is no system for informing staff when policies change. Many times, stakeholders are unaware of changes. This can result in mistakes, inefficiencies, and even failed audits. Maria wishes there was an approach that would notify everyone about significant changes so that they could remain informed and in line.

Finally, Maria has trouble with the manual procedures. Everything concerning quality policies is done manually, which requires a significant amount of time and work. Mistakes such as incorrect data entry and formatting problems are common while working manually. As the organization grows, the number of policies increases, making the manual process much more difficult to administer.

Every day, Maria works hard to maintain high-quality standards, but these challenges make her job frustrating. She hopes that one day, improvements will be made so she can focus on ensuring quality without so many obstacles.

ITRO

In a busy office, the ITRO team worked hard to organize and monitor important papers. They ensured that policies were properly posted, keeping records up to date, and made sure that everyone had access to the most recent information. However, their job was not easy.

One of the most difficult issues they encountered was insufficient automation in policy uploads. Every time a new policy was added, the team had to ensure that everything was in order. This required a significant amount of time and effort, and little errors would often get through. These mistakes could generate confusion, requiring them to go back and fix the documents, delaying the process even further.

Another issue was the absence of real-time notifications. When a policy was uploaded, there was no automated notice to inform all the right people. This meant that team members had to send emails out to every employee in APC. The team also struggled with version control concerns. Without a clear system for tracking revisions, old policies could be left in place while new versions were overlooked. This caused uncertainty because employees weren't sure if they were following the most recent policies. Mistakes like these can have a wide-ranging impact on the organization, making consistency and efficiency difficult to achieve.

Despite these difficulties, the ITRO team stayed committed. They understood how important their hard work was to the organization's running smoothly. They wanted better tools and systems to help them manage policies more efficiently. Until then, they worked carefully to ensure that every document they handled was accurate and well-organized.

Staff

Nathaniel sat at his desk, looking at the system on his screen. One of his main challenges was keeping track of policy versions. Policies changed regularly and without a clear structure for managing updates, Nathaniel was unsure if he was looking at the most recent version. Sometimes old policies remained in the system, causing confusion among staff. This made it difficult to ensure that everyone followed the appropriate rules and procedures.

Another difficulty he experienced was the delay in receiving policy update notices. Changes were implemented, but there is a lack of an automated system for informing everyone. Nathaniel had to rely on manual communications from administrators, which frequently caused delays. Some employees continued to follow outdated policies because they were unaware of the changes. This caused unnecessary compliance concerns and conflicts.

The approval process was another nightmare. Every new or changed policy required multiple layers of review and signatures. Because everything was done manually, approval took longer than expected. Nathaniel frequently had to wait for upgrades that took weeks to appear in the system. These delays hindered operations and made it impossible to make the necessary modifications on time.

Nathaniel noted that a better system was needed. Many of these issues might be solved by implementing a centralized platform for managing policy versions, automating update notifications, and defining roles.

2.2. Literature Reviews

2.2.1 Quality Management System

A Quality Management System (QMS) is a structured framework of policies, processes, and procedures designed to ensure that an organization can consistently deliver products or services that meet customer and regulatory requirements. By implementing a QMS, organizations can standardize and improve their quality control measures, efficiency, and compliance (ASQ, n.d.).

For educational institutions like Asia Pacific College, a QMS ensures the standardization of policies and processes, thereby supporting transparency, accountability, and consistent decision-making. This is particularly critical in managing academic and administrative policies, which affect multiple stakeholders, including students, staff, and regulatory bodies.

2.2.2 Advantages of Web-Based Systems

The types of policy management systems are classified into three: paper-based, mixed media, and digital. Each system presents its unique advantages and evident disadvantages.

During the early ages, policies were initially managed using paper-based systems, where they were written or printed onto physical documents and stored in file cabinets or binders. This method was soon acknowledged as an expensive process, given that the costs of paper and printing may quickly increase. Additionally, managing such files requires significant effort and patience to record, organize, and retrieve information. Paper-based systems are also vulnerable to risks such as fire, loss, theft, or tampering.

Mixed media systems, which combine paper-based methods with digital storage on internal drives, help reduce some of these challenges. This, however, may lead to storage capacity issues. Furthermore, signing documents in such systems can be time-consuming, as it often involves printing, signing, and scanning to maintain a digital record.

Digital systems, while initially more expensive to implement, offer significant long-term savings in time and money. They are recognized to streamline processes, improve accessibility, and reduce errors. Moreover, digital systems also allow for easier collaboration, enabling multiple users to work on the same document simultaneously. Still and all, despite these advantages, digital systems are not without risks, as they are still prone to drawbacks such as hacking or server outages (NEOGOV, 2020).

The introduction of computer technology has encouraged many organizations to transition from paper-based to digital systems. This shift has made it easier to store and organize policies, while also promoting collaboration. Implementing digital systems is particularly beneficial for companies, businesses, and institutions. Asia Pacific College is no exception—adopting digital systems can greatly enhance the institution's efficiency while reducing or eliminating the environmental impact associated with paper-based systems (Haleem, 2022).

2.2.3 Comparison of existing systems

In recent years, educational institutions have implemented Quality Management Systems (QMS) to ensure the quality of their service and further develop it. These institutions have adopted different styles to implement quality management systems. In a study conducted by Santos R. (2019), the applicability of a QMS based on the European Foundation for Quality Model (EFQM) was analyzed to assess its impact on the organizational performance of a Higher Education Institution (HEI).

In the same study, the EFQM model was designed to support organizations in Europe, with the objective of continuous improvement of performance. The principles of the EFQM model are based on these fundamental concepts:

- Add value to clients.
- Building a sustainable future.
- Develop Organizational Capacity.
- Take advantage of Creativity and Innovation
- Leading with Vision, Inspiration and Integrity.
- Manage with agility.

- Succeed through the talent of people.
- Maintain outstanding results.

Based on these concepts, the model presented in figure 2.5 proposes nine criteria to analyze the relationship between an organization's actions and the results it achieves.

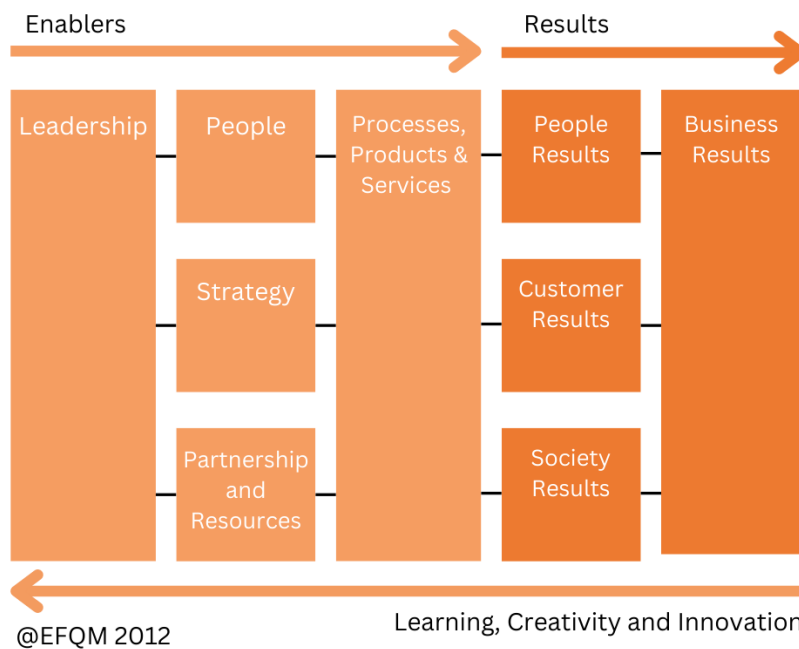


Figure 2.5 – The 9 criteria of the EFQM 2013 Model

The implementation of the EFQM model in this HEI has led to numerous advantages, such as the creation of “quality groups” in each functional area within the institution. Continuously improving processes according to the EFQM principles and creating regular meetings to discuss techniques for improvement and promoting internal standards. Another advantage identified is the operational efficiency of the organization, particularly through cost reduction with supplies and functional departments. As a result, this would then increase output within the institution.

While the merits of the EFQM model are significant, they are not without their disadvantages, as challenges within this specific HEI have arisen. To be specific, this HEI lacks technique towards plans and management to meet the needs of the EFQM model. This challenge was attributed to the institution being part of the public sector, as the EFQM model was originally designed for private-sector organizations. Another challenge encountered was employee resistance to change, which is closely tied to the previous issue. The EFQM model incorporates techniques commonly used in the private sector, making it difficult for public HEI employees to adapt to new procedures, ultimately leading to delays in implementation.

An analysis conducted by Trinidad (2021), examined the impact of ISO 9001: 2015 on the performance of personnel within schools in Imus City. The study specifically focused on the improvement of personnel performance in terms of quality and productivity after ISO 9001: 2015 implementation, the problems that are being faced, and the solutions that are applied to address these problems.

In the context of this study the ISO 9001: 2015 refers to a globally recognized set of standards designed to ensure products and services that meet customer expectations. It incorporates the PDCA cycle and risk-based thinking process approach to promote continuous improvement and effective quality management.

The Plan-Do-Check-Act (PDCA) cycle is a project planning procedure that is used for continuous improvement. The steps of the cycles are as follows: The **Plan** involves identifying the objectives and preparing strategies to achieve them. **Do** is executing what was planned. **Check** is monitoring the results, evaluating the outcomes, and identifying what was learned. And **Act** is implementing improvements based on what was learned.

Risk-based thinking as defined in the study, involves identifying the factors that could affect the organization's processes and quality management system, potentially causing deviations from its goals. It also includes the implementation of actions to minimize the negative effects and maximize the potential opportunities that can be gained.

With a clear understanding of the ISO 9001: 2015 and its process approach, the study now turns to the analysis of findings. The following section presents the results, highlighting the impact of ISO 9001:2015 on personnel performance, the challenges encountered, and the solutions implemented to address these issues.

Based on the findings of Trinidad et al, the implementation of ISO 9001: 2015 has improved the performance of personnel and promoted learning within the organization to foster continuous improvement by focusing on the needs of the clients and stakeholders.

The factors that hinder personnel from conforming to ISO 9001: 2015 standards are the lack of proper training and unclear roles within the organization. Acknowledging these shortcomings can help identify solutions to improve workplace effectiveness.

To address the factors that hinder the implementation of ISO 9001: 2015, personnel conduct regular meetings to discuss findings and continuously monitor outcomes. This approach helps maintain consistency and ensures quality within the organization.

The implementation of quality management systems in educational institutions has significantly enhanced the quality of their services. Although institutions may adopt different quality management models, they share the common goal of meeting the needs of their stakeholders. Research highlights the crucial role of effective management, strong leadership, and clear communication in driving the continuous improvement of service quality.

With that in mind, even though APC has established good quality management practices, its services can still be enhanced by fostering better feedback mechanisms and improving communication among personnel. These efforts will support continuous improvement and ensure that the organization effectively addresses the needs and concerns of all stakeholders.

2.2.4 Use of API

The development of technology has helped our society grow in numerous ways, enhancing efficiency across various industries such as manufacturing, education, and transportation. One of the key innovative technological tools that is widely used is APIs.

According to Efuntade (2023) , an application programming interface (API) is a set of functions that allow the developers to access features and data of the software. Enabling interconnection between programs and other applications. By allowing the connection of different applications, there would be no need to create a module that is only used to execute a specific task. With APIs, the integration of apps for navigation, e-commerce, and social media, easy exchange of data, enabling communications and enhancing overall user experience.

The use of APIs and integration of applications can enable users of the proposed system to implement policies and other methods that support the implementation process within Asia Pacific College.

2.3. Solution Analysis

2.3.1 Functional Requirements

The proposed Quality Management System (QMS) aims to simplify policy management, increase transparency, and ensure responsibility throughout the organization. The proposed solutions will address the issues pointed out in the current system by automating processes, improving role-based access, and ensuring efficient policy recording and management.

a) User Account Creation and Management

The system will allow users to create accounts using their APC emails, requiring verification through a one-time PIN and confirmation by the administrator. Upon successful account creation, the system will provide users with default access to certain functionalities based on their roles. This role-based access ensures that users can only engage with functionalities required for their roles, thereby enhancing system security. Only administrator, QA director, and QA personnel can assign roles, but only the administrator can assign the role of QA director.

b) Policy Management

The system shall enable users to submit, review, verify, approve, and manage institutional policies. Users will be able to develop new policies using established templates, ensuring consistency in format and content. Users will also have access to a repository containing all approved policies, making it easier to organize and retrieve documents as needed. The viewing of policy documents will contain watermarks to prevent unauthorized distribution. Furthermore, the upload of documents will feature an automated generation of version numbering for major or minor revisions and an editable control code managed by the QA personnel and Director.

c) Task Management

This module will show a detailed list of tasks assigned to each user and include the nature of the work. These tasks may vary from user to user as some of these tasks are but not limited to review, verification, approval, etc. Users will be notified of new tasks by system notifications, ensuring timely awareness and action. Task progress will be tracked, with real-time updates supplied to both the assigned user and their supervisors. Furthermore, the system will create reports summarizing task completion rates and outstanding duties, enabling team leaders to optimize burden distribution. This feature guarantees that tasks are adequately managed, deadlines are met, and overall organizational objectives are met efficiently.

d) Policy Review, Verification and Approval

The system will simplify the policy review and approval process by allowing quality assurance (QA) personnel to assign subject matter experts (SMEs) to assist with policy verification and approval. The system will keep track of the status of each policy proposal, whether it is being reviewed, approved, rejected, or must be revised. If a policy is rejected, its creator will receive comments on why the policy does not meet the specified standards and what revisions are needed. This ensures that all policies are thoroughly reviewed before approval, promoting consistency and adherence to organization standards. Furthermore, a unique digital code will be generated to serve as an individual's signature. This code can be referenced within the system database to verify the individual's identity and authentication.

e) Feedback Mechanism

A feedback mechanism will be implemented into the system, allowing users to make recommendations, and comments, and raise issues about specific policies at any point of the process. Users can provide feedback, which will be examined by the QA personnel. Users will receive notifications when their feedback is acted upon, providing transparency and keeping them informed of any changes made in response to their input. This will promote an environment of open communication in which users' perspectives are valued and considered during policy creation and revision.

f) Notification System

The system will contain an automated notification system to alert users to significant events such as new policy proposals, assigned tasks, and policy changes. Users will receive automatic notices when their input has been assessed, responsibilities are due, or the status of a policy in which they are involved changes. These notifications will help users remain on top of their tasks, increasing efficiency and ensuring important activities are not forgotten.

g) Audit Trail Facility

The module will display the revision history of each policy. The audit trail will include detailed information such as timestamps, the individuals involved, and an explanation of the policy modifications. This feature promotes accountability by making it simple to track the history of any policy and review the activities that have occurred over time.

h) Search, Filter, and Sort Functions

The system will allow users to quickly locate specific policies, staff, or relevant data by entering keywords or phrases. Users can filter or sort policies based on a variety of parameters, including date of release, department, policy owner, and policy category. This search and filter functionality will help users navigate the system and get the information they need, increasing overall efficiency and user experience.

i) Reports

The system will provide the QA director with an automated generation of reports about the quality management system. This will allow the QA director to promote the continuous improvement of the system by making decisions that would enhance the effectiveness of the system and alignment with the institutional goals.

2.3.2 Non-Functional Requirements

Beyond its core functionalities, the system's effectiveness can also be determined by its non-functional requirements. These requirements may shape how users perceive and interact with the system. The Non-Functional Requirements of this system are as follows user friendly, ease of use, system performance, system scalability, and system availability.

a) User-Friendly

In terms of user-friendliness, the system's design must be familiar or easy to understand. This ensures that the system can accommodate a wide range of users such as the elderly, younger generations, individuals who are new to the system, less tech-savvy users, and more experienced individuals. If done correctly the system can be easily interacted with and understood by new or technology-challenged individuals while providing a pleasant experience for more seasoned users.

b) Ease of use

The system's ease of use plays a large part. Although it may sound like user friendly, it is entirely different. User-friendly refers to the overall design of the system, ease of use focuses on simplicity and efficiency of task accomplishment of the system. This refers to the ability of the system to allow users to complete certain tasks of the system with less effort and avoid complex procedures.

c) System Performance

Like all websites, it is desired to have a good system performance. It must be able to respond to user actions and requests as quickly as possible. The system must be able to minimize load times and execute actions swiftly. This would ensure a smoother experience for the users of the system.

d) System Scalability

Directly related to system performance is system scalability. It is good to have a responsive system but as time passes by and the system is used by an increasing number of users then it is appropriate to say more and more data is going to be added to the system. This data may be in the form of user data, file data, etc. So, the system must be able to handle the increasing data load efficiently.

e) Availability

System availability refers to the system being accessible to the user from any location and time as long that they have uninterrupted access to the internet and appropriate credentials. The system must be available 24/7 ensuring uninterrupted access to users and providing convenience and reliability.

2.3.3 Technical Requirements

The development of the proposed web-based system must consider several technical requirements. The requirements included in the system are key aspects such as the internet, database, computers, programming languages and platforms, and lastly data storage.

a) Internet

Given that the proposed system is web-based, it will need an adequate internet connection to function properly. To ensure that the system runs smoothly, users should have access to a stable and high-speed internet connection. A strong internet connection ensures that the website remains stable and responsive, allowing users to upload and download files. Furthermore, it will ensure that users can communicate within the system, delivering the best possible experience for all users. With a good internet connection, the system's full functionality will be available without interruption, allowing for real-time interactions and the smooth completion of all tasks.

b) Database

The website's functioning is built on its database, which serves as the core repository for any important information. The project's Database Management System (DBMS) will be MySQL. MySQL will manage essential records such as user accounts, active policies, policy submissions, feedback, and organizational data. By employing MySQL, the system will keep data structured and secure, providing the basis for all the platform's features and functionalities.

c) Computers/Laptops

The system's development will be taking place mostly on PCs or laptops, which will be utilized to produce both the website design and functionality, as well as the underlying database. These devices will be the primary platform for writing the system's code, ensuring that all functionalities are properly integrated and tested. The website will be developed and optimized for desktop use, with a layout and user interface aimed at larger screens with improved navigation.

d) Programming Platform and Language

The system will be implemented utilizing a combination of HTML, CSS, and JavaScript, with each language playing a specific purpose in the website's design. HTML will be utilized to arrange the system's content, ensuring that all text, graphics, and other elements are well-organized within the webpage. CSS will be used to style the system, regulating design elements such as layout, colors, fonts, and overall visual look to produce a user-friendly and visually appealing interface. JavaScript will handle the majority of the system's functionality, including interactive elements like form evaluation, constantly changing content updates, and other behavior-driven execution.

e) Data Storage

The system will oversee maintaining huge quantities of data, including user accounts, policy files, feedback, and other essential data. Given the volume and complexity of this data, it is necessary to have a reliable and secure storage solution. To achieve these requirements, the system will use the AIR HUB Server Storage services. This storage solution will provide the infrastructure required to safely store and manage the system's huge amounts of data. AIR HUB Server Storage ensures that the data is quickly available, well-organized, and secure, allowing the system to run smoothly while protecting the necessary data.

2.3.4 User Requirements

To ensure effectiveness, accessibility, and efficiency, the proposed web-based system must be established with a variety of user requirements. These requirements are focused on ensuring consistent user experience by supporting the different requirements of every user within the system. The system must allow users to efficiently do their activities, such as managing policies, evaluating documents, tracking policy status, and facilitating communication. By meeting these user needs, the system aims to increase the workflow of the overall quality management system.

Administrator

The Administrator is expected to manage user roles, ensuring that each user has the authorizations needed to accomplish their task. They have only power to designate the Quality Assurance (QA) Director role, which serves to ensure efficient oversight and system compliance. The role is restricted to approved staff members, such as the QA Director, ITRO, or other certain individuals, to prevent unauthorized access and maintain security. The Administrator helps to keep the system structured by managing user access and permissions, simplifying role assignments, and ensuring that all users can carry out their roles properly.

Quality Assurance Director

The QA Director oversees quality assurance processes and ensures that policies are carried out effectively. Their tasks include providing input on policies to improve their alignment with institutional goals, as well as obtaining thorough feedback reports to identify and address areas for improvement. In addition, the QA Director gets access to all policy analytics, allowing for data-driven decisions to improve policy performance. They also supervise the entire QA team to ensure collaboration and efficiency in meeting the organization's quality goals.

Quality Assurance Personnel

Quality assurance personnel ensures that the institution's activities consistently meet established standards while fostering continuous improvement. This includes managing policies to assure compliance and relevance, supervising Staff to ensure consistency in performance, and coordinating task management to correspond with institutional goals.

Staff

Staff have an important role in the effective development, revision, and implementation of institutional policies. To keep the institution's operational and academic guidelines updated, they oversee developing new policies, updating existing ones, and submitting them for evaluation. They also ensure that policies are evaluated for accuracy and compliance with institutional standards before adoption, ensuring uniformity and openness at all levels of the company. A collaborative feedback mechanism is also established, allowing academics and staff to provide constructive feedback on proposed policies, ensuring that they match the institution's aims and stakeholders' needs. These tasks are managed within a standardized framework, resulting in a smooth policy lifecycle from conception to approval.

2.3.5 Project Management Requirements

2.3.5.1 Time

The development of the system should be completed at the end of the 3rd Term of the academic year 2025 – 2026. Within the given timeframe, it allows thorough completion of the project and meets the requirements. The project development includes GUI Development, Functionality Development, Database Management, Integration of Functionality with Database, Testing and Documentation.

2.3.5.2 Manpower

To ensure the successful creation and implementation of our proposed system, we thoroughly reviewed the project's specific goals and scope. Based on this assessment, we have determined the main responsibilities required to carry out the development process effectively. These roles will oversee several parts of the project, such as system design, development, testing, and deployment. Each will be assigned specific tasks and responsibilities to ensure that the system fulfills the desired functionality, performance, and user expectations. By establishing a well-structured team with clearly defined tasks, we hope to speed up the development process, effectively manage any obstacles, and provide a high-quality solution that meets our project goals. The manpower needed to develop the system consists of the following:

Front-end Developer	In charge of the front-end side of the system and implementing the user interface
Back-end Developer	In charge of the back-end side of the system including the integration of the database and other external services as well as the logic and functionality of the system.
UI/UX Designer	We must design the overall look and feel of the website as well as ensure that the system provides positive user experience.
Project Manager	Responsible for project planning, conflict resolution, and monitoring progress. This role involves overseeing all aspects of the project and providing overall leadership to ensure its successful execution.
Software Tester	In charge of implementing test procedures and test cases, identifying bugs and issues, and helping the developers solve issues.
Database Developer	In charge of designing the entity relationship diagram, developing the database, and integrating the database into the server.
Documenter	In charge of the documentation for the system

2.3.5.3 Resources

The resources for developing the system will include a range of tools that support different aspects of the project, from design to implementation. These resources will enable the creation of the system's graphic interfaces, structure, and workflows while establishing communication and collaboration among team members. Additionally, it will assist in documenting key information and decisions throughout the project, ensuring everything is well-organized and accessible. Finally, the resources will facilitate the development and management of the system's functionality and data, ensuring that the system is built efficiently and operates smoothly.

Design Tools	Utilized for creating initial designs and prototypes for graphic interfaces.
Diagramming Tools	Used to design and visualize diagrams that represent the structure, processes, and workflows of the system.
Collaboration Tools	Used to facilitate communication and teamwork among team members, enabling seamless sharing of ideas, updates, and feedback throughout the project.
Documentation Tools	Used to record findings, decisions, and solutions related to the system, ensuring all project information is well-organized and easily accessible for future reference.
Program Development Tools	Used to build and develop the system utilizing various programming languages and frameworks, ensuring efficient and accurate implementation of the project's functionality.
Database Management Tools	Used to create, manage, and maintain databases, ensuring efficient storage, organization, and retrieval of data for the system.

Chapter 3

DESIGN AND DEVELOPMENT METHOD

3.1 Technical Process Plan

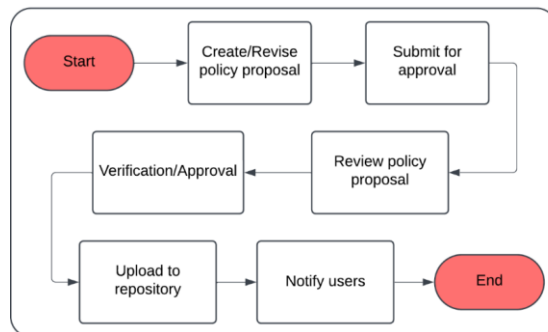


Figure 3.1 Proposed Policy Implementation Process

The policy implementation process outlined in Figure 3.1 begins with staff either creating a new policy or downloading an existing one from the system for revision outside the system. Once revised, the proposed policy is submitted into the system for approval. The first step in the approval process involves a review by the quality assurance (QA) personnel.

After determining whether the proposed policy meets the necessary standards and requirements, the QA personnel either authenticate it using the system's digital signature for further approval or request revisions before proceeding. After passing the quality assurance review, the proposed policy is sent to subject matter experts selected by the QA personnel for verification and approval.

Upon verification and approval by the subject matter experts, the proposed policy is authorized for upload to the system. The QA personnel will store the approved policy in the policy repository for proper organization and user access. The system automatically sends notifications to alert users about new or revised policies.

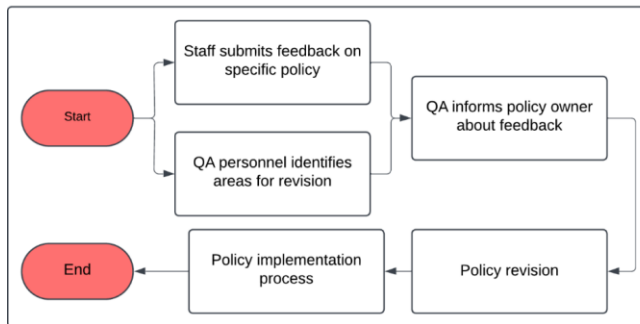


Figure 3.2 *Proposed Feedback Process*

In Figure 3.2, the proposed feedback process can begin with staff submitting suggestions, comments, or concerns on specific policies or with the QA personnel identifying policies, processes, or procedures requiring revision. The QA personnel will notify the policy owner regarding necessary revisions or updates. After that, the policy owner reviews and revises the concerned policy as needed. Finally, the revised policy will go through the policy verification and approval process.

3.1.1 Proposed Functional Description

The proposed system is a web-based policy management repository designed for the Asia Pacific College staff to provide a platform for storing, managing, and retrieving institutional policies. This system aims to ensure that all relevant documents are organized, up-to-date, and available to users within the organization. The following are the core functionalities of the proposed system:

User Access, Authentication, and Roles

The system would provide a login system where staff within Asia Pacific College could access the system. Authentication of login will be based on the user's APC email verifying their credentials before gaining access to the website. For security measures, the system will incorporate one-time pins which will be received via email. The system also supports role-based access, assigning users to specific roles that define their access level and permitted features within the system.

Policy Implementation

The system allows users to view the most recent and active policies in the repository, ensuring they have access to the latest institutional guidelines. In addition to viewing policies, the system offers the capability for users to create or update policies. This is done through an editable file template provided through the system, which gives users the ability to modify policies as needed. Once the user has finished creating or updating a policy, they can submit the file directly into the system for review.

Upon submission, the file will be received by the Quality Assurance Personnel, who will conduct an initial review of the proposed policy to ensure that it meets the required standards and aligns with institutional objectives. After the Quality Assurance Personnel has completed their review, the policy will be forwarded to subject matter experts within APC, who will verify the relevance of the content. The subject matter experts will also assess the policy for any necessary changes or improvements before granting final approval. Once the proposed policy has gone through the review, verification, and approval processes, it will be stored and organized in appropriate folders within the repository. This ensures that the policy is properly categorized and easily accessible for future implementation.

These core functionalities form the system's foundation, ensuring an efficient and secure approach to policy management. In addition to this foundation, the system will also need external tools. It will require database integration to store and manage policy documents and other relevant data. Additionally, APIs will be used for security features and watermarking to protect document integrity and prevent unauthorized distribution.

3.2. Data Model

3.2.1 Entity Relationship Diagram

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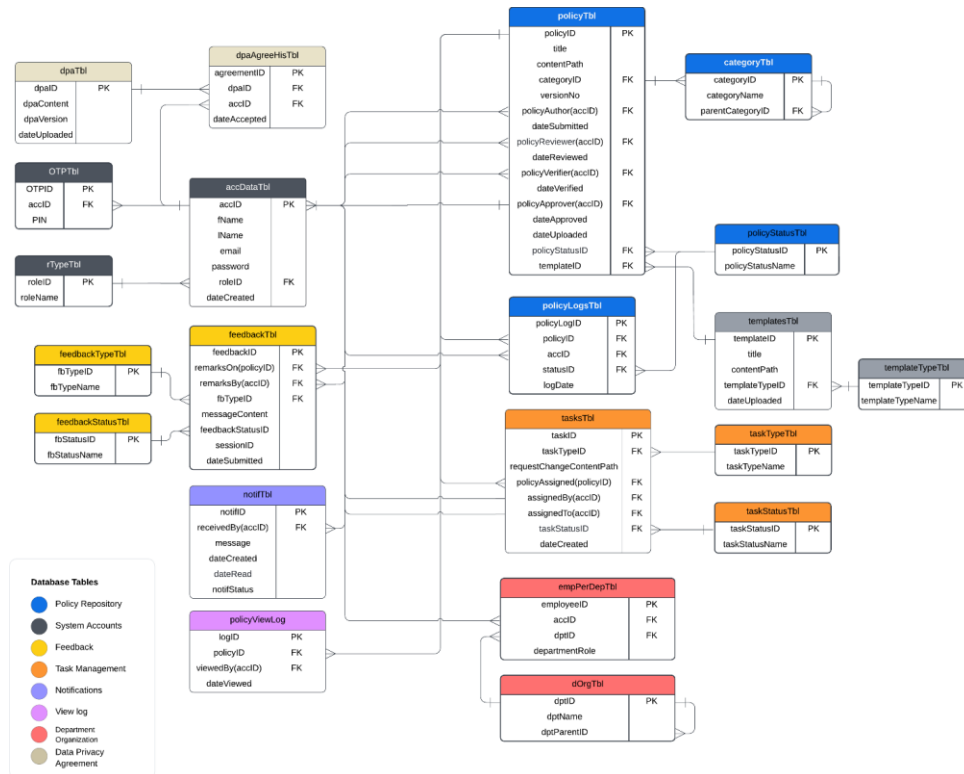


Figure 3.3 QMS Entity Relation Diagram

In Figure 3.3, the Entity-Relationship Diagram of the Quality Policy Management System is presented. The diagram is color-coded for easier identification and illustrates ten entities. These entities store critical data, such as names, email addresses, files, etc. Each table holds specific information and interacts with other tables, ensuring a structured and cohesive data flow throughout the system. The Document History table records policy versions, including submission, verification, and approval details. It tracks the document's author, verifier, and approver, along with their respective dates of submission, verification, and approval as well as their digital signatures. The Feedback table records remarks made by users, including the person who provided the feedback, the entity it is directed at, the content of the remark, and the date it was submitted.

The Notification table stores system-generated messages sent to users, including the recipient, message content, and the date the notification was sent. The Policy Repository table stores information about policies, including the policy ID, file name, content, version number, and the folder where it is located. The System Accounts table holds the user information this includes the account ID, the user's name and last name, email, password, and assigned role within the system. The Submissions Table holds the proposed policy details which include the filed policy ID, file name, content, author, and submission date. It also keeps track of the verifier and approver, the dates of verification and approval, and the policy status.

The Status table stores status-related information, including the status ID and the statuses of policies and tasks. Policy statuses include 'Reviewed,' 'Verified,' and 'Approved,' while task statuses include 'In Progress' and 'Not Started.' These statuses vary by module and help track the progress of policies and tasks within the system. The Task Management table includes task-related data, such as the task ID, assigned user, task creator, and task type. It also holds the policy ID to direct the user on what policy they will be working on.

The View Log table records user activity related to policy access, including the user who viewed a policy, the date it was viewed, and whether the policy was downloaded. The Digital Signature History Table holds the records of the signature of the users, and the policies where the signature was applied. Lastly, User Organization organizes account ID through employee ID and to the respective departments.

3.2.2 Data Dictionary

The proposed system has 15 tables of data with each varies in number and their uses. Each table has a field name to identify the name of the data, data type to identify the kind of data values that can be stored in the database these maybe an integer, character etc., constraints enforce rules in the database that prevents invalid data types. Field size refers to the maximum amount of data that can be stored, data format defines how a data is stored and displayed depending on the data type. The tables also provide descriptions and examples to further clarify the use of the data in the table.

Commented [JR2]: Same with erd

Table 3.1 Account Data Table

accDataTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
accID	INT	PK, NN, AI	10	Unique Identifier for Account Data	12
fName	VARCHAR	NN	45	First Name of account holder	Dave
lName	VARCHAR	NN	45	Last name of account holder	Trillanes
email	VARCHAR	NN	45	APC email used by the acc holder	example@faculty.apc.edu.ph
password	VARCHAR	NN	45	password of acc holder	nahhh
role	INT	FK, NN	2	Foreign key used to identify which role the user is assigned to	1
dateCreated	Date	NN	10	The date the account was created	12-07-27

Tables 3.1, 3.2 and 3.3 holds related data regarding user account detail, roles and security. The accDataTbl or the Account Data table holds the user data which are the account ID (accID), their first and last name (fname & lname), their email, password, role and the date of account creation (date created). The rTypeTable or the Role Type Table holds the role type a user is assigned to, and the role ID. The OTPTbl or the One Time Pin Table temporarily stores the 6-digit pin sent to user for security verification.

Table 3.2 *Role Type Table*

rTypeTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
roleID	INT	PK, NN, AI	2	Unique identifier for the Role Type Table	2
roleName	VARCHAR	NN	45	Name for the Role type	Quality Assurance

Table 3.3 *One Time Pin Table*

OTPTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
OTPID	INT	PK, NN, AI	6	Unique Identifier for the One Time Pin data	100123
requestBy	INT	FK	10	Foreign Key that identifies who requested an OTP	200567
PIN	CHAR	PK, NN, AI	6	The generated One Time Pin used for authentication	845812

Table 3.4 *Employee per Department Table*

empPerDepTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
employeeID	INT	PK, NN, AI	10	Unique identifier for each entry in the Employees per department table	100123
accID	INT	FK, NN	10	Foreign key used to identify the employee	200567
dptID	INT	FK, NN	10	The foreign key used to identify which department the employee is under	301
departmentRole	VARCHAR	NN	45	The role of the user within the aforementioned department	Head

Table 3.4 Employee per Department Table or empPerDepTbl holds data regarding the department organization. This is where user account data is organized to their respective departments within Asia Pacific College by linking the employee ID to the department ID. The data held here are the employee's ID(employeeID), the account ID(accountID), department ID(dprID), and the role of the employee within the department(departmentRole).

Table 3.5 *Departments Organization Table*

dOrgTbl						
Field name	Data Type	Constraints	Field Size	Data Format	Description	Example
dptID	INT	PK, NN, AI	6	111111	Unique identifier for each entry in the Departments table	301
dptName	VARCHAR	NN	45	CCCCCCCCC...	Name of the Department	Building Management Office
dptParentID	INT	FK, NN	6	111111	indicates which department it is under	100
dptHeadID	INT	FK, NN	10	1111111111	Identify who is the head of the department	200567

Table 3.5 Department Organization Table or dOrgTbl holds data which are department ID (dptID), department name (dptName), department parent ID (dptParentID) and department head ID (dptHeadID). This table show the organization of the departments in Asia Pacific College within the system.

Table 3.6 *Feedback Table*

feedbackTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
feedbackID	INT	PK, NN ,AI	10	Unique identifier for Feedback	500
remarksOn(policyID)	INT	FK, NN	10	Foreign key used to identify which policy was the feedback directed to	12345
remarksBy(acclID)	INT	FK, NN	10	Foreign key used to identify who wrote the remarks about a certain policy	12345
fbType	INT	FK, NN	2	Foreign Key to that identifies the type of feedback	2
content	LONGTEXT	NN	250	Message of the commenter for the Quality Assurance and/or the Policy Owner	Regarding the Policy....
sessionID	INT	NN	10	Refers to a unique identifier assigned to a specific session in a system	21
dateSubmitted	DATE	NN		Used to identify when the feedback was submitted	12-12-25

Table 3.6 Feedback Table or feedbackTbl handles feedback organization and storage. The table holds data such as the feedback ID(feedbackID), the policy the feedback is for(remarksOn), the author of the feedback(remarksBy), the type of feedback(fbType), the content of the feedback(content), the session number(sessionID), and the date the policy is submitted(dateSubmitted).

Table 3.7 *Feedback Type Table*

feedbackTypeTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
fbTypeID	INT	PK, NN ,AI	10	Unique identifier for Feedback Type	500
feedbackTypeName	VARCHAR	NN	45	The name of the feedback type	Report

Table 3.8 *Feedback Status Table*

feedbackStatusTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
fbStatusID	INT	PK, NN ,AI	10	Unique identifier for Feedback Status	500
fbStatusName	VARCHAR	NN	45	The name of the status feedback	Read

Table 3.7 Feedback Type Table or feedbackTypeTbl stores and handles the types of feedback done. The data stored here are the feedback type ID(fbTypeID) and the name of the type of feedback(feedbackTypeName). Meanwhile Table 3.8 Feedback Status Table stores holds the different policy statuses.

Table 3.9 Task Table

taskTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
taskID	INT	PK	10	Unique identifier for tasks	1001
taskTypeID	FK	FK	10	Foreign Key identifying the type of task	2001
requestChangeContentPath	VARCHAR	NN	45	A link that redirects user to the policy to be changed	/files/request.docx
policyAssigned(policyID)	INT	FK	10	Foreign Key referencing the assigned policy	3001
assignedBy(accID)	INT	FK	10	Foreign key that identifies who assigned the task	201
assignedTo(accID)	INT	FK	10	Foreign key that identifies who was assigned a task	301
taskStatusID	INT	FK	10	Foreign Key identifying the status of the task	1001
dateCreated	DATE	NN	10	The date when the task was created	18-02-2005

Table 3.10 Task Type Table

taskTypeTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
taskTypeID	INT	PK	10	Unique identifier for the different types of tasks	1001
taskTypeName	VARCHAR	NN	45	Name of the task type	Review

Table 3.11 *Task Status Table*

taskStatusTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
taskStatusID	PK	PK	10	Unique identifier for the status of the task	1001
taskStatusName	VARCHAR	NN	45	Name of the status of the tasks	Pending

Table 3.8 Task Table or taskTbl is responsible for handling data that regulates the task within the system. It records data such the ID of the task(taskID), the ID of the type of task(taskTypeID), a link to redirect user to the policy to be changed(requestChangeContentPath), the assigned policy(policyAssigned), the person who assigned the task(assignedBy), the person the task was assigned to(assignedTo), the ID of the task status(taskStatusID), and the date the task was created(dateCreated). Tables 3.9 and 3.10 are referenced in Table 3.8. Table 3.9 Task Type Table or taskTypeTbl is responsible for storing the type of task while Table 3.10 Task Status Table or taskStatusTbl is responsible for for storing the statuses of a policy.

Table 3.12 Policy Table

policyTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
policyId	INT	PK, NN, AI	10	Unique Identifier for the policy documents	1
title	VARCHAR	NN	45	The title of the policy	Quality_Policy.docx
contentPath	VARCHAR	NN	45	A link that redirects user to the content of the file	/files/policy_content.docx
categoryId	INT	NN	10	Category Identifier of the file	2
versionNo	INT	NN	10	Version number of the policy	3
policyAuthor(accID)	INT	FK	10	Foreign key that identifies the author of the proposed policy	102
dateSubmitted	DATE	NN	8	The date the proposed policy is submitted	12-02-25
policyReviewer(accID)	INT	FK	45	Foreign key that identifies the reviewer of the proposed policy	205
dateReviewed	DATE	NN	10	The date the proposed policy is reviewed	14-02-2025
policyVerifier(accID)	INT	FK	10	Foreign Key that identifies the verifier of the proposed policy	310
dateVerified	DATE	NN	10	The date the proposed policy is verified	15-02-2025
policyApprover(accID)	INT	FK	10	Foreign key that identifies the approver of the proposed policy	412
dateApproved	DATE	NN	10	The date the proposed policy is approved	16-02-2025
dateUploaded	DATE	NN	10	The date the policy is uploaded.	17-02-2025
policyStatusID	INT	FK	2	Foreign key that identifies the status of the proposed policy	1
templateID	INT	FK	2	Foreign key that identifies the template used	1

Table 3.13 *Category Table*

categoryTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
categoryID	INT	PK, NN, AI	10	Unique Identifier for the policy category	1
categoryName	VARCHAR	NN	45	Name of the policy category	HR Policies
parentCategoryID	INT	FK	10	Foreign Key referencing parent category	2

Table 3.14 *Policy Logs Table*

policyLogsTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
policyLogID	INT	PK, NN, AI	10	Unique Identifier for each policy log entry	1001
policyID	INT	FK	10	Foreign Key referencing the policy	1
accID	INT	FK	10	Foreign Key referencing the user responsible for policy updates	205
statusID	INT	FK	2	Foreign Key referencing the policy status	1
logDate	INT	NN	10	The date when the log entry was created	17-02-2025

Table 3.15 *Policy Status Table*

policyStatusTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
policyStatusID	INT	PK, NN, AI	10	Unique Identifier for each policy status	1
policyStatusName	VARCHAR	NN	45	Name of the different statuses of the policy	Approved

Table 3.11 Policy Table or policyTbl is responsible for holding information regarding a policy. The table holds data such as the ID of a policy(policyID), the title of the policy(title), the link that redirects user to the content of the policy(contentPath), the ID of the policy category which is referenced from Table 3.12,(categoryID), the version number of the policy(versionNo), the author, reviewer, verifier and the approver of the policy(policyAuthor, policyReviewer, policyVerifier, policyApprover) as well as the dates each task was done(dateSubmitted, dateReviewed, dateVerified, dateApproved). It also references the status of the policy(policyStatusID) from Table 3.14 Policy Status Table and the templateID.

Table 3.16 *Notification Table*

notifTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
notifID	INT	PK, NN, AI	10	Unique identifier for notifications	3001
recievedBy(accID)	INT	FK, NN	10	Foreign key that identifies who received the message	501
message	SHORTTEXT	FK, NN	120	The actual content of the message	
timestamp	DATE	NN	26	The date and time the notification was sent	2025-02-17-12-30-00
dateRead	DATE	NN	10	The date of the notification was read.	2025-03-17
notifStatus	INT	NN, DEFAULT, BOOLEAN	1	Checks if the notification was read or unread by the user	0

Table 3.13 Notification Table or notifTbl handles the organization and assigning of notifications to the users of policy updates or tasks. The table holds information such as the notification ID(notifID), the users that received the notification(receivedBy), the content of the notification(message), the date and time of the notification was sent(timestamp), the date the notification was read by the user(dateRead) and the status of the notification if it was read by the user(notifStatus).

Table 3.17 Policy Viewing Table

<i>polViewTbl</i>					
<i>Field name</i>	<i>Data Type</i>	<i>Constraints</i>	<i>Field Size</i>	<i>Description</i>	<i>Example</i>
<i>logID</i>	<i>INT</i>	<i>PK, NN, AI</i>	<i>10</i>	<i>Unique identifier for policy viewing log</i>	<i>2001</i>
<i>policyID</i>	<i>INT</i>	<i>FK, NN</i>	<i>12</i>	<i>Foreign key that identifies what policy was viewed</i>	<i>12345</i>
<i>viewedBy(accID)</i>	<i>INT</i>	<i>FK, NN</i>	<i>10</i>	<i>Foreign Key that identifies who viewed a certain policy</i>	<i>301</i>
<i>dateViewed</i>	<i>DATE</i>	<i>NN</i>	<i>1</i>	<i>The date the policy was viewed</i>	<i>15-02-2025</i>

Table 3.17 Policy Viewing Table or *polViewTbl* manages and records the users that view a certain policy for auditing purposes. The table holds data such as the log ID(*logID*), the person who views a policy(*viewedBy*), the policy that was viewed(*policyID*), and the date the policy was accessed(*dateViewed*).

Table 3.18 Data Privacy Act Table

<i>dpaTbl</i>					
<i>Field name</i>	<i>Data Type</i>	<i>Constraints</i>	<i>Field Size</i>	<i>Description</i>	<i>Example</i>
<i>dpaID</i>	<i>INT</i>	<i>PK, NN, AI</i>	<i>10</i>	<i>Unique identifier for Data Privacy Act</i>	<i>1</i>
<i>dpaContent</i>	<i>SHORTTEXT</i>	<i>NN</i>	<i>120</i>	<i>The actual content of the Data Privacy Act</i>	<i>Republic Act 10173, or the Data Privacy Act (DPA) of 2012.</i>
<i>dateUploaded</i>	<i>DATE</i>	<i>NN</i>		<i>The date the DPA was uploaded</i>	<i>2025-03-01</i>
<i>dpaVersion</i>	<i>INT</i>	<i>NN</i>	<i>10</i>	<i>The version of the DPA</i>	<i>7</i>

Table 3.18 Data Privacy Act Table or dpaTbl manages and records the Data Privacy Act that has been set. If the Data Privacy Act is ever be updated or changed then this table is responsible for holding the changes. The table holds information such as the DPA ID(dpaID), the content of the DPA(dpaContent), the date it was uploaded(dateUploaded and the version of the DPA(dapVersion).

Table 3.19 Data Privacy Act Agreement History Table

fdpaAgreeHisTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
agreementID	INT	PK, NN, AI	10	Unique identifier for Data Privacy Act	3
acctID	INT	Fk	10	The actual content of the Data Privacy Act	23
dpaID	INT	FK	10	The date the DPA was uploaded	12
dateAccepted	DATE	NN	10	The version of the DPA	2025-03-01

Table 3.19 Data Privacy Act Agreement History table or dpaAgreeHisTbl records the user's agreement of the Data Privacy Act. The table holds information such as the agreement ID(agreementID), the account ID(acctID), the DPA ID(dpaID), and the date the user accepted the terms and conditions of the DPA(dateAccepted)

Table 3.20 *Templates Table*

templatesTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
templateID	INT	PK, NN, AI	6	Unique Identifier for downloadable templates	100123
title	VARCHAR	NN	50	The title of the downloadable template	Policy_Template
contentPath	VARCHAR	NN	45	A link that redirects user to a specified template	/templates/policy.docx
templateTypeID	INT	FK	10	Foreign key referencing the template type	1001123
dateUploaded	DATE	NN	10	The date when the template was uploaded	19-02-2005

Table 3.21 *Template Type Table*

templatesTypeTbl					
Field name	Data Type	Constraints	Field Size	Description	Example
templateTypeID	INT	PK, NN, AI	6	Unique Identifier for types of templates	100123
templateTypeName	VARCHAR	NN	50	The name of the type of template	Change Request Form

Table 3.20 Templates Table or templatesTbl is responsible for holding and organizing the downloadable templates for user's to use. The data held here are the ID of the template(templateID), the title of the template(title), the link that redirects user to the donloadable template(contentPath), the IDs of the types of templates, which was referenced in Table 3.21 Template Type Table or templatesTypeTable,(templateTypeID) and the date the template was uploaded(dateUploaded).

3.3. Process Model

3.3.1 Data Flow Diagram

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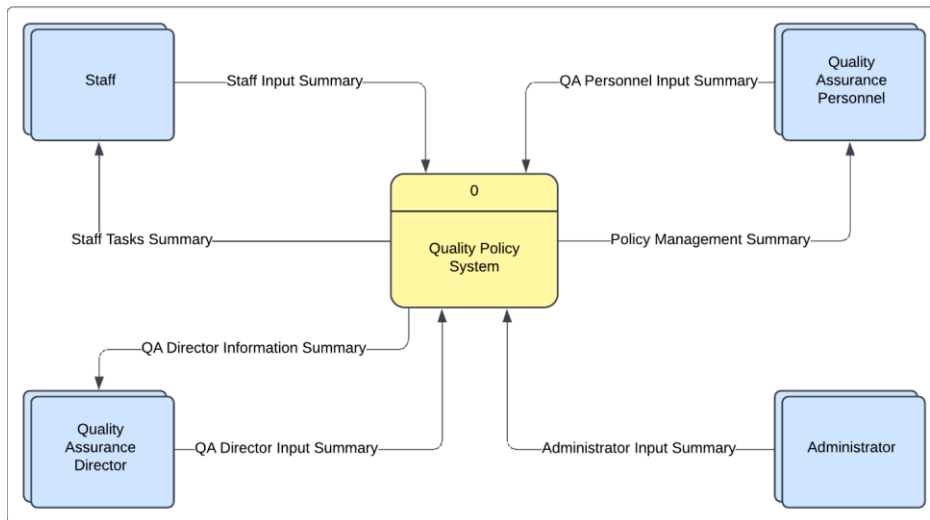


Figure 3.4 Quality Management System Context Diagram

Figure 3.4 shows the context diagram of the system, giving an overview of the data shared between the system and its users. The main users involved are the Staff, QA Personnel, QA Director, and the Administrator. The figure also shows a summary of the flow of data between the users and the system. The Staff provides the system with policy information, verification and approval information, and feedback information. Then, the QA personnel provide the system with review information, upload information, and feedback information. As for the QA Director, they provide the system with feedback information based on the reports that are generated by the system. They also provide policy categorization and department information for the organization of data. The Administrator provides role information for the designation of roles to each user and content of the Data Privacy Agreement.

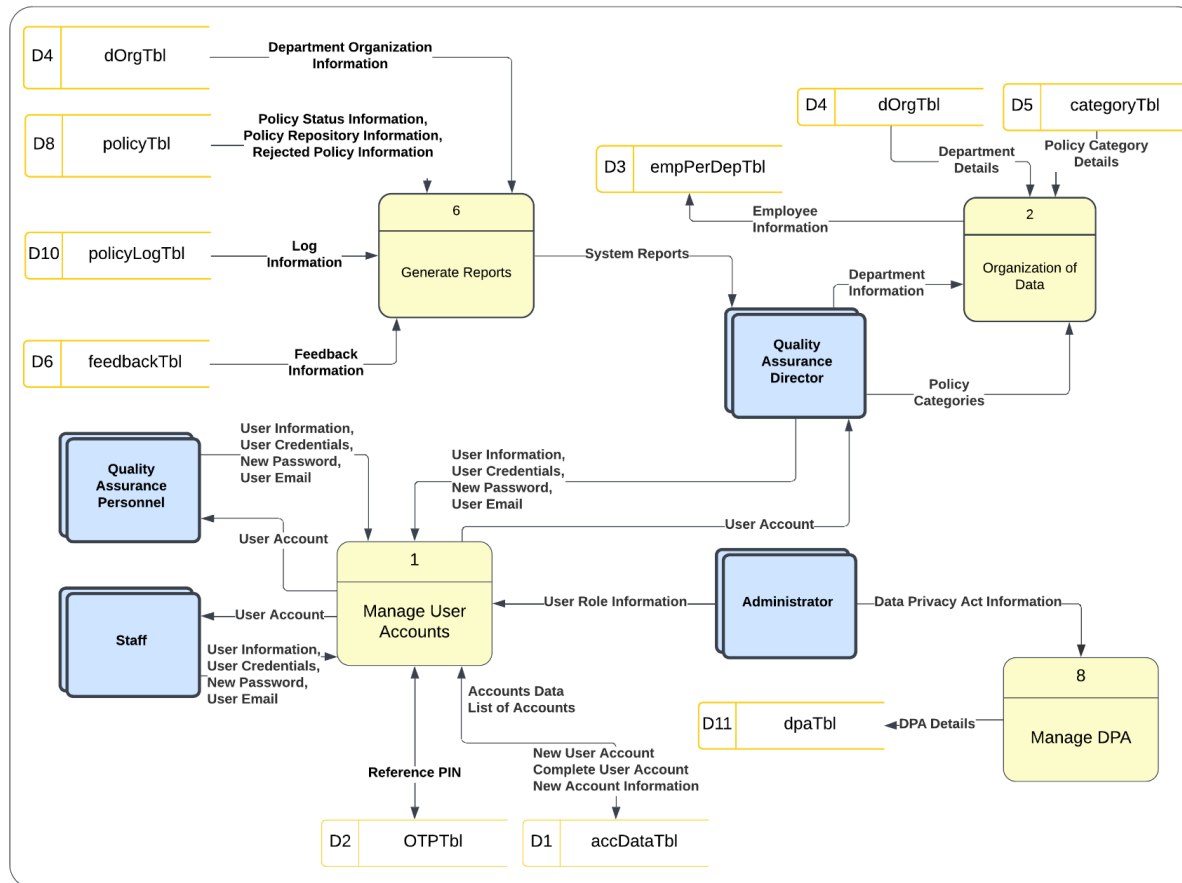


Figure 3.5 QMS Level 0 Data Flow Diagram Part 1

The Level 0 Data Flow Diagram consists of a single Data Flow; however, it has been divided into two parts which are Figure 3.5 and 3.6 to enhance image clarity. The processes and data stores are defined as follows:

Processes

- 1.0 Manage User Accounts:** The process accepts user information and role type details from the administrator to designate role access to the users.
- 2.0 Organization of Data:** The process receives policy category information and department category information from the QA Director for the management of policy folders and department organization.
- 3.0 Policy Submission:** This process accepts policy proposals and generates tasks for QA personnel to review. It also includes the process of requesting revisions, either voluntarily or as assigned.
- 4.0 Policy Implementation Process:** This process receives details from notifications and tasks, including review, verification, approval, and categorization information, to facilitate the implementation of the policy.
- 5.0 View Policy Process:** This process takes input from users, to find desired policies. It also receives feedback input from the users to create tasks for revision.
- 6.0 Generate Reports:** This process takes policy, feedback and department information to generate reports for the QA Director to analyze
- 7.0 Notification Process:** This process is triggered whenever there is a new policy, feedback, task, or update. It generates notifications to inform users accordingly.
- 8.0 Manage DPA:** This process accepts input from the Administrator to make changes to the data privacy agreement that will be displayed

Data Stores

D1 accDataTbl: Stores user account information

D2 OTPTbl: This temporarily stores system-generated PINs for user verification.

D3 empPerDepTbl tbl: Stores information about employees assigned to a specific department.

D4 dOrgTbl: Stores department information for organization.

D5 categoryTbl: Stores information about policy categories.

D6 feedbackTbl: Stores the feedback information submitted by the users.

D7 tasksTbl: Stores information about tasks that were generated.

D8 policyTbl: stores all policy information.

D9 notifTbl: Stores information about notification messages.

D10 polViewTbl: Stores logs of policy updates for tracking and reference.

D11 dpaTbl: Stores versions of data privacy agreements.

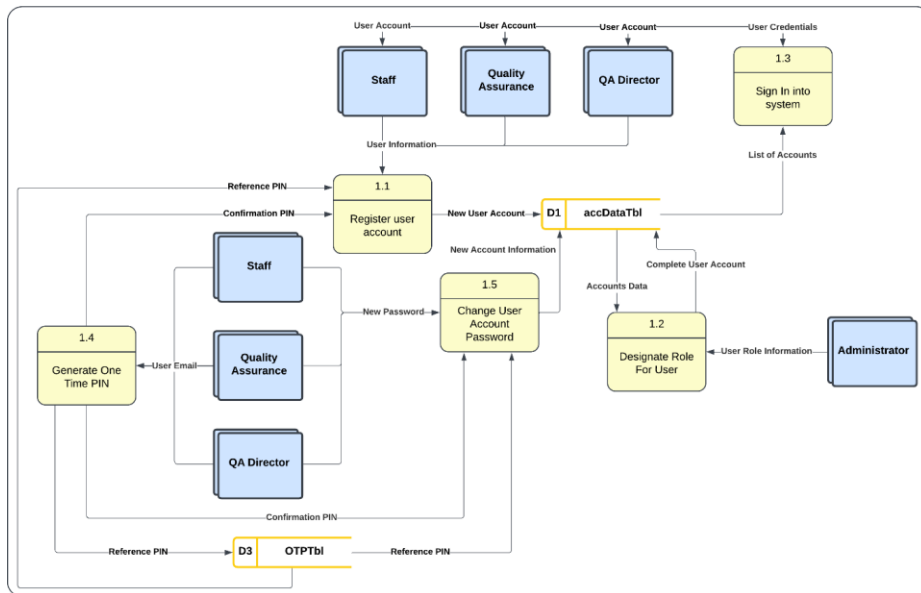


Figure 3.7 Level 1 Diagram -1.0 Manage User Accounts

Figure 3.7 shows the Level 1 Data Flow Diagram for "Manage User Accounts," which describes the processes involved in user account administration. This diagram illustrates the creation of accounts, role assignment, system login, one-time PIN generating, and password changes. The registration procedure includes security features such as Reference and Confirmation PINs, and the data collected is stored in specified data repositories. The diagram additionally shows the assignment of user roles by administrators, which determines system access. It also displays the usual user login procedure, the creation of temporary PINs for increased security or password resets, and the password-changing process.

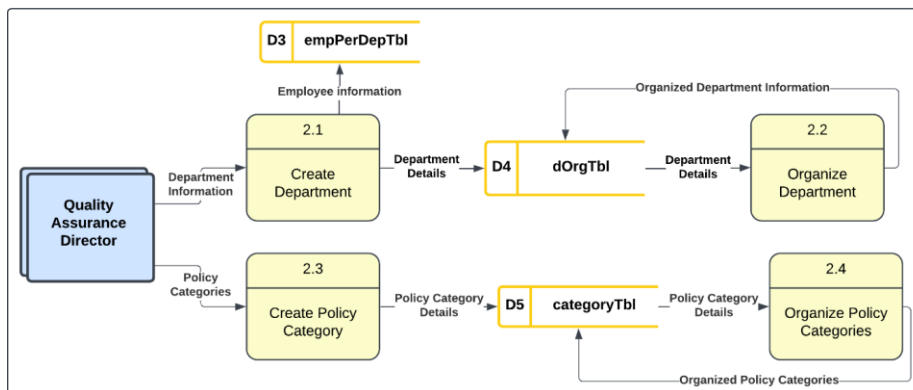


Figure 3.8 Level 1 Diagram – 2.0 Organization of Data

Figure 3.8 depicts the Level 1 Data Flow Diagram for "Organization of Data," illustrating the processes involved in structuring department and policy category information. The process is initiated by the Quality Assurance Director. First, From the input of the Quality Assurance Director "Employee Information" from the "empPerDepTbl" database (D3) is used as input by the "Create Department" process (2.1) to generate "Department Details." These details are then stored in the "dOrgTbl" database (D4) and subsequently used by the "Organize Department" process (2.2) to create "Organized Department Information." In parallel, the Quality Assurance Director provides "Policy Categories" as input to the "Create Policy Category" process (2.3). This process generates "Policy Category Details," which are stored in the "categoryTbl" database (D5). The "Organize Policy Categories" process (2.4) then uses these details to produce "Organized Policy Categories." This diagram effectively illustrates how the Quality Assurance Director manages the organization of department and policy category data through the creation and organization of detailed information

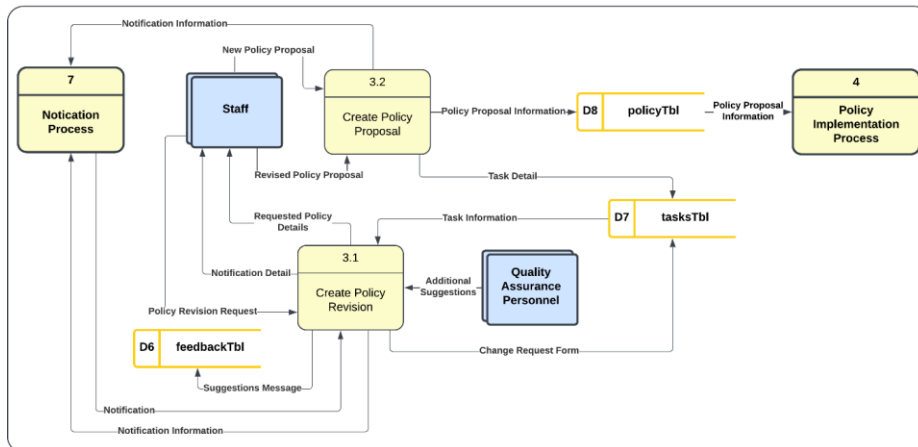


Figure 3.9 Level 1 Diagram -3.0 Policy Submission

Figure 3.9 illustrates the Level 1 Data Flow Diagram for "Policy Submission," detailing the steps involved in proposing, revising, and implementing policies within an organization. The process begins with "Staff" initiating a "New Policy Proposal" which is processed by the "Create Policy Proposal" process (3.2). This process generates "Policy Proposal Information" that is stored in the "policyTbl" database (D8). The "Policy Proposal Information" is then passed to the "Policy Implementation Process" (4). If revisions are needed, the "Policy Implementation Process" (4) sends "Requested Policy Details" back to the "Staff." The "Staff" then initiates the "Create Policy Revision" process (3.1), which stores "Task Detail" in the "tasksTbl" database (D7). "Quality Assurance Personnel" can provide "Additional Suggestions" for the revision, which are incorporated into a "Change Request Form." This form updates the "feedbackTbl" database (D6) with "Suggestions Message." The "Staff" receives a "Revised Policy Proposal" to resubmit to the "Policy Implementation Process" (4). Throughout the process, the "Notification Process" (7) sends "Notification Information" to the "Staff" and receives "Notification Detail" from the "Create Policy Revision" process (3.1), ensuring all stakeholders are informed about the status of the policy proposal.

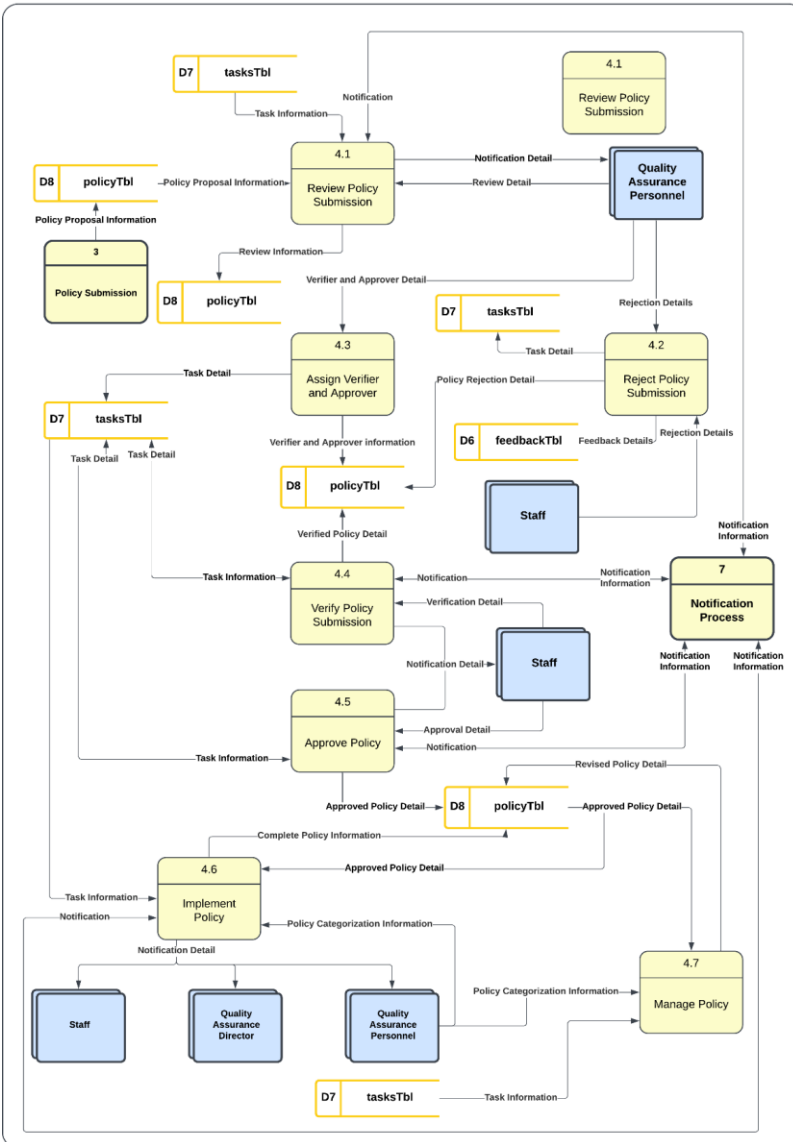


Figure 3.10 Level 1 Diagram - 4.0 Policy Implementation Process

Figure 3.10 illustrates the comprehensive "Policy Implementation Process," outlining the systematic steps involved in reviewing, verifying, approving, and implementing new policies. Initially, a submitted policy undergoes a thorough review, utilizing information from the 'tasksTbl' table. This review generates a detailed report that is disseminated to both the Quality Assurance Personnel and the Staff. Subsequently, the process designates individuals to verify and approve the policy, recording this assignment information in the 'policyTbl' table and generating corresponding tasks in the 'tasksTbl' table. If the policy is deemed unsatisfactory during the initial review, the Quality Assurance Personnel receive detailed feedback and rejection information, which is systematically documented and stored in the 'tasksTbl' and 'feedbackTbl' tables. Conversely, if the policy meets the initial review criteria, it proceeds to a meticulous verification stage, utilizing information from the 'tasksTbl' table. The comprehensive verification results are then communicated to the Staff. Upon successful verification, the policy advances to the approval stage, culminating in the recording of all approved policy details within the 'policyTbl' table. Finally, the policy is formally implemented, triggering notifications to the Staff and furnishing the Quality Assurance team with pertinent policy categorization information. Furthermore, the process encompasses a mechanism for managing existing policies, enabling the Quality Assurance Personnel to mark policies as obsolete and to transfer policies between different departments or folders as required.

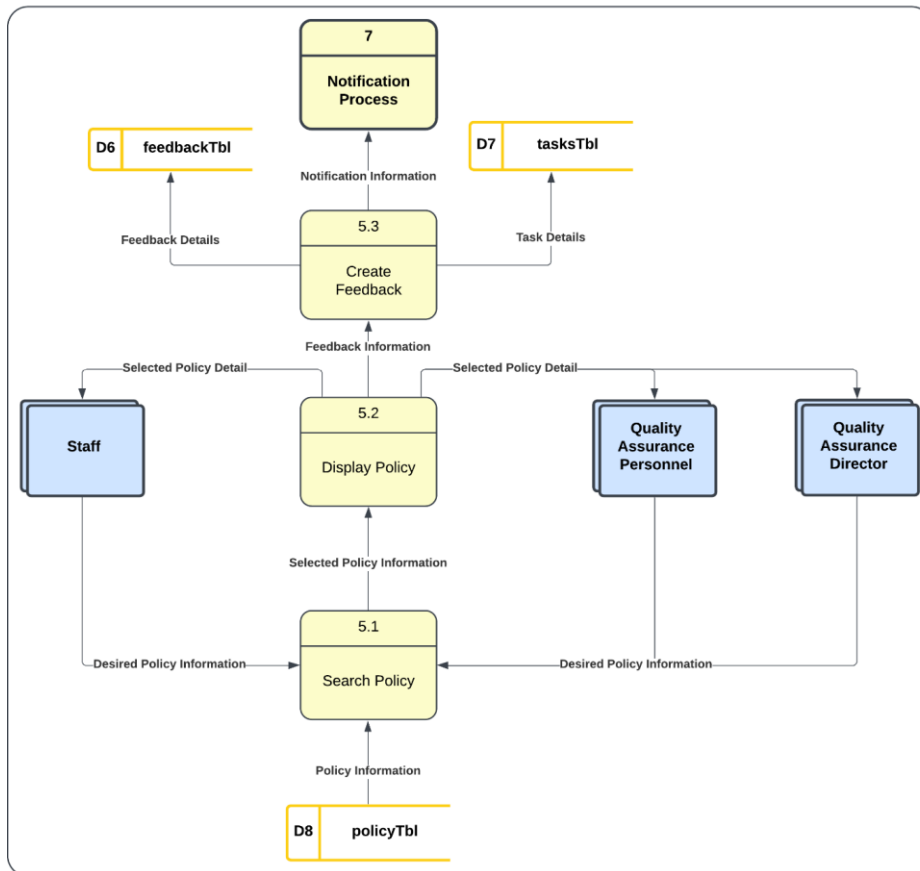


Figure 3.11 Level 1 Diagram -5.0 View Policy Process

Figure 3.11 illustrates the "View Policy Process," outlining the procedures involved in accessing, reviewing, and providing feedback on policies within the system. This process commences when a user, typically a staff member, initiates a search for a specific policy using the "Search Policy" process (5.1). The user inputs their desired search criteria, and the system queries the "policyTbl" table (D8), which houses all policy-related data. Upon locating the relevant policy, the system retrieves the comprehensive policy details and presents them to the user through the "Display Policy" process (5.2), facilitating thorough examination.

Should the user identify areas for improvement or wish to provide suggestions, they can utilize the "Create Feedback" process (5.3) to articulate their feedback in detail. This feedback is then systematically recorded in the "feedbackTbl" table (D6) for future reference and analysis. Concurrently, the system generates a notification containing the feedback details and pertinent task information from the "tasksTbl" table (D7), which is then transmitted to the Quality Assurance Personnel and the Quality Assurance Director, ensuring prompt awareness and enabling appropriate action.

Furthermore, the "Notification Process" (7) is for disseminating information and maintaining transparency by notifying the Staff about the submitted feedback, ensuring that relevant stakeholders remain informed and involved throughout the feedback process.

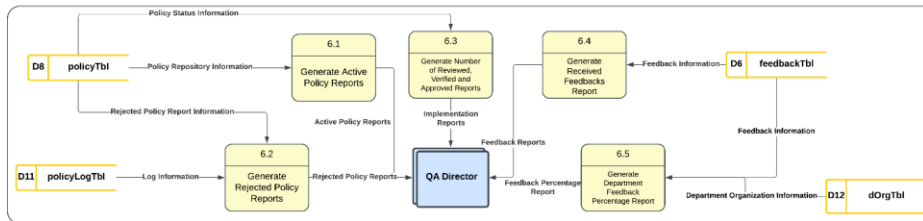


Figure 3.12 Level 1 Diagram - 6.0 Generate Reports

Figure 3.12 depicts the process of creating several reports for the Quality Assurance (QA) Director, which include insights on policy statuses and feedback trends. The procedure uses data from several tables to provide reports, making it easier to make informed decisions and maintain policies. First, the system uses the "policyTbl" table (D8), which contains detailed information about each policy, to generate two key reports: a report on active policies (6.1), which provides an overview of currently implemented policies, and a report on rejected policies (6.2), which provides insights into policies that did not meet the required standards. The second report additionally includes log information from the "policyLogTbl" database (D11), which provides a historical record of rejected policies.

In addition to policy status reports, the system creates reports on policy progress and feedback. The "Policy Progress Report" (6.3) uses data from the "tasksTbl" table (D7) to display the number of policies that have successfully passed through the review, verification, and approval stages, emphasizing the effectiveness and efficiency of the policy implementation process. Meanwhile, the "Feedback Report" (6.4) uses data from the "feedbackTbl" table (D6) to gather and present all feedback received, allowing the QA Director to obtain a thorough understanding of user viewpoints and identify potential policy improvements.

Furthermore, the system includes a "Feedback by Department Report" (6.5), which uses data from both the "feedbackTbl" database (D6) and the "dOrgTbl" table (D12) to analyze and report the percentage of feedback generated by each department.

3.3.2 Data Flow Summary

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Table 3.18 shows the summary of data and flow of data that would be thrown around in the proposed system the Quality Management System. The visualization of the data flow within the system is separated into two. The first is the context diagram, it is the most general representation of the flow of data. The second visualization is more detailed but separated into two which are the Level 0 and Level 1. Level 0 is the more detailed representation of the context diagram it defines the major system and process, while Level 1 further delves into the subsystem and subprocess of the major system.

Table 3.18 Data Flow Summary

No.	Data Flow	Database Attributes
Context Diagram		
A	Staff Input Summary	
A.1	User Information	See data flow 1.1
A.2	User Credentials	See data flow 1.8
A.3	User Email	See data flow 1.9
A.4	New Password	See data flow 1.10
A.5	Policy Revision Request	See data flow 3.1
A.6	New Policy Proposal	See data flow 3.6
A.7	Policy Revision Proposal	See data flow 3.7
A.8	Rejection Details	See data flow 4.7
A.9	Verification Detail	See data flow 4.14
A.10	Approval Detail	See data flow 4.16
A.11	Desired Policy Information	See data flow 5.2
B	Staff Tasks Summary	
B.1	User Account	See data flow 1.8
B.2	Requested Policy Detail	See data flow 3.3
B.3	Notification Detail	See data flow 3.12
B.4	Selected Policy Detail	See data flow 5.2
B.5	Policy Details	See data flow 4.3
B.7	Feedback Details	See data flow 4.10
C	QA Personnel Input Summary	

C.1	User Information	See data flow 1.1
C.2	User Credentials	See data flow 1.8
C.3	User Email	See data flow 1.9
C.4	New Password	See data flow 1.10
C.5	Additional Suggestions	See data flow 3.2
C.6	Review Detail	See data flow 4.5
C.7	Rejection Details	See data flow 4.7
C.8	Verifier and Approver Detail	See data flow 4.11
C.9	Desired Policy Information	See data flow 5.2
C.10	Policy Categorization Information	See data flow 5.7
D	Policy Management Summary	
D.1	User Account	See data flow 1.8
D.2	Policy Proposal Detail	See data flow 4.2
D.3	Notification Detail	See data flow 3.12
D.4	Selected Policy Detail	See data flow 5.2
D.5	Feedback Details	See data flow 4.10
E	QA Director Input Summary	
E.1	User Information	See data flow 1.1
E.2	User Credentials	See data flow 1.8
E.3	User Email	See data flow 1.9
E.4	New Password	See data flow 1.10
E.5	Department Information	See data flow 2.1
E.6	Policy Categories	See data flow 2.6
E.7	Desired Policy Information	See data flow 5.2
F	QA Director Information Summary	
F.1	User Account	See data flow 1.8
F.2	Notification Detail	See data flow 3.12
F.3	Selected Policy Detail	See data flow 5.2
F.4	Active Policy Reports	See data flow 6.7
F.5	Rejected Policy Reports	See data flow 6.8
F.6	Implementation Reports	See data flow 6.9
F.7	Feedback Reports	See data flow 6.10

F.8	Feedback Percentage Report	See data flow 6.11
G	User Role Information	See data flow 1.13
H	Data Privacy Act Information	See data flow 8.1
Level 0 and Level 1 Data Flow		
1	Manage User Accounts	
1.1	User Information	fName lName email password
1.2	New User Account	acclID fName lName email password dateCreated
1.3	Accounts Data	acclID fName lName email password roleID
1.5	Complete User Account	
1.6	List of Accounts	
1.7	User Account	
1.8	User Credentials	email password
1.9	User Email	email
1.10	New Password	password
1.11	New Account Information	acclID password
1.12	User Role Information	roleID
1.13	Confirmation PIN	PIN
1.14	Reference PIN	OTPID requestBy PIN
2	Organization of Data	
2.1	Department Information	departmentRole departmentName

2.2	Employee information	dptName dptParentID employeeID accID dptID
2.3	Department Details	dptID dptName
2.4	Organized Department Information	dptParentID
2.6	Policy Categories	categoryName
2.7	Policy Category Details	categoryID categoryName
2.8	Organized Policy Categories	parentCategoryID
3	Policy Submission	
3.1	Policy Revision Request	requestChangeContentPath assignedBy(accID)
3.2	Additional Suggestions	remarksOn(policyID) remarksBy(accID) messageContent dateSubmitted
3.3	Requested Policy Details	contentPath
3.4	Change Request Form	requestChangeContentPath assignedBy(accID)
3.5	Suggestions Message	feedbackID remarksOn(policyID) remarksBy(accID) fbTypeID messageContent sessionID dateSubmitted
3.6	New Policy Proposal	title contentPath dateSubmitted
3.7	Policy Revision Proposal	accID
3.8	Policy Proposal Information	policyID title contentPath policyAuthor dateSubmitted policyStatusID
3.9	Task Detail	taskID taskTypeID requestChangeContentPath

		policyAssigned(policyID) assignedBy(accID) assignedTo(accID) taskStatusID dateCreated
3.10	Notification	message notifStatus
3.11	Notification Information	message dateCreated
3.12	Notification Detail	message notifStatus
4	Policy Implementation Process	
4.1	Policy Proposal Information	See data flow 3.8
4.2	Task Information	taskTypeID policyAssigned(policyID) assignedBy(accID) assignedTo(accID) taskStatusID
4.3	Notification	receivedBy(accID) message notifStatus
4.4	Notification Detail	See Data Flow 3.12
4.5	Review Detail	dateReviewed
4.6	Review Information	policyID policyReviewer(accID) dateReviewed policyStatusID
4.7	Rejection Details	taskTypeID policyID accID dateCreated message messageContent
4.8	Task Detail	See data flow 3.9
4.9	Policy Rejection Detail	policyID policyStatusID
4.10	Feedback Details	feedbackID remarksOn(policyID) remarksBy(accID) fbTypeID messageContent sessionID dateSubmitted
4.11	Verifier and Approver Detail	policyVerifier(accID) policyApprover(accID)

4.13	Verifier and Approver Information	policyID policyVerifier(accID) policyApprover(accID)
4.14	Verification Detail	dateVerified
4.15	Verified Policy Detail	policyID policyVerifier(accID) dateVerified policyStatusID
4.16	Approval Detail	dateApproved
4.17	Approved Policy Detail	policyID policyApprover(accID) dateApproved policyStatusID
4.18	Policy Categorization Information	categoryID
4.19	Complete Policy Information	categoryID versionNo policyStatusID
4.20	Revised Policy Detail	
5	View Policy Process	
5.1	Policy Information	policyID title contentPath categoryID versionNo policyAuthor(accID) dateSubmitted
5.2	Selected Policy Detail	policyReviewer(accID) dateReviewed policyVerifier(accID) dateVerified policyApprover(accID) dateApproved dateUploaded
5.3	Desired Policy Information	title categoryID versionNo
5.4	Selected Policy Information	policyID categoryID versionNo
5.5	Feedback Information	policyID messageContent
5.6	Feedback Details	See Data Flow 4.10
5.7	Task Detail	See data flow 3.9

5.8	Notification Information	See Data Flow 3.11
6	Reports	
6.1	Policy Repository Information	policyID title policyStatusID
6.2	Rejected Policy Information	See Data Flow 6.1
6.3	Log Information	policyID accID statusID logDate
6.4	Policy Status Information	See Data Flow 6.1
6.5	Feedback Information	See Data Flow 5.5
6.6	Department Organization Information	dptID dptName dptParentID
6.7	Active Policy Reports	title policyStatusID
6.8	Rejected Policy Reports	See Data Flow 6.2
6.9	Implementation Reports	policyID policyStatusID
6.10	Feedback Reports	feedbackID remarksOn(accID) remarksBy(accID) fbTypeID feedbackStatusID dateSubmitted
6.11	Feedback Percentage Report	feedbackID dptID
7	Notification Process	
7.1	Notification	See Data Flow 3.11
7.2	Notification Information	See Data Flow 3.11
7.3	Notification Detail	See Data Flow 3.12
7.4	Notification Info	notifID receivedBy(accID) message dateCreated dateRead notifStatus
8	Manage DPA	
8.1	Data Privacy Act Information	dpaContent
8.2	DPA Details	dpaID dpaContent dpaVersion dateUploaded

3.4. Proposed System

3.4.1 User Interface Hierarchical Diagram

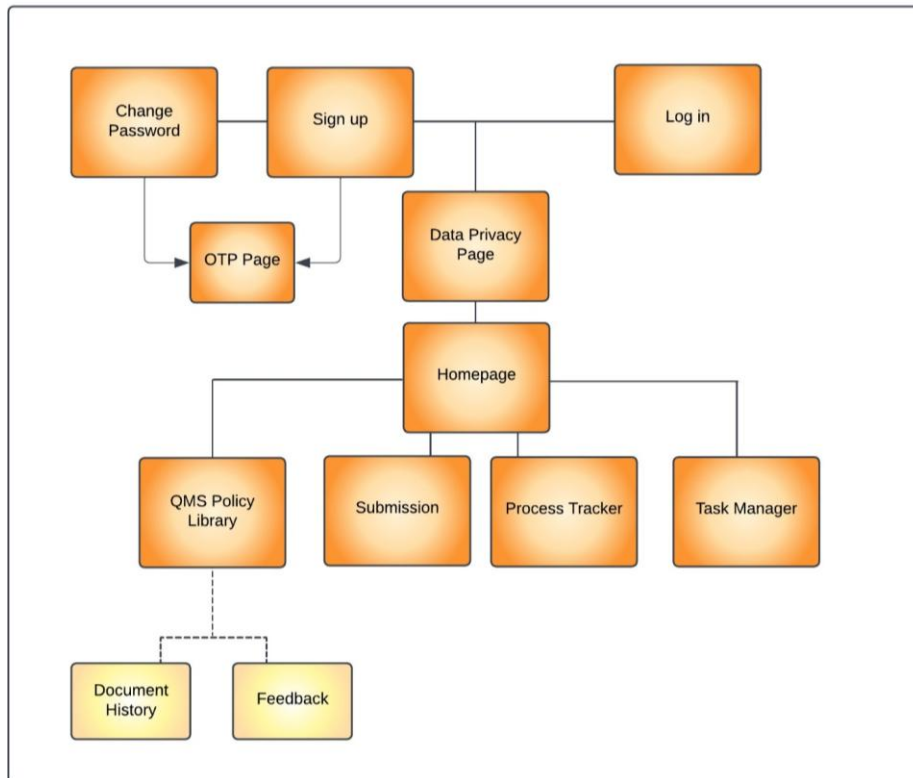


Figure 3.15 User Flow for Staff

Figure 3.15 presents the hierarchy of pages for users with a “Staff” role. Users will first encounter the login, sign-up, OTP, and change password pages upon accessing the system. After signing up or logging in, they will be directed to the Data Privacy Act homepage, where they must agree to proceed. Once they accept, the homepage is displayed, offering four key features: the Policy Library, Submission of Policies, Process Tracker, and Task Manager. Users with a “Staff” role have access only to the Policy Library, which includes the Document History and a feedback feature.

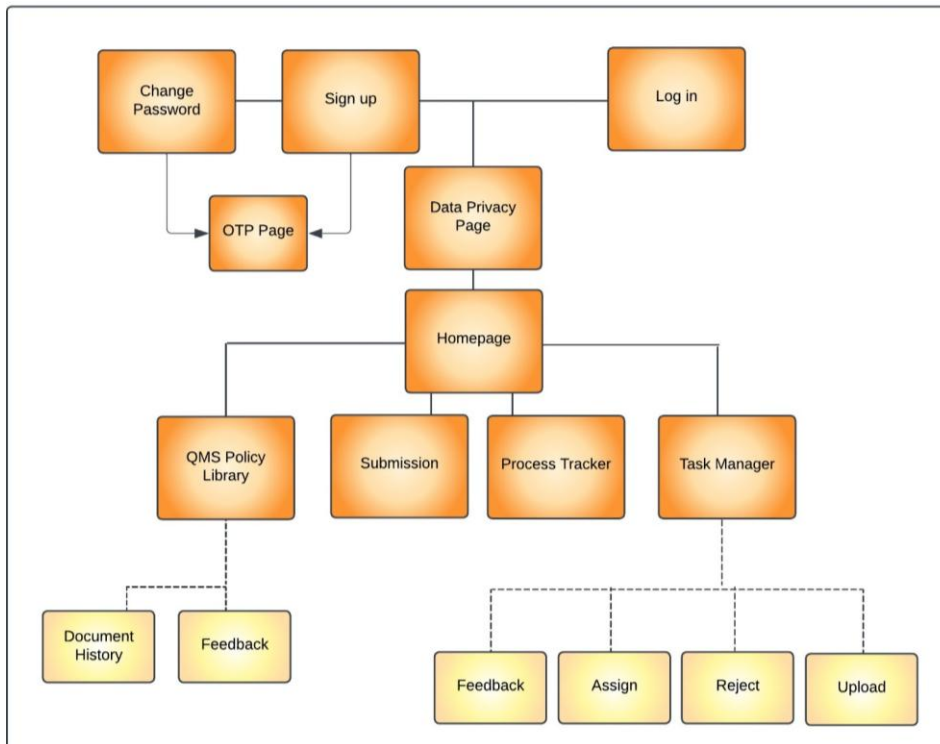


Figure 3.16 User Flow for Quality Assurance Personnel

Figure 3.16 illustrates the hierarchy of pages for users with a “Quality Assurance Personnel” role. Similar to users with a “Staff” role, the hierarchy includes the first set of pages, including the login, sign-up, OTP, and change password pages. After signing up or logging in, they will be directed to the Data Privacy Act homepage, where they must agree to proceed. Once they accept, the homepage is displayed, providing access to the Policy Library, Submission of Policies, Process Tracker, and Task Manager. In addition to the access granted to staff, which includes the Policy Library containing the Document History and Feedback feature, Quality Assurance personnel also have access to the Task Manager, which allows them to assign tasks, reject submissions, upload documents, and provide feedback.

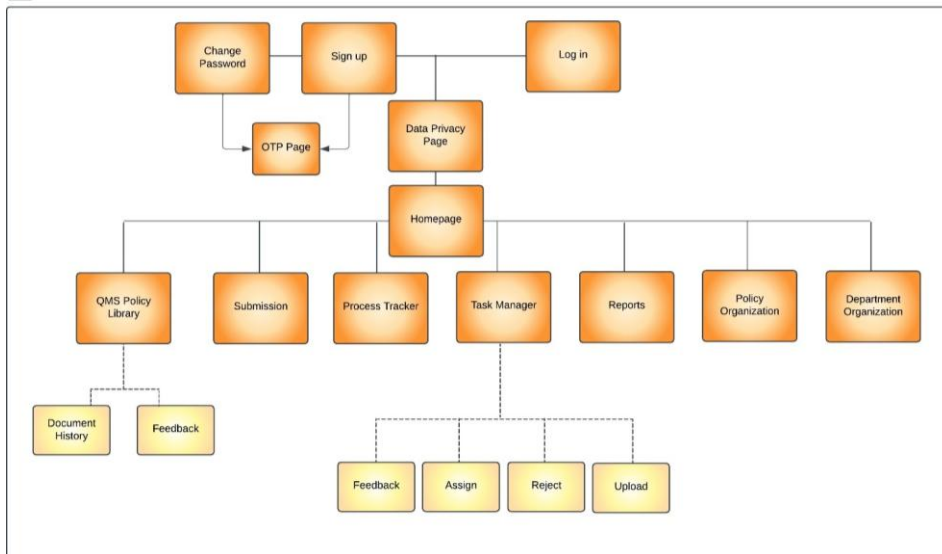


Figure 3.17 User Flow for Quality Assurance Director

Figure 3.17 illustrates the hierarchy of pages for users with a “Quality Assurance Director” role. Like users with a “Quality Assurance Personnel” role, the hierarchy includes the first set of pages, including the login, sign-up, OTP, and change password pages. After signing up or logging in, they will be directed to the Data Privacy Act homepage, where they must agree to proceed. Once they accept, the homepage is displayed, providing access to the Policy Library, Submission of Policies, Process Tracker, Task Manager, Reports, Policy Organization, and Department Organization. In addition to the access granted to Quality Assurance Personnel, which includes the Policy Library containing the Document History and Feedback feature, as well as the Task Manager, which allows them to assign tasks, reject submissions, upload documents, and provide feedback, the Quality Assurance Director has additional access to the Reports, Policy Organization, and Department Organization page.

3.4.2 User Interface Description

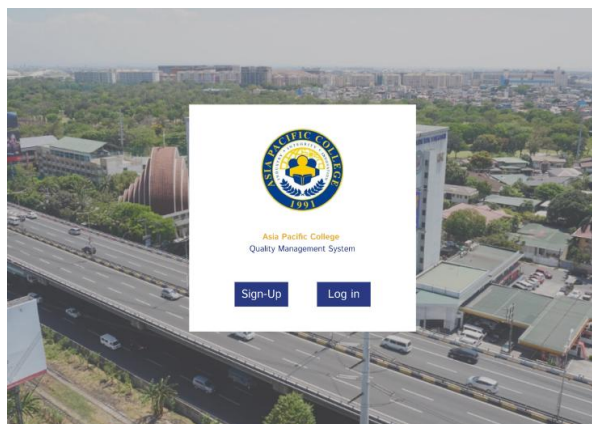


Figure 3.18 Landing Page

The Quality Management System landing page serves as the entry point to the system, introducing users to its purpose and features. It prominently displays the Asia Pacific College (APC) logo along with a background image of the campus, providing a familiar interface. The system's name is displayed at the center of the page, accompanied by two primary buttons: Sign-Up and Log-In. The Sign-Up button allows new users to register an account, while the Log-In button provides access to those who have already created an account. When the user clicks the sign-up button, it will first show the Data Privacy Act Page to inform users before allowing them to proceed. (see Figure 3.19 A)

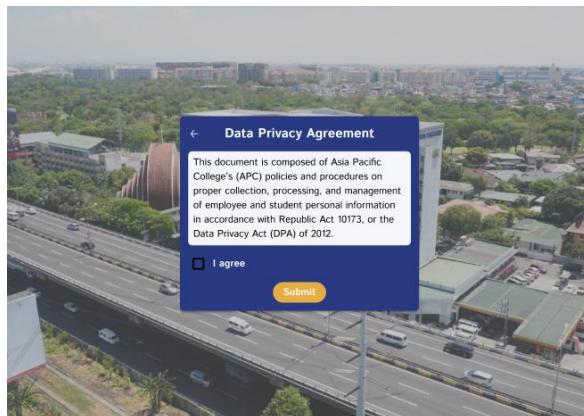


Figure 3.19 A Data Privacy Page

The Data Privacy Agreement (DPA) page explains Asia Pacific College's (APC) policies on collecting, processing, and managing personal information from employees and students. This follows the Data Privacy Act (Republic Act 10173) of 2012 to protect user data. On this page, users will see an arrow icon in the top left corner. Clicking this icon will take them back to the Landing page if they choose not to agree with the policy. To continue, users must click the black box labeled "I agree" to confirm their acceptance of the Data Privacy terms. After checking the box, they can click the Submit button to proceed (see Figure 3.19.A). Once submitted, the user can now fill out the account registration details in the sign-up page.

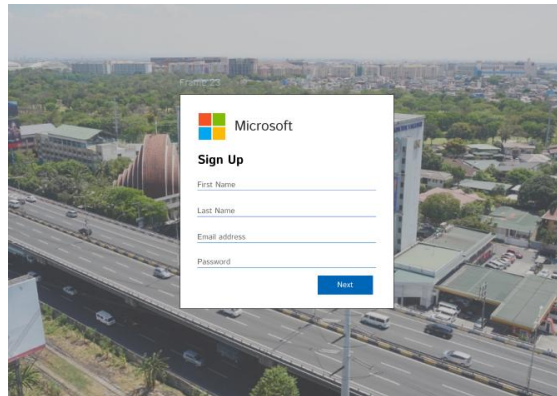


Figure 3.19 B Sign-up Page

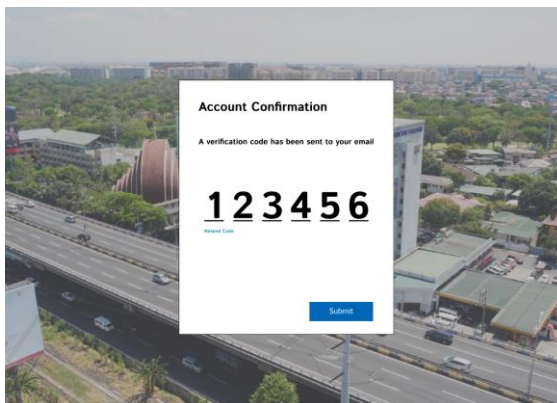


Figure 3.19 C OTP Verification Page

The Sign-Up page (see Figure 3.19 B) is designed to help new users create an account by providing their basic information. Users are required to enter their first and last name, Asia Pacific College email address, and password to ensure a secure account setup. Inputting a valid APC email is essential for verifying user identity and granting access to the system. Once all the required information is filled out, users can proceed by clicking the next button to verify their account with a one-time pin that was sent to their inputted APC email. After that they can then click the submit button to complete their registration (see Figure 3.19 C). After successfully signing up, they will be redirected to the Landing Page, where they can use their newly account to log in and access the system.

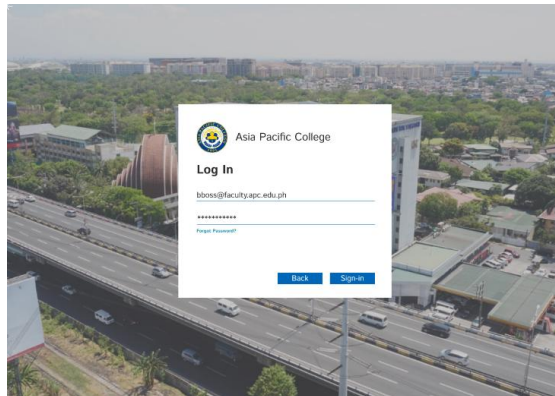


Figure 3.20 A Log-in Page

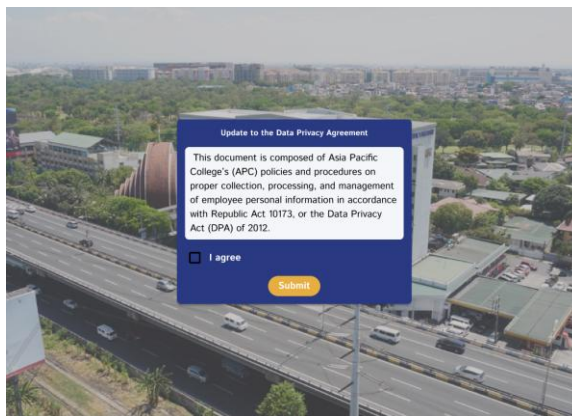


Figure 3.20 B Data Privacy Page Update

The Log-In page allows users to securely access the system by entering their credentials. To begin, users must input their APC Email Address in the designated field, and they will need to enter their password to proceed. If a user forgets their password, they can reset it by clicking the blue text located below the password field. This will initiate a verification process, where a code will be sent to their APC Email Address to confirm their identity before allowing them to set a new password. The data privacy terms may change, and any updates will be displayed to users once after logging into their account, before they proceed to the home page. (see Figure 3.19.B).

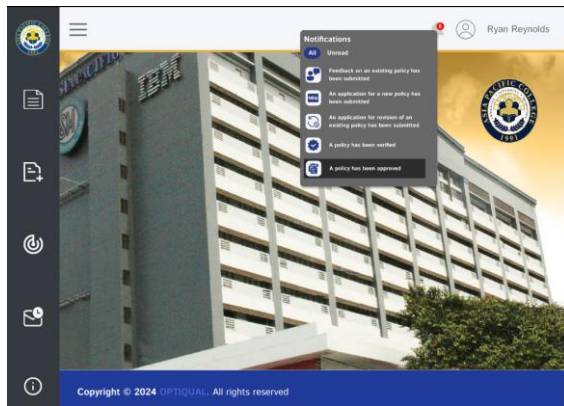


Figure 3.21 A Notification Feature

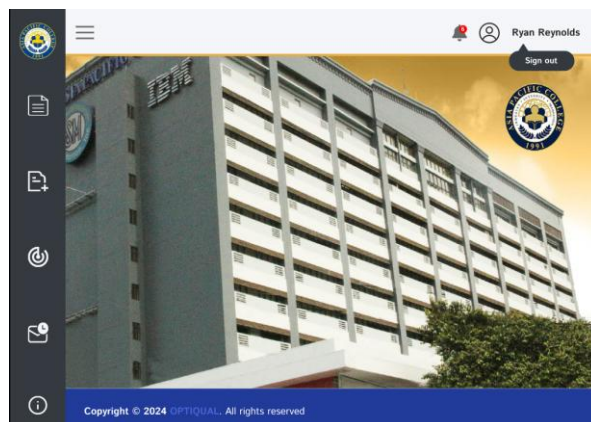


Figure 3.21 B Sign Out

Once logged in, the user can now access the system. The sidebar, located on the left side, displays the available modules based on the user's role. Some features, such as the Notification feature and the Profile icon, located in the top right corner of the system, are available to all users. Clicking the Notification icon allows users to view important updates, such as newly uploaded policies or changes made to existing ones.

In the notification feature, users can see the number of notifications they have and choose to view either all notifications or only unread ones (see Figure 3.21 A). Meanwhile, clicking the profile icon provides the option to sign out of the account (see Figure 3.21 B). After signing out, the user will be redirected back to the landing page (see Figure 3.19).

Staff Point of View

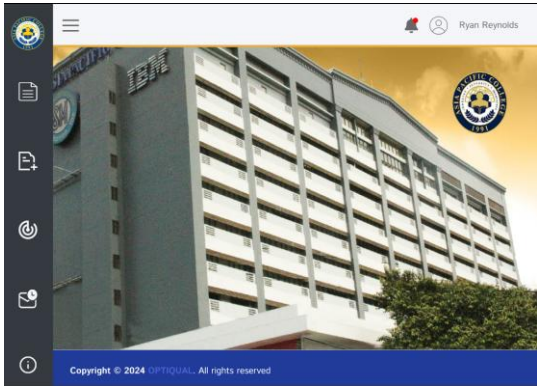


Figure 3.22 Staff Home Page

When users log in, they are directed to the system's homepage, which varies based on their assigned role. For staff members, the dark gray sidebar on the left contains four main features: Policy Repository, Policy Creation, Process Tracker, and Task Manager (see Figure 3.24).

Policy Repository Module

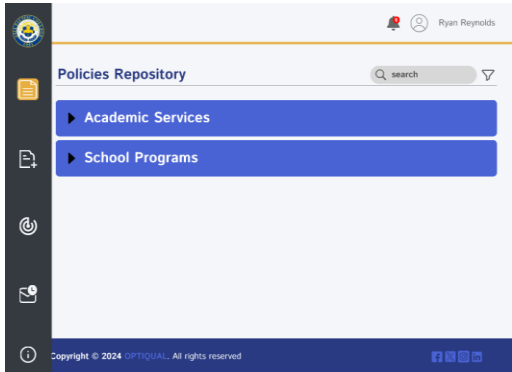


Figure 3.23 A Policy Repository Page

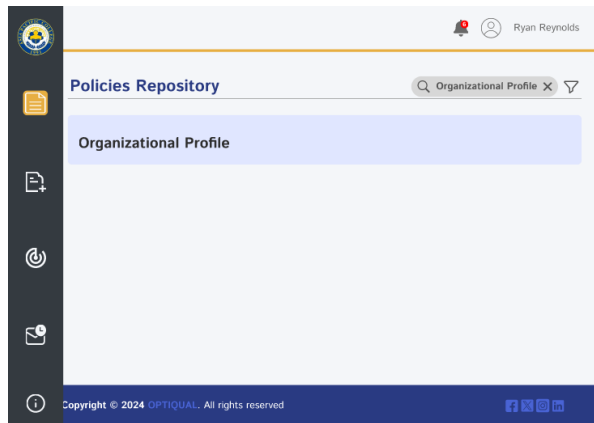


Figure 3.23 B Search Function



Figure 3.23 C Filter & Sort Function

The Policy Repository is a digital library where all school policies are stored in organized folders. To make searching easier, there is a search bar and a filter button at the top right of the page (see Figure 3.23 B). If users want to find a specific policy, they can type keywords in the search bar. If users want to view specific types of policies, they can use the filter button to refine the results. Additionally, they can sort policies alphabetically or by the date of upload (see Figure 3.23C).



Figure 3.23 D Policy Viewing

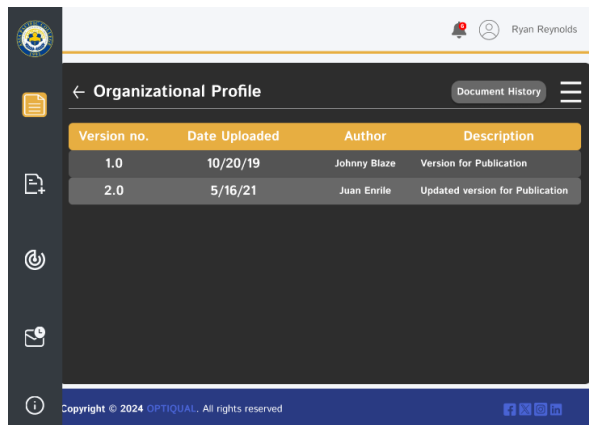


Figure 3.23 E Document History Button

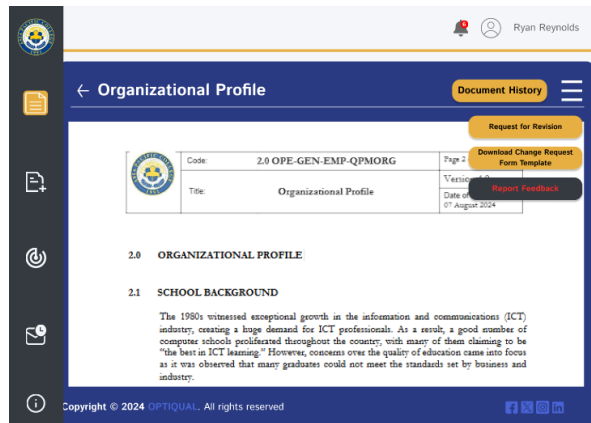


Figure 3.23 F Panel Icon

When a user clicks on a policy, its full content is displayed on the screen, each policy in the repository has a watermark to prevent unauthorized distribution or duplication (see Figure 3.23D). Additionally, two important buttons are available: the document history button and the panel icon. The document history button shows the entire history of the policy, including past updates and changes (see Figure 3.23 E). The panel icon provides additional options, including the request for revision button, download change request form, and the report feedback button (see Figure 3.23 F).

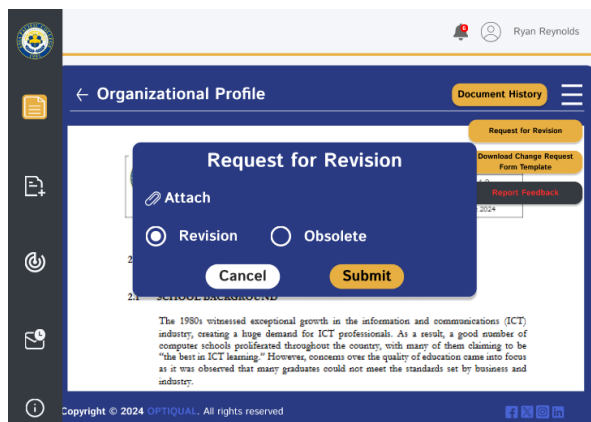


Figure 3.23 G Request for Revision Prompt

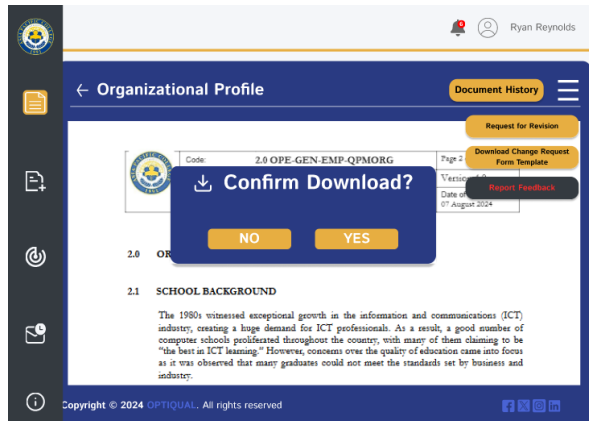


Figure 3.23 H Download Change Request Form Template

If a user thinks that a policy needs changes, they can click the request for revision button to submit a change request form, which is used for revising or updating a certain policy. They will need to attach the form file and choose between two options: Revision (if they want to update the policy) or Obsolete (if they think the policy should be removed). If they don't have the form yet, they can download the template from the system before submitting their request. (see Figure 3.23 G and Figure 3.23 H).

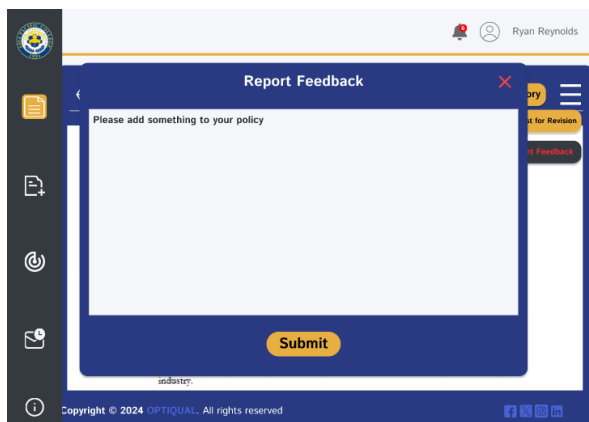


Figure 3.23 I Report Feedback Function

If a user just wants to give feedback or share their thoughts about a policy, they can use the report feedback button (see Figure 3.23 I). After submitting the report feedback, the Quality Assurance Personnel will review it. Users will then receive a notification once their feedback has been reviewed and addressed.

Policy Submission Module

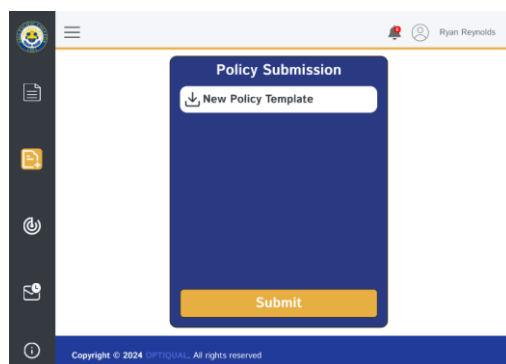


Figure 3.24 A Policy Submission Page

The policy submission module allows users to create and upload new policies within the system. On the policy submission page, users will see a button labeled new policy template (see Figure 3.24 A). When they click this button, a message will appear asking, "Confirm download?" with two options: Yes and No (see Figure 3.24 B).

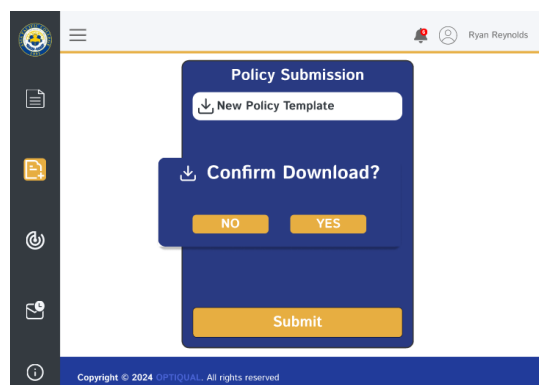


Figure 3.24 B File Download Confirmation

If the user selects "Yes", the system will download the policy template, which must be completed with all the required information. Once completed, the user must save the document as a PDF. They can then enter the policy title in the designated field and attach the completed file using the attach button.

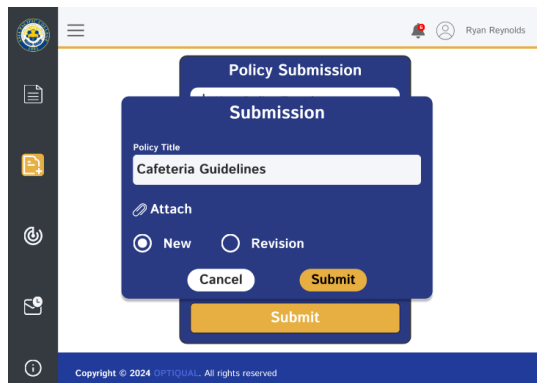
The screenshot shows a web application interface for 'Policy Submission'. A dark blue sidebar on the left contains icons for home, documents, attachments, settings, notifications, and a user profile. The main content area has a light blue header with a hamburger menu, a notification bell, and the user name 'Ryan Reynolds'. A modal form titled 'Policy Submission' is displayed. It has a 'Submission' section with a 'Policy Title' field containing 'Cafeteria Guidelines'. Below this is an 'Attach' section with two radio buttons: 'New' (selected) and 'Revision'. At the bottom of the modal are 'Cancel' and 'Submit' buttons. A larger 'Submit' button is also visible at the bottom of the modal's content area. The footer of the page reads 'Copyright © 2024 OPTIQUAL. All rights reserved.'

Figure 3.24 C File Upload

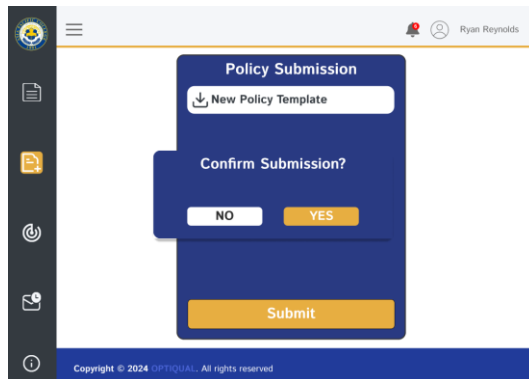
The screenshot shows the same web application interface as Figure 3.24 B. The modal form titled 'Policy Submission' now shows a 'New Policy Template' button with a download icon. Below this is a 'Confirm Submission?' dialog with 'NO' and 'YES' buttons. At the bottom of the modal is a 'Submit' button. The footer of the page reads 'Copyright © 2024 OPTIQUAL. All rights reserved.'

Figure 3.24 D Confirm Submission

Once the file is attached, the system will display another confirmation message with two options: Cancel and Submit (see Figure 3.24 D) If the user selects Cancel, they will be redirected back to the main page of the module without submitting the policy. However, if they select "Yes", the policy will be submitted and will enter the policy review process for evaluation and approval.

Process Tracker Module

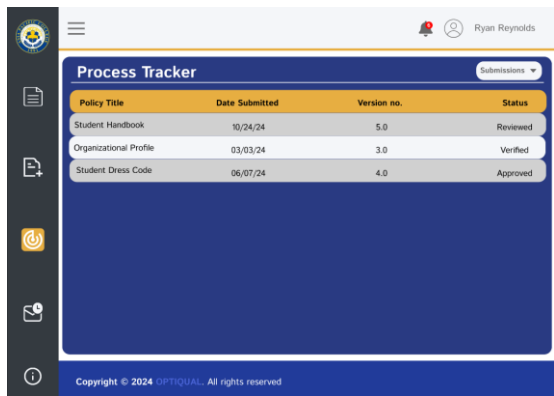


Figure 3.25 A Process Tracker: Policy Submissions

The Process Tracker module helps users monitor the progress of their submitted policies and feedback reports. At the top right of the page, users will find buttons allowing them to choose whether they want to track the status of their proposed policy (see Figure 3.25 A) or their report feedback (see Figure 3.5 B)

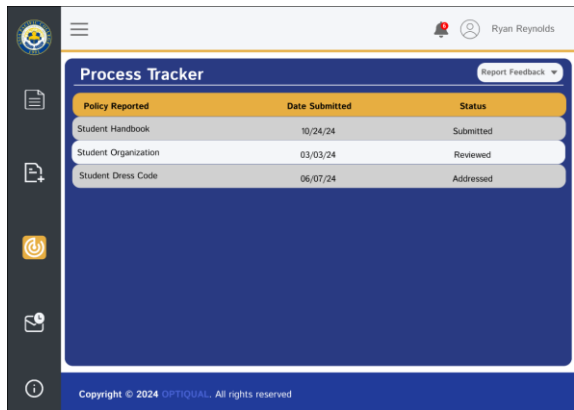


Figure 3.25 B Process Tracker: Report Feedback

Once they select a process, the system will display the full document and its current progress. The policy implementation process consists of four stages: submission, review, verification, and approval. For each stage, users can see who signed off on it and the date it was approved (see Figure 3.25 C). However, the feedback process consists of three stages: Submission, Review, and Addressed. For each stage, users can see who signed off on it and the date it was approved (see Figure 3.25 D). This allows users to easily track where their policy or feedback report stands and understand the necessary steps before completion.



Figure 3.25 C View Policy Implementation Process

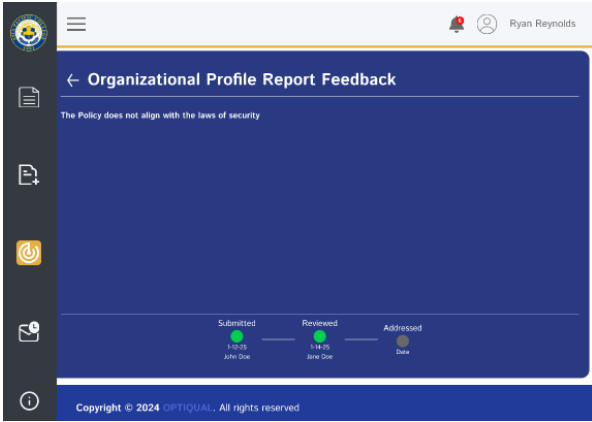


Figure 3.25 D View Report Feedback Process

Task Manager Module

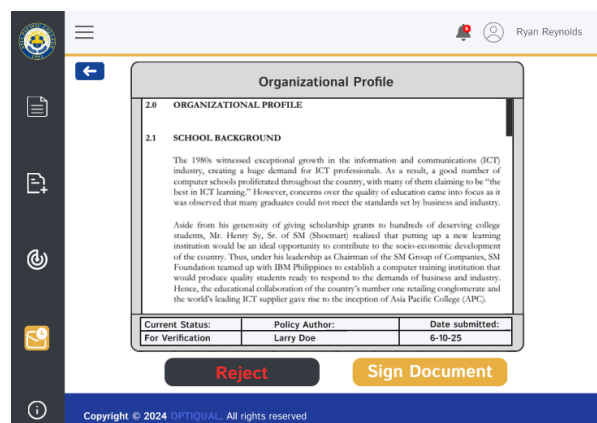


The screenshot shows the Task Manager page with a table of tasks. The table has columns for Policy Title, Author, Date Submitted, Version no., and Status. The tasks listed are Student Handbook, Student Dress Code, Introduction, Organizational Profile, and Calistoga Guidelines.

Policy Title	Author	Date Submitted	Version no.	Status
Student Handbook	John Doe	1/2/25	5.0	For Verification
Student Dress Code	Amy Dowe	7/5/25	2.0	For Approval
Introduction	Larry Doe	6/10/25	3.0	For Revision
Organizational Profile	John Dow	7/2/25	1.0	Revision Request
Calistoga Guidelines	Ryan Doe	6/10/25	3.0	Rejected

Figure 3.26 A Task Manager Page

The Task Manager page helps users keep track of their assigned tasks, making it easier to manage and review policies. On this page, users will see a list of tasks, each displaying important details such as the policy title, author, date submitted, version number, and the status of the task (see Figure 3.26 A).



The screenshot shows the Organizational Profile task verification page. It displays the title 'Organizational Profile' and the section '2.0 ORGANIZATIONAL PROFILE'. Below this is the section '2.1 SCHOOL BACKGROUND' with a paragraph of text. At the bottom, there is a table with three columns: Current Status, Policy Author, and Date submitted. The Current Status is 'For Verification', the Policy Author is 'Larry Doe', and the Date submitted is '6-10-25'. Below the table are two buttons: 'Reject' and 'Sign Document'.

Current Status:	Policy Author:	Date submitted:
For Verification	Larry Doe	6-10-25

Figure 3.26 B For Verification Task

If a user wants to verify/approve a policy, they can simply click on it to open the full document. Once opened, they will see the author of the policy and when it was submitted (see Figure 3.26 B). At the bottom, there are two buttons: "Reject" and "Sign Document." If the user rejects the policy, a text field will appear where they must enter the reason for rejection. After filling it out, they can submit the rejection feedback (see Figure 3.26 C), which will then be sent back to the author. If the user approves the policy, they can click the "Sign Document" button, prompting the system to generate a digital code as a signature and imprint it onto the document. This feature utilizes an API to embed the digital signature into the document. (see Figure 3.26 B)

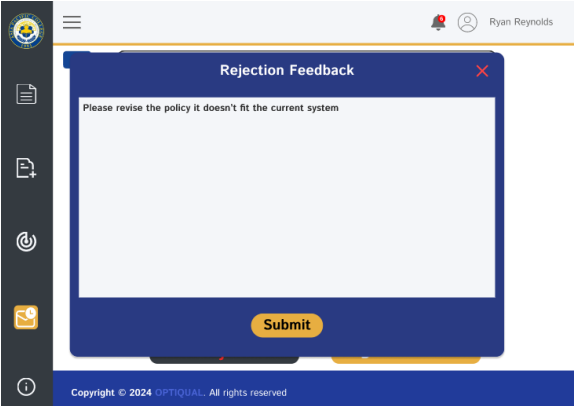


Figure 3.26 C Rejection Feedback Prompt

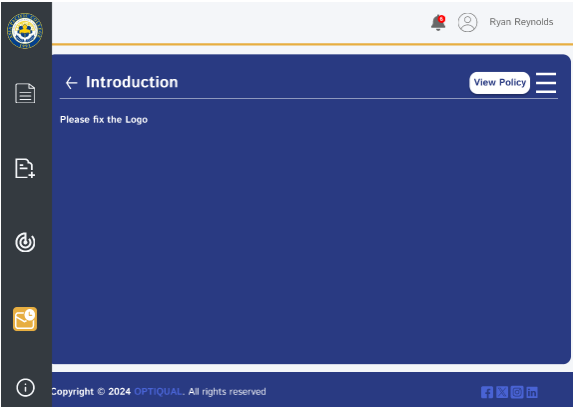


Figure 3.26 D For Revision Task

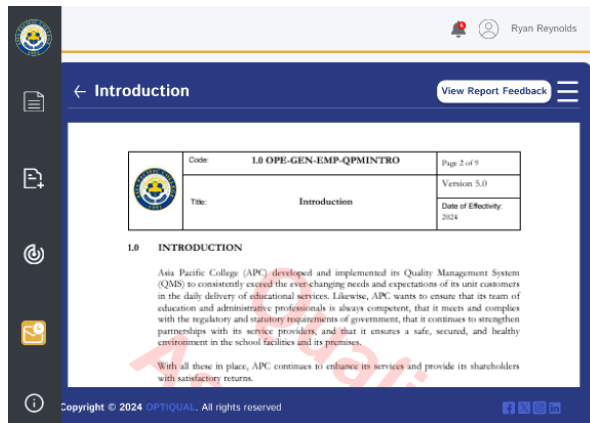


Figure 3.26 E View Policy

If a user clicks on a task with the status "For Revision," they will automatically be directed to the report feedback provided (see Figure 3.26 D). In the top right corner, there is a "View Policy" button that allows the user to see the entire document (see Figure 3.26 E). Beside this button is a panel icon, which contains three options when clicked: the reply button, which allows the user to respond to the comments (see Figure 3.26 F).

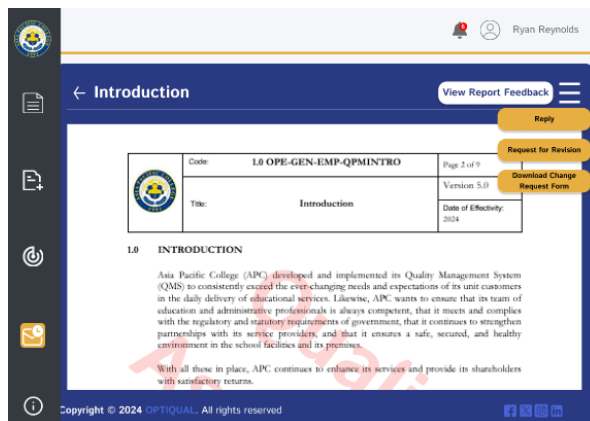


Figure 3.26 F View Policy Panel Icon

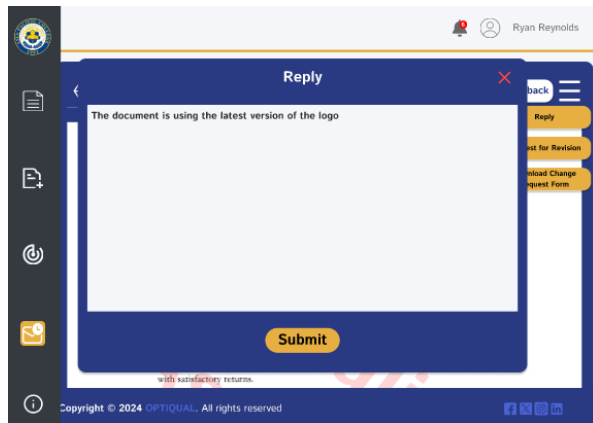


Figure 3.26.G Reply Prompt

The request for revision button, enables the user to submit a revised document. When this button is clicked, a field will appear where the user can attach the file they want to submit. (see Figure 3.26 H). Lastly, if the user does not have the required form for the change request, they can download it using the download change of form request form button inside the panel icon (see Figure 3.26 I).

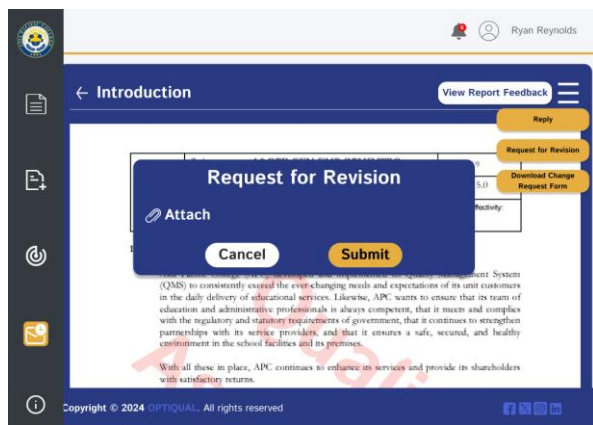


Figure 3.26 H Request for Revision Prompt



Figure 3.26 I Download Change of Request Form Prompt

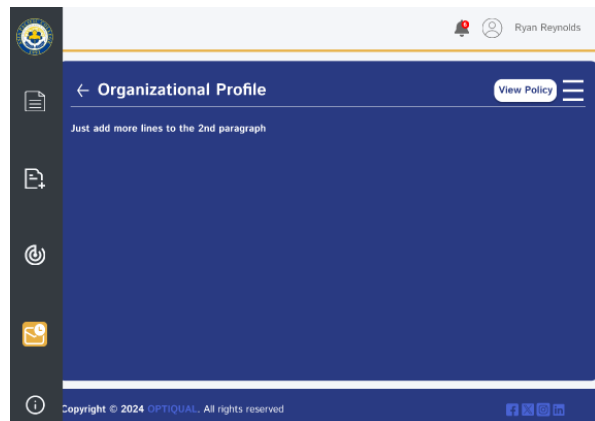


Figure 3.26 J Request for Revision Task

Once a user clicks on a task with a "Request for Revision" status, they will automatically be directed to the comments or suggestions provided regarding their proposed policy revision (see Figure 3.26 J). There is also a "View Policy" button that allows the user to see the entire document (see Figure 3.26 K). Beside this button is a panel icon, which contains two options: the submit revision button and the download editable file button (see Figure 3.26 L).



Figure 3.26 K View Policy Prompt (Request for Revision Task)



Figure 3.26 L Hamburger Button

The submit revision button allows the user to submit a revised policy by filling in the required information, such as the policy title, and attaching the revised policy document (see Figure 3.26 M). The download editable file button enables the user to download the file, followed by a confirmation message to ensure the download was successful (see Figure 3.26 N).

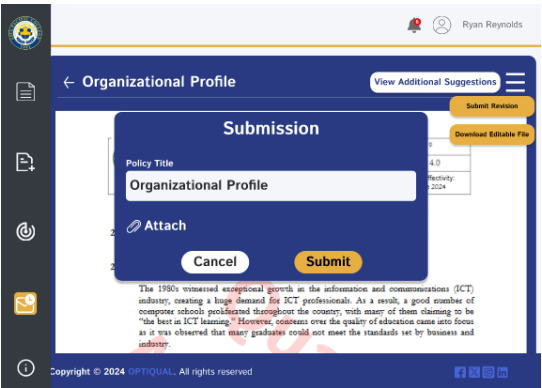


Figure 3.26 M Submit Revision Prompt



Figure 3.26 N Download Editable File Prompt

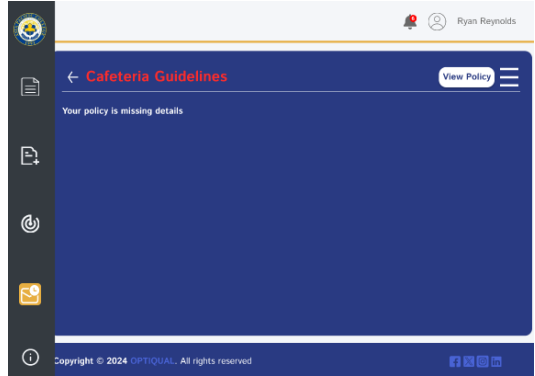


Figure 3.26 O Rejected Status Task

Once a task is given a "Rejected" status, its title will turn red to indicate the rejection. When the user clicks on a task with this status, they will be directed to the feedback explaining why the policy was rejected (see Figure 3.26 O). In the top right corner, there is also a "View Policy" button that allows the user to see the entire document (see Figure 3.26 P). Beside this button is a panel icon, which contains two options: submit another request and cancel request. The submit another request button allows the user to submit a new or updated file while also providing the option to remove the old file. (see Figure 3.26 Q) The cancel request button completely deletes and cancels the entire request. (see Figure 3.26 R).

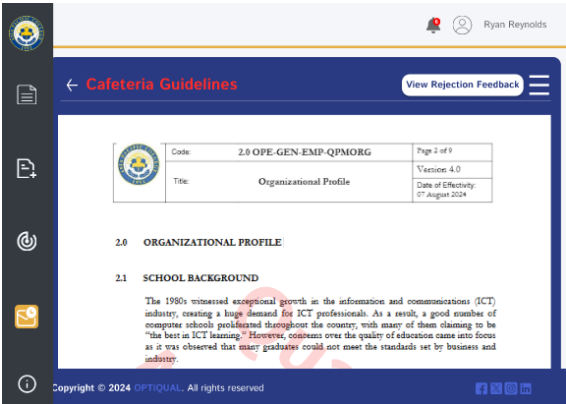


Figure 3.26.P View Policy (Rejected Status Task)

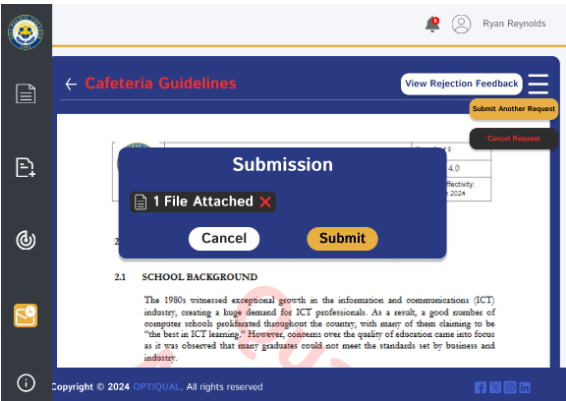


Figure 3.26 Q Submit Another Request Prompt

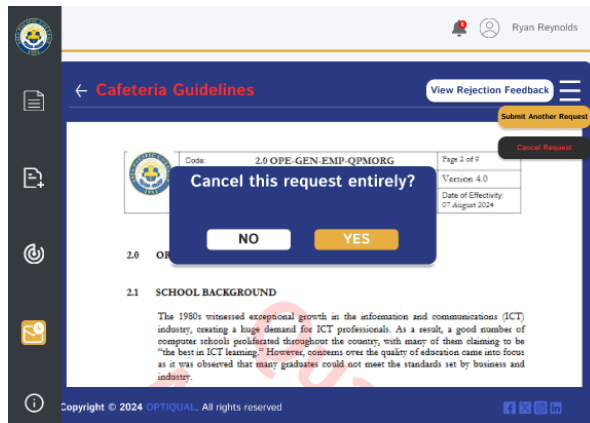


Figure 3.26 R Cancel Request Prompt

Information Module

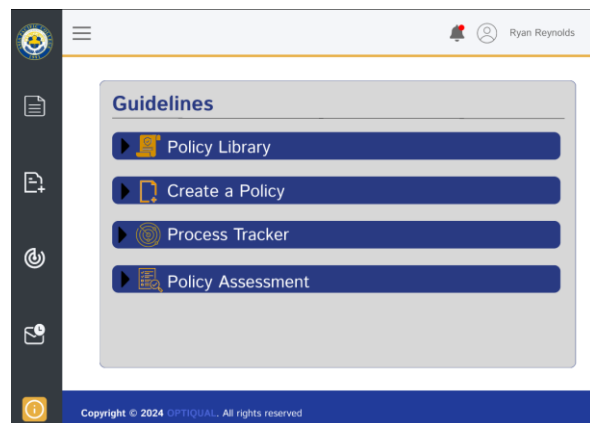


Figure 3.27 A Information Module

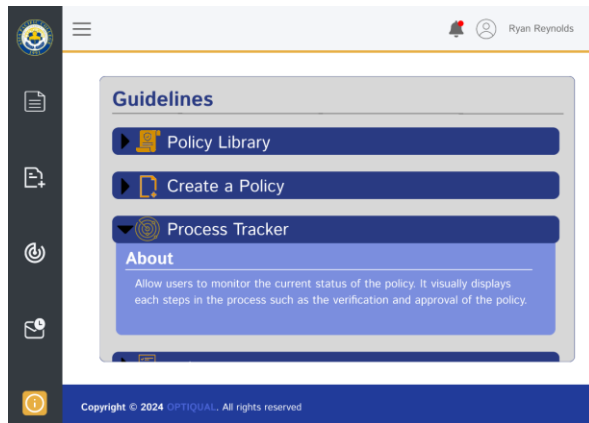


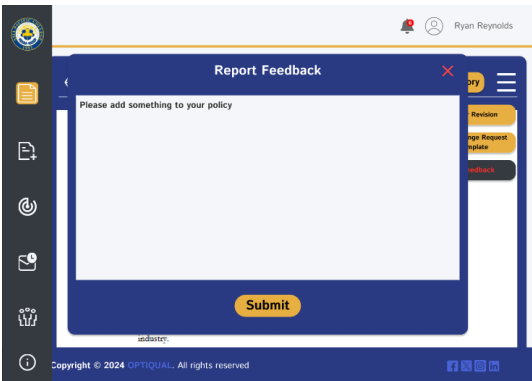
Figure 3.27 B Information Page

Upon accessing the Information module, users will find a structured list of all the modules within the system, each accompanied by a brief explanation of its function. This allows users to familiarize themselves quickly with the different parts of the platform without having to navigate through each module individually. Whether a user is new to the system or just needs a refresher, the information page offers a convenient way to understand the core functionalities of the system immediately. (See Figure 3.27 A and Figure 3.27 B).

Quality Assurance Personnel's Point of View

For Quality Assurance (QA) Personnel, most of the modules function the same as in the Staff view. However, there is a slight difference in function for the QA personnel in the policy repository, and task manager. Additionally, there is also an additional module called the role manager that provides the QA Personnel with the ability to manage their team (see Figure 3.30 A)

Policy Repository



3.28 A Report Feedback Prompt

The report feedback button has an extra feature in the policy repository module for QA Personnel. The user can give feedback on a policy (see Figure 3.28 A) and assign which department should handle the revision of that policy. Once the QA Personnel has submitted their feedback, a new field will appear. Here, they will need to select the department responsible for revising the policy. The department is chosen based on which team or office the policy belongs to or relates to (see Figure 3.28 B). This ensures that the right people will review and update the policy properly.

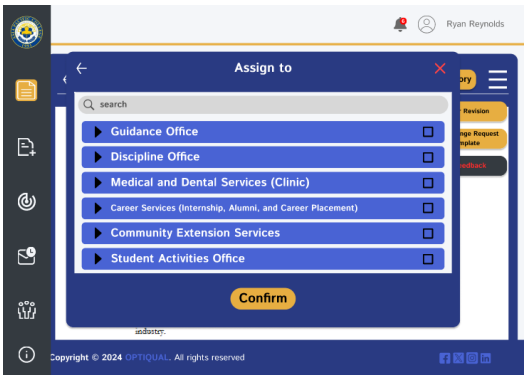
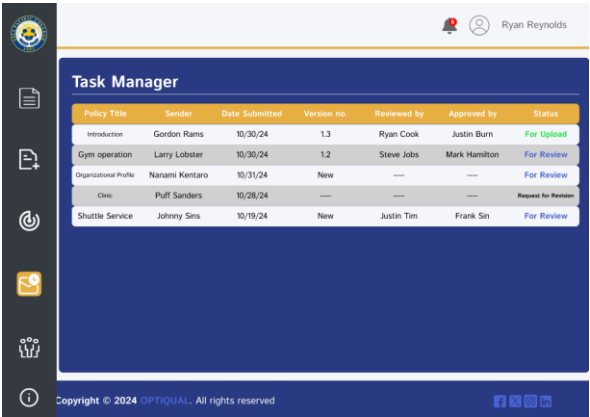


Figure 3.28 B Assign Function

Task Manager



The screenshot shows the 'Task Manager' interface. At the top, there is a header bar with a notification bell and a user profile icon for 'Ryan Reynolds'. Below the header, the title 'Task Manager' is displayed. A table lists several policies with columns for Policy Title, Sender, Date Submitted, Version no., Reviewed by, Approved by, and Status. The policies listed are: Introduction (Gordon Rams, 10/30/24, 1.3, Ryan Cook, Justin Burn, For Upload), Gym operation (Larry Lobster, 10/30/24, 1.2, Steve Jobs, Mark Hamilton, For Review), Organizational Profile (Nanami Kentaro, 10/31/24, New, ---, ---, For Review), Clinic (Puff Sanders, 10/28/24, ---, ---, ---, Request for Revision), and Shuttle Service (Johnny Sims, 10/19/24, New, Justin Tim, Frank Sin, For Review). A sidebar on the left contains icons for home, documents, a plus sign, a refresh icon, a mail icon, a group of people icon, and an information icon. The footer includes a copyright notice 'Copyright © 2024 OPTIQUAL. All rights reserved' and social media icons.

Policy Title	Sender	Date Submitted	Version no.	Reviewed by	Approved by	Status
Introduction	Gordon Rams	10/30/24	1.3	Ryan Cook	Justin Burn	For Upload
Gym operation	Larry Lobster	10/30/24	1.2	Steve Jobs	Mark Hamilton	For Review
Organizational Profile	Nanami Kentaro	10/31/24	New	---	---	For Review
Clinic	Puff Sanders	10/28/24	---	---	---	Request for Revision
Shuttle Service	Johnny Sims	10/19/24	New	Justin Tim	Frank Sin	For Review

Figure 3.29 A Quality Assurance Personnel Task manager



The screenshot shows the 'Organizational Profile' page. At the top, there is a header bar with a notification bell and a user profile icon for 'Ryan Reynolds'. Below the header, the title '← Organizational Profile' is displayed. A table shows the profile details: Code (1.0 OPE-GEN-EMP-QPMINTRO), Page (2 of 9), Title (Introduction), Version (5.0), and Date of Effectivity (2024). Below the table, the section '1.0 INTRODUCTION' is displayed, followed by a paragraph of text. The footer includes a copyright notice 'Copyright © 2024 OPTIQUAL. All rights reserved' and social media icons.

Code	Page
1.0 OPE-GEN-EMP-QPMINTRO	2 of 9

Title	Version	Date of Effectivity
Introduction	5.0	2024

1.0 INTRODUCTION

Asia Pacific College (APC) developed and implemented its Quality Management System (QMS) to consistently exceed the ever-changing needs and expectations of its unit customers in the daily delivery of educational services. Likewise, APC wants to ensure that its team of education and administrative professionals is always competent, that it meets and complies with the regulatory and statutory requirements of government, that it continues to strengthen partnerships with its service providers, and that it ensures a safe, secured, and healthy environment in the school facilities and its premises.

With all these in place, APC continues to enhance its services and provide its shareholders with satisfaction returns.

Figure 3.29 B Task Manager Page (Policy Viewing)

The Quality Assurance (QA) Personnel also have access to the Task Manager Page but with a few additional features unique to their role. Unlike regular users, QA Personnel are responsible for reviewing proposed policies before they are sent for verification and approval, after which they are officially added to the system.

For Review Status

When a new proposed policy is submitted, QA Personnel can view it in the Task Manager module (see Figure 3.29 A). Clicking on a policy submission opens the document for a detailed review (see Figure 3.29 B). At the top-right corner of the page, three key action buttons allow the QA team to manage the policy submission effectively:

Rejecting a Policy Submission

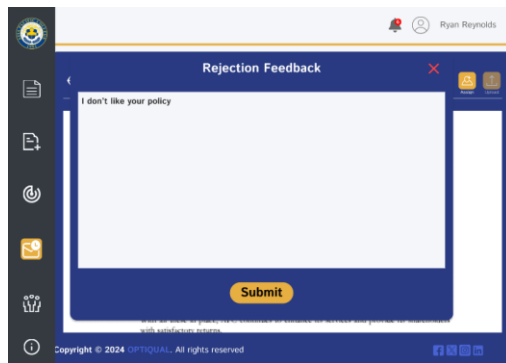


Figure 3.29 C Task Manager Page (Rejection Prompt)

If a submitted policy fails to meet the requirements or does not comply with existing guidelines, QA Personnel can click the rejection button to decline it (see Figure 3.31.C). This action removes the policy from further review, and the user who submitted it will be notified about the rejection.

Assigning Evaluators (Verifier & Approver)

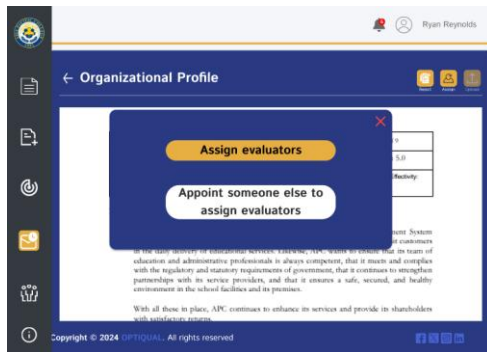


Figure 3.29 D Task Manager Page (Assigning evaluators)

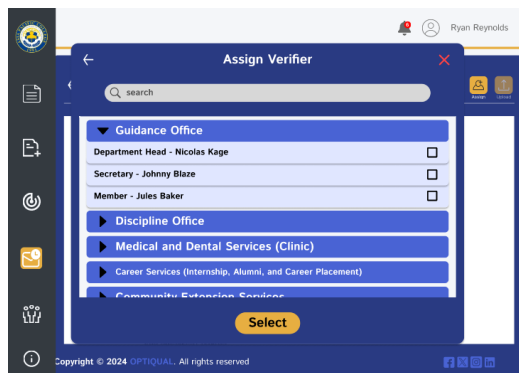


Figure 3.29 E Task Manager Page (Assigning Verifier)

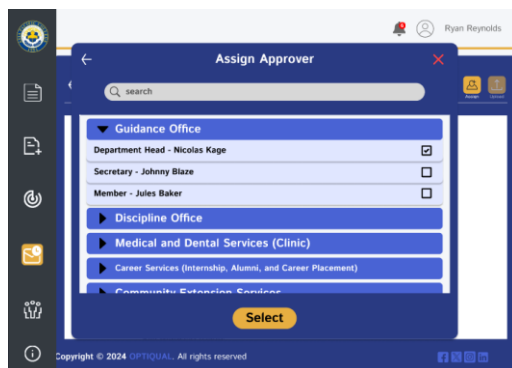


Figure 3.29 F Task Manager Page (Assigning Approver)

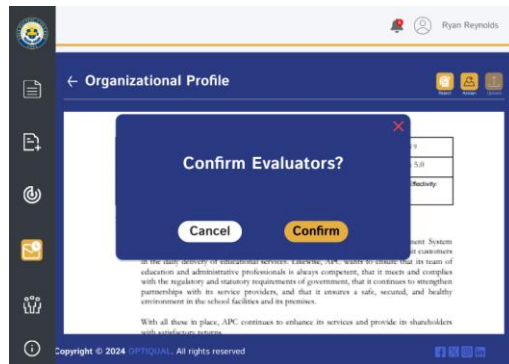


Figure 3.29 G Task Manager Page (Confirm Evaluators)

QA Personnel can also assign evaluators who will further assess the policy. This is done by clicking the assign evaluators button, where they can designate a verifier, and an approver based on their expertise in the subject matter (see Figures 3.29 E-G). The assigned individuals will then receive the policy for evaluation.

Request for Revision status:

 A screenshot of a web application interface. The header bar shows the 'Ryan Reynolds' user profile. The navigation bar has a back arrow and the text 'Change Request Form for Letter of Guarantee and Authorization'. The main content area contains a form with a logo on the left. To the right of the logo are fields for 'Date Received:' and 'Received by:'. Below these is a field for 'Change Request Form No.'. At the bottom, there is a table with document details.

Document Code:	CLINIC-PMT28		
Document Title:	Annex T28 - Letter of Guarantee and Authorization		
Version Number:	2.0	Effective Date:	2 January 2013
Classification:	☐ Quality Policy Manual (QPM)	☐ Procedures Manual (PM)	

 The footer shows 'Copyright © 2024 OPTIQUAL. All rights reserved.' and social media icons.

Figure 3.29 H Task Manager Page (Change of Request Form)

For policies with a status of "Request for Revision," the user can click on a task from the list to view the entire document along with the requested changes (see Figure 3.29 H). This allows the user to review the proposed revisions before actual revision of the document. At the top right corner of the screen, there are two buttons: Reject and Accept.

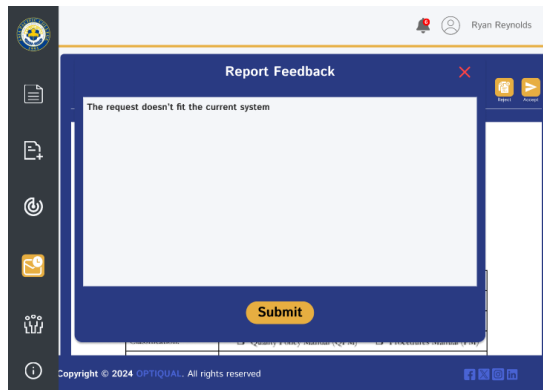


Figure 3.29 I Task Manager Page (Reject Button)

If the policy does not meet the required standards, the user can click the “Reject” button. When this button is clicked, a text field will appear, allowing the user to enter the reason for rejection. This ensures that the necessary feedback is provided so the policy can be improved accordingly (see Figure 3.29 I).

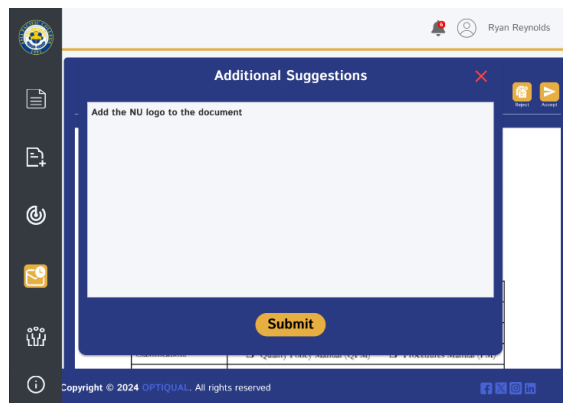


Figure 3.29 J Task Manager Page (Accept Button)

If the policy meets the quality standards, the user can click the “Accept” button to approve the proposed revision. This action opens an “Additional Suggestions” prompt where the QA personnel can input additional suggestions for the revision of the document. (see Figure 3.29 J).

For upload status:

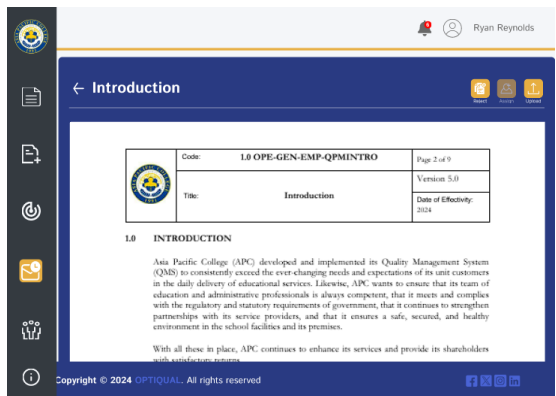


Figure 3.29 K Task Manager Page (Uploading)

Once a policy has successfully passed the evaluation and approval process, QA Personnel can proceed with uploading it to the Policy Repository (see Figure 3.29 K). This process requires filling in key details such as the policy title, document code, the classification of revision specifies whether the update is minor or major and automatically updates the policy document's version. and a changes made description summarizing its contents. After completing all the required fields, the QA Personnel can click the "Confirm" button to officially upload the policy into the system (see Figure 3.29 L). This ensures that the approved policy is properly documented and accessible to all relevant users.

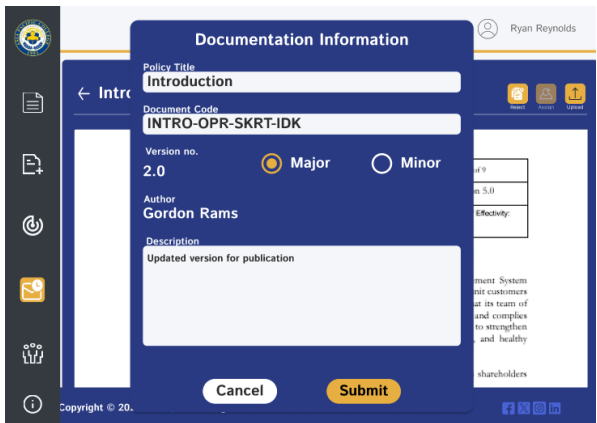


Figure 3.29 L Task Manager Page (Documentation Information)

Role Manager

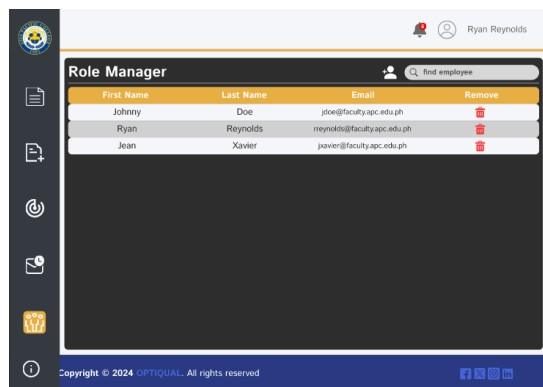


Figure 3.30 A Role Manager

Adding a New Member

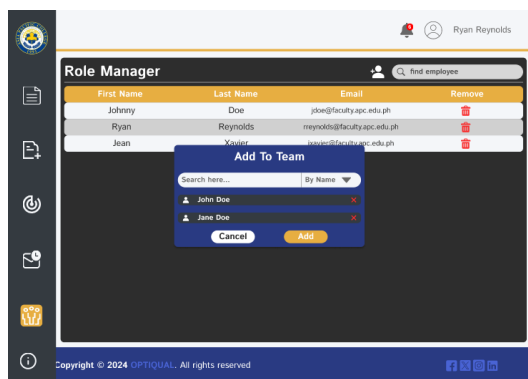


Figure 3.30 B Role Manager (Add Member)

QA Personnel can add new members by clicking the profile icon with a plus sign, which is located next to the search field. Once clicked, a prompt will appear where they can search the new member by name or email address (see Figure 3.32.B). Below this input field, there are two buttons: “Cancel” and “Add”. If the user decides not to proceed, they can click Cancel to exit otherwise, clicking “Add” will successfully add the member, and their name will appear on the list of QA team members.

Removing a member

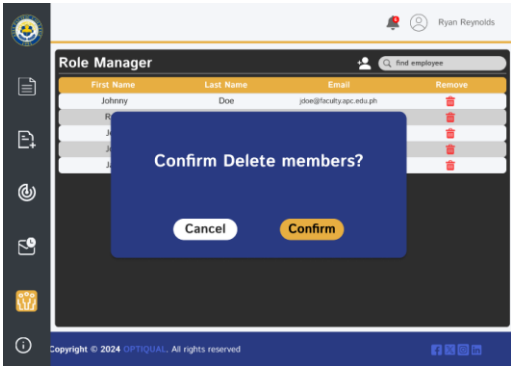


Figure 3.30 C Role Manager (Delete Member)

If the QA Personnel needs to remove a member from the team, they can easily do so by clicking the delete icon next to the member’s details in the list. The list includes the first name, last name, email, and a remove option. Once the delete icon is clicked, a confirmation message will appear asking, "Confirm delete member?" with options to either “cancel” or “confirm”. If the user clicks "Cancel," the deletion will be canceled. If "Confirm" is selected, the chosen member will be removed from the QA team. (See Figure 3.30 C)

Quality Assurance Director's Point of View

Department Manager

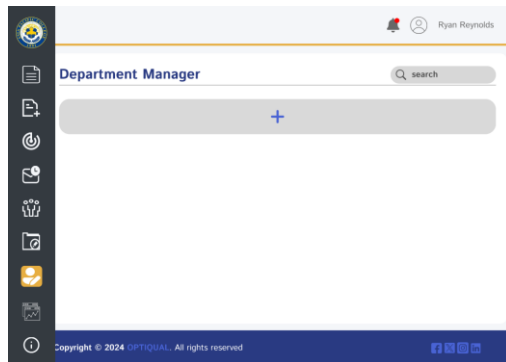


Figure 3.31 A Department Manager Page

The Quality Assurance Director's Point of View is similar to the Quality Assurance Personnel but includes two additional modules for organization. The first additional module is the “Department Manager”, which allows the QA Director to construct the organizational chart that the system will use for reference. They can create folders by clicking the Pluss button and enter the department name (see Figure 3.31 B). Once created, the folder helps in managing the departments. The QA Director can also assign personnel by inputting their name and role within the department (see Figure 3.31 C). To maintain organization, they have the option to delete folders when a department is no longer active (see Figure 3.31 D) and rename folders if a department’s name needs to be updated (see Figure 3.31 E).

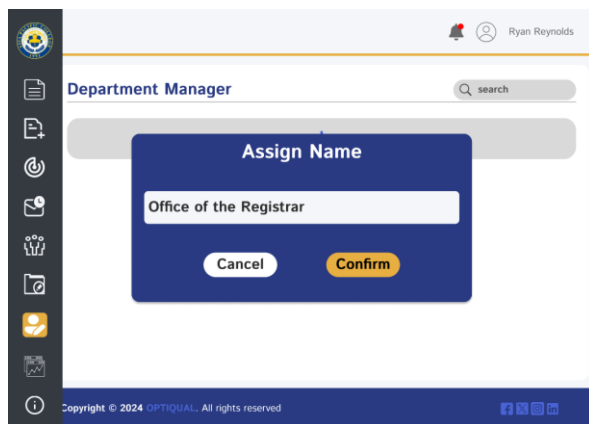


Figure 3.31 B Department Manager Page (Assign Name)

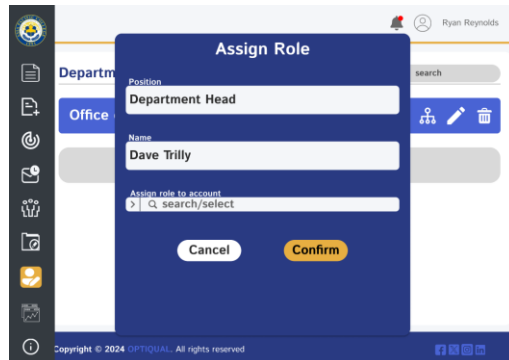


Figure 3.31 C Department Manager Page (Assign of Role)

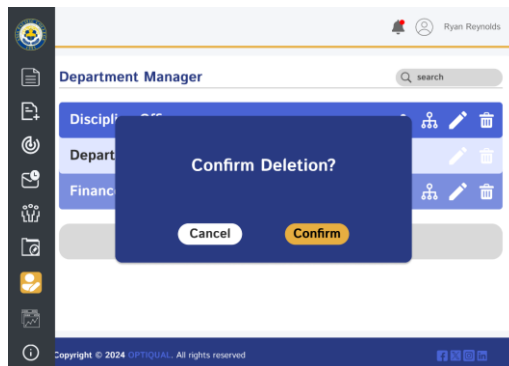


Figure 3.31 D Department Manager Page (Deleting Folder)

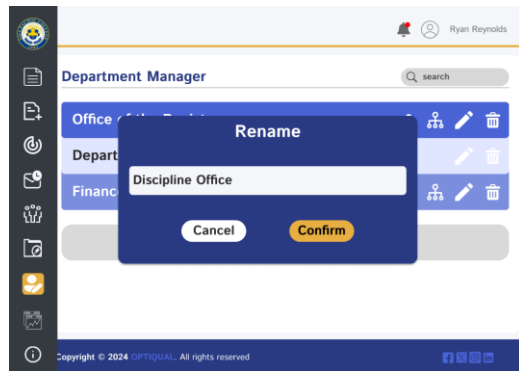


Figure 3.31 E Department Manager Page (Rename Folder)

Policies Manager

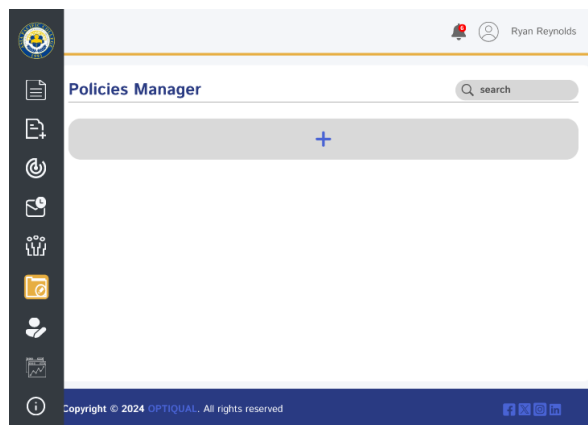


Figure 3.32 A Policy Manager Page

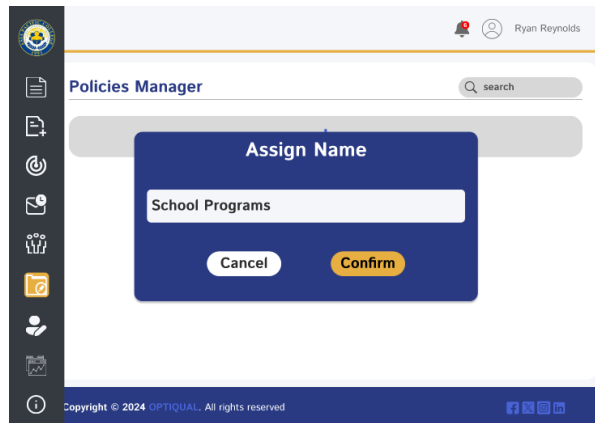


Figure 3.32 B Policy manager Page (Assign Folder Name)

The Policies Manager module allows the QA Director to create folders to keep policies organized and easy to find (see Figure 3.32 A). To create a folder, the QA Director must type a name for it (see Figure 3.32 B). After entering the name, there are two buttons: “Cancel” and “Confirm”. If the user clicks “Cancel”, the folder will not be created. But if they click Confirm, the folder will appear on the screen, ready to be used.

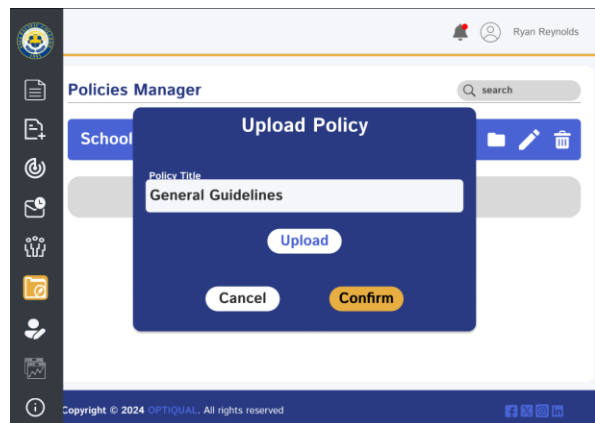


Figure 3.32 C Policy manager Page (Upload Policy)

Once the folder is created, the QA Director can upload policies by clicking the “Upload” button. This will show two ways to add a file: browsing for a file (selecting one from the computer) or dragging and dropping a file into the folder (see Figure 3.32 C). If the user clicks “Cancel”, the upload will stop. If they click “Confirm”, the policy will be saved inside the folder. The QA Director can also create subfolders inside the main folder.

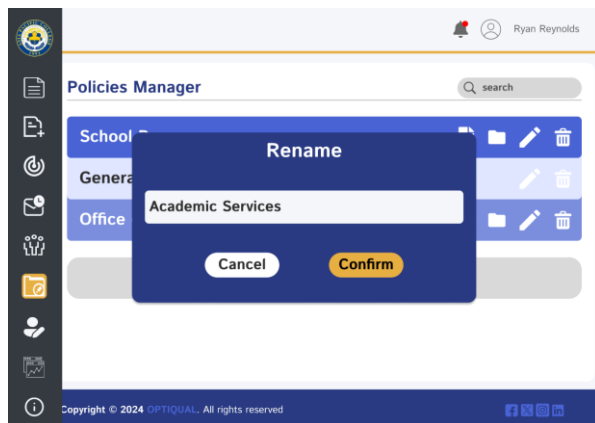


Figure 3.32 D Policy manager Page (Rename Folder)

The QA Director can also rename folders if they want to change the name (see Figure 3.32 D). After typing the new name, the system will show two buttons: “Cancel” and “Confirm”. Clicking “Cancel” will discard the changes, while clicking “Confirm” will save the new name.

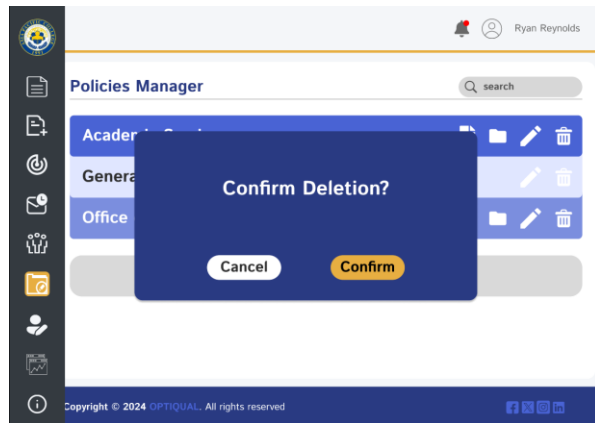


Figure 3.32 E Policy Manager Page (Delete Folder)

If a folder is no longer needed, The QA Director can delete it (see Figure 3.32 E). Before deletion, a confirmation prompt appears with “Cancel” and “Confirm” buttons. Clicking “Cancel” will keep the folder, while clicking “Confirm” will permanently remove it.

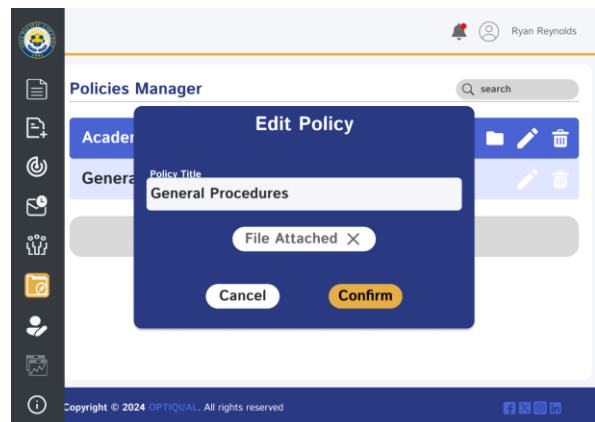


Figure 3.32. F Policy Manager Page (Edit Policy)

The QA Director can also edit policies to update or modify the contents (see Figure 3.32 F). After making changes, they must choose between “Cancel” and “Confirm”. Clicking “Cancel” will discard the changes, while clicking “Confirm” will save the updated policy.

Reports Dashboard



Figure 3.33 A Reports Module

The Reports module is a special tool designed for the Quality Assurance Director to easily monitor policy-related activities (see Figure 3.33 A). It provides a clear overview of policy statuses, starting with the Number of Active Policies per Department, where green represents approved policies and grey (or unfilled) represents the total submitted policies. There is also a date wherein the QA Director can click, which shows the monthly policy reports (see Figure 3.33 B).



Figure 3.33 B Number of Active Policies

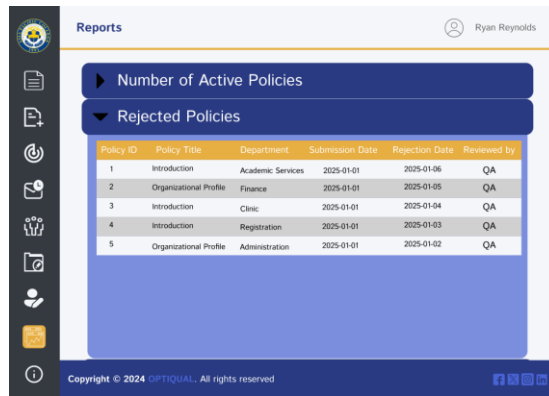


Figure 3.33 C List of Rejected Policies



Figure 3.33 D Number of Reviewed, Verified, Approved Policies

Next, the List of Rejected Policies displays details such as the policy id, title, department name, submission and rejection dates, and the reviewer's name (see Figure 3.33.C). The reports also display the total number of submitted policies, along with their status: red for reviewed, blue for verified, and green for approved. There is also a function wherein the QA Director can pick which monthly report they want to see (see Figure 3.33.D).

Additionally, the List of Feedback Received and Actions Taken helps track responses to feedback per department. It includes the department name, the policy receiving feedback, the time taken to respond, and the action status, which can be undecided, under review, or completed (see Figure 3.33 E). Lastly, the Percentage of People per Department that provides Feedback section helps identify which department is actively provide feedback, (see Figure 3.33 F). This module simplifies monitoring, ensuring policies are reviewed and handled efficiently.



Department	Feedback on	Average time taken	Actions
SOE	Program requirements	10.7 hrs	Undecided
Registrar	Enrollment Process	2 hrs	Under Review
SOMA	Program requirements	10.4 hrs	Under Review
Admissions	Enrollment Process	1.2 hrs	Completed
SOCIT	Program requirements	12.5 hrs	Completed
Registrar	Enrollment Process	10.7 hrs	Completed
SOA	Program requirements	6.2 hrs	Completed
Admission	Enrollment Process	23.7 hrs	Undecided
Registrar	Enrollment Process	10.7 hrs	Undecided
SOM	Program requirements	14.2	Under Review

Figure 3.33 E Feedback Received and Actions Taken



Department	Total Number of Employees	Number of Employees that Provide Feedback	%
Registrar	10	5	50%
Admissions	10	3	33.33%
Finance	10	2	20%
IT	15	4	26.67%
BMO	20	1	5%

Figure 3.33 F Percentage of People per Department that Provides Feedback

Admin Point of View

Update Data Privacy Act Module

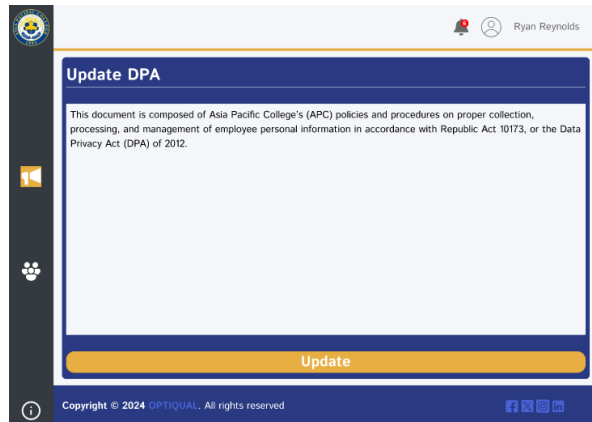


Figure 3.34 Updating of Data Privacy Act

From the admin's point of view, they can update the Data Privacy Act, which appears on the screen before users can enter their information on the Sign-Up Page. This ensures that users are aware of the latest privacy policies before they provide any personal details. Whenever there are updates or changes to the Data Privacy Act, the admin is responsible for uploading the new version to the system. To do this, the admin accesses the Update to Data Privacy Act Module, where they can review, and edit the existing Data Privacy Act with the latest version (see Figure 3.34). Once the updated data privacy is uploaded, it will automatically appear on the Sign-Up Page, requiring users to read and acknowledge it before they can proceed with account registration. The Updated DPA notice will also appear once when a user logs in.

Account Manager Module

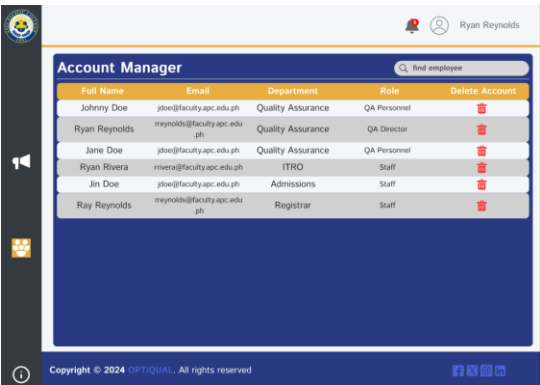


Figure 3.35 A Account Manager

In the Account Manager Module, the admin can view a list of all registered accounts in the system. This list includes essential details such as the full name, email, role, and a delete option. If the admin needs to find a specific user, they can use the Search Bar at the top of the page. By typing a name or email, the system will filter the list and display only the matching accounts, making it easier to locate a specific person without manually searching through all entries (see Figure 3.37.A).

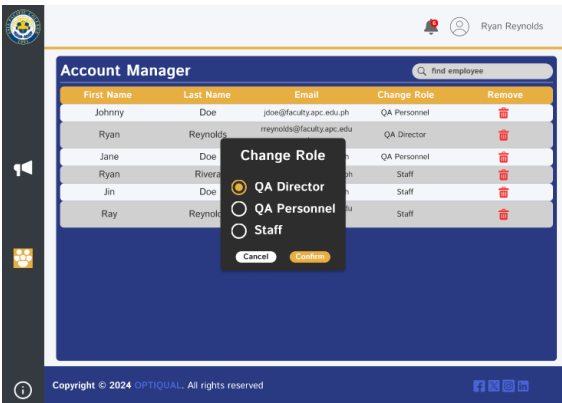


Figure 3.35 B Change Role Function

If the admin wants to change a user's role, they can click on the person's current role. When clicked, a Change Role field will appear, displaying three available options: QA Director, QA Personnel, and Staff. The QA Director is the highest role, responsible for overseeing the entire Quality Assurance process. The QA Personnel role is assigned to those managing policies and ensuring compliance with quality standards. The Staff role is for regular users with limited access, mainly for viewing or submitting policies. The admin can select the appropriate role, and once confirmed, the system will update the user's role accordingly (see Figure 3.35 B).

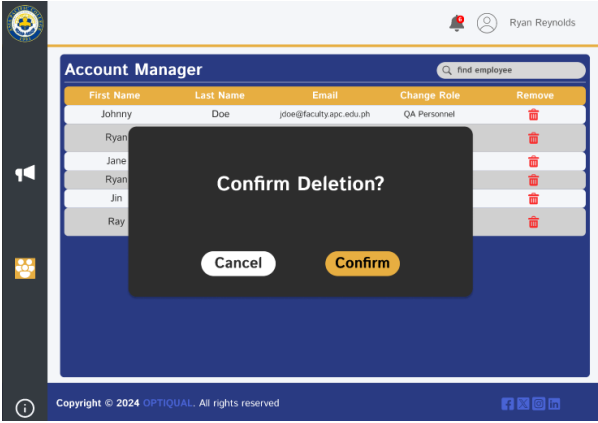


Figure 3.35 C Deletion of Account

If the admin wants to remove or delete an account, they can click the Trash Bin icon next to the user's details. Before the account is deleted, a confirmation message will appear, asking the admin to confirm or cancel the action. If the Cancel button is clicked, the deletion process will be stopped, and the account will remain in the system. If the Confirm button is clicked, the account will be permanently removed (see Figure 3.35 C).

3.4.3 Reports Visualization

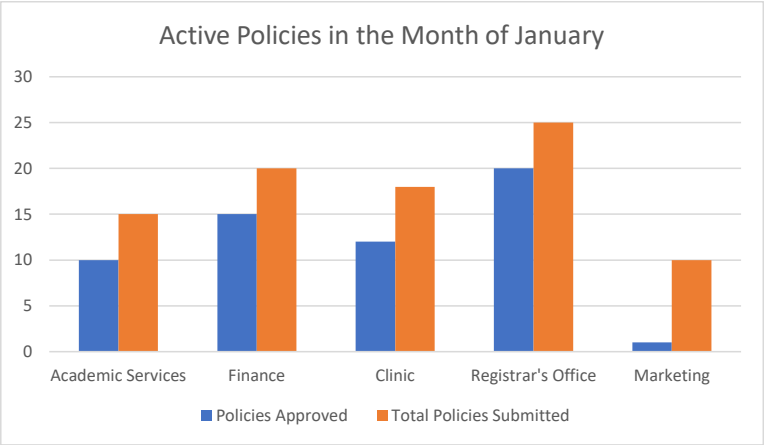


Figure 3.36 Number of Active Policies

Figure 3.36 shows the number of active policies submitted across different departments. It provides a clear breakdown of policies per month, with each department listed alongside the number of policies submitted and the number of policies approved. This helps track how efficiently each department is managing and processing its policies. For example, in the month of January the Academic Services has submitted a total of 15 policies with 10 of have been approved and implemented.

Table 3.19 Rejected Policies

Rejected Policies					
Policy ID	Policy Title	Department	Submission Date	Rejection Date	Rejected By
1	Remote Work Policy	Academic Services	2025-01-10	2025-01-15	QA Personnel
2	Budget Allocation	Finance	2025-01-12	2025-01-18	QA Personnel
3	Operational Guidelines	Clinic	2025-01-14	2025-01-20	QA Personnel
4	Registration Process	Registrar’s Office	2025-01-17	2025-01-22	QA Personnel
5	Marketing Strategy	Marketing	2025-01-18	2025-01-23	QA Personnel

Table 3.19 offers a comprehensive overview of the policies that were proposed but ultimately rejected during the review and approval stages. It includes crucial details for each rejected policy, such as the Policy ID, Policy Title, the Department that submitted it, and the specific dates of both the policy submission and rejection. It also shows the person who rejected the proposal.

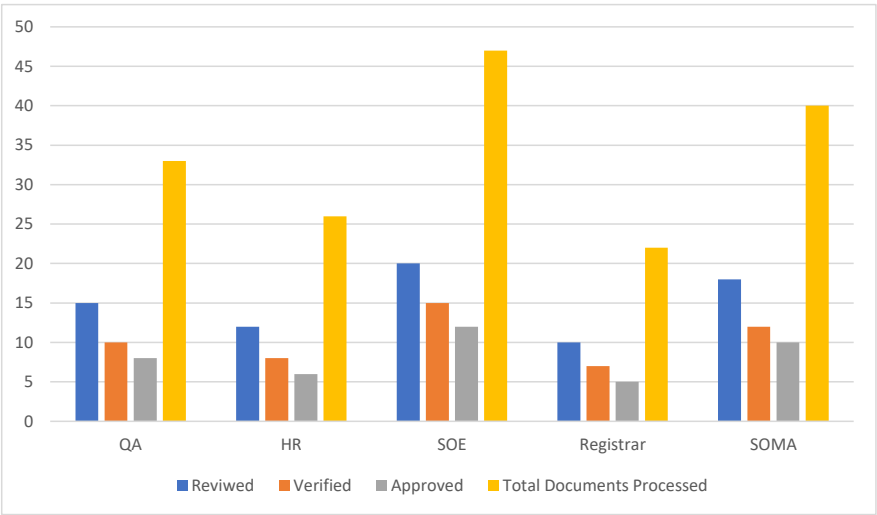


Figure 3.37 Number of Reviewed Verified and Approved Policy

Figure 3.37 presents an organized report on the policies that have been reviewed, verified, and approved by various employees across departments. It provides a detailed breakdown of each department’s contributions to the policy management process. The columns include the Department name, the number of Documents Reviewed, Documents Verified, Documents Approved, and the Total Documents Processed.

For example, an employee from the QA department reviewed 15 documents, verified 10 of them, approved 8, and processed a total of 33 documents. Similarly, employees from HR, SOE, Registrar, and SOMA departments are also listed, showing their efforts in handling documents throughout the process. This table highlights the efficiency and workload distribution among employees, giving insight into how many documents each department and individual has processed, verified, and approved.

Table 3.20 Feedback Received and Actions Taken

Feedback Received and Actions Taken			
Policy Title	Process Stage	Average Time Taken	Actions
Cafeteria Guidelines	Submission	2 days	Undecided
Registration Guidelines	Review	3 days	Under Review
Administration Guidelines	Verification	2.5 days	Under Review
Academics Guideline	Approval	4 days	Completed
Admissions Guidelines	Revision	3.5 days	Completed
Dress Code Guidelines	Feedback Response	1 day	Completed

Table 3.20 offers a detailed report on the feedback process for specific policies, focusing on the average time taken during each stage and the actions that have been or are being taken in response to the feedback received. It includes the stages of the process, such as Submission, Review, Verification, Approval, Revision, and Feedback Response, and provides key insights into how long each stage typically takes on average.

For example, the Submission stage takes an average of 2 days, but the action status is still "Undecided." The Review and Verification stages are both "Under Review," with an average time of 3 and 2.5 days, respectively. On the other hand, the Approval and Revision stages have been marked as "Completed," with an average time of 4 and 3.5 days. The Feedback Response stage has been completed in just 1 day.

Table 3.21 Percentage of people per department who always provide feedback

Percentage of people who always provide feedback			
Department	Total Employees	Employees Who Provide Feedback Regularly	Percentage (%)
HR	50	35	70.00%
Finance	40	20	50.00%
IT	45	30	66.67%
Operations	60	45	75.00%
Marketing	30	20	66.67%
Total	225	150	66.67%

Table 3.21 shows how many people from each department always give their feedback, or ideas, regularly. It tells us how many people work in each department and how many of them always share their thoughts. For example, in HR, there are 50 people, and 35 of them always give feedback, which is 70 out of 100 people. In Finance, 20 out of 40 people give feedback regularly, so that's 50 out of 100 people. The table helps us see which department has more people sharing their ideas and which ones might need a little more encouragement. In total, 150 people out of 225 are always giving feedback, which means most people in the institution are helping by sharing their thoughts.

3.5. Methods, tools, and techniques

3.5.1 Development Method

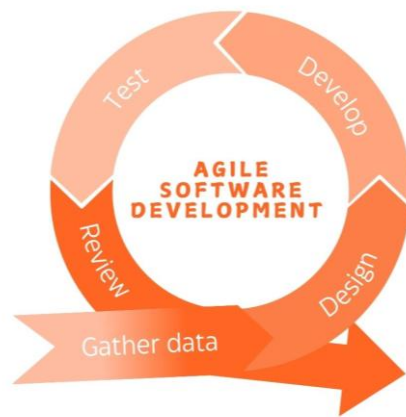


Figure 3.38 Agile Software Development Model

The Agile Software Development Methodology was used as a framework to create the software system. Agile is an iterative and adaptable method for software development and project management. This process divides the project into short, manageable parts known as iterations or sprints, which typically span a few weeks. Each iteration follows the Agile cycle, as shown in Figure 3.38.

Before the cycle begins, the "Gather Data" phase collects and examines data to ensure the best possible user experience. The cycle is divided into various phases, the first of which is the "Design" phase, which involves planning and structuring the system architecture and user interface. This is followed by the "Develop" phase, which includes code and database configuration. Next, the "Test" process guarantees that the system is operational and error-free. Once testing is completed, the system is evaluated in the "Review" step, during which the client provides input, and any necessary changes are made.

Following the review, the cycle resumes to the "Design" step to begin development of the next version. This process continues until the entire software project is completed.

3.5.2 Tools and Techniques

3.5.2.1 Software Development Tools

- **Visual Studio Code** - An open-source code editor created by Microsoft, will be the primary development and testing environment for the software system. It includes essential features like collaboration tools, code development support, and database management, as well as built-in support for HTML, CSS, JavaScript, and PHP, allowing developers to efficiently write and manage project code.
- **IntelliJ IDEA Community Edition**- This integrated development environment (IDE) will be used for frontend development, specifically to create the system's user interface with HTML. IntelliJ IDEA features intelligent code completion, debugging tools, and version control integration, allowing developers to simplify frontend design and implementation while maintaining efficiency and accuracy.
- **MySQL Workbench**- before implementing the system's database, MySQL Workbench will be used for database design and testing. Developers will use its tools, which include SQL query execution, database modeling for visualization, and the construction of database views, to ensure that the database is well-structured and optimized before deployment.
- **phpMyAdmin** - the system's database management application, will be used to manage data from our MySQL Database. This web-based platform provides a user-friendly interface that allows developers to conduct critical activities such as database, editing, SQL query execution, and data import/export, resulting in efficient database management.
- **LucidChart** - This diagramming tool will help developers create the system's Entity Relationship Diagram (ERD). Lucidchart's easy interface and real-time collaboration tools enable efficient and structured database construction, as well as the clear representation of relationships between data entities.

- **Figma** - The user interface for the software system will be designed using Figma, a cloud-based design and prototyping tool. Developers will use its features to prototype UI layouts, set design principles, and verify proper placement and scaling of interface elements, hence improving the front-end development workflow.

3.5.2.2 Project Management Tools

- **Microsoft Teams** - Microsoft Teams serves as a collaborative platform that enhances communication and coordination among the project's developers. It is also the most accessible option, as it is widely used by students at Asia Pacific College. The platform offers various communication tools, including file sharing and storage, project announcements, messaging, online meetings, and task assignments. These features are already familiar to the developers, making it an efficient choice for collaboration.
- **Microsoft Excel** - This spreadsheet program allows for easy data storage as well as the creation of tables and charts for project management. The project manager will utilize this tool to generate Gantt charts for scheduling and to document data and events related to project management, taking advantage of its features for efficiently creating graphs and tables.

3.5.2.3 Software Development Techniques

- **Data Flow Diagram:** This software development technique is used by the system developers to visualize the movement of the data within the system. This includes the entities or users of the system as the source and receiver of data, the major databases to store the data, and the major and specific processes in the system.
- **Entity Relationship Diagram:** The system developers also used an Entity Relationship Diagram to visualize the tables of the system's database; the names of the fields, the foreign keys, and the primary keys, as well as the relationship between the tables.

3.5.2.4 Project Management Techniques

Work Breakdown Structure: To identify the specific tasks that need to be performed to accomplish the development of the system, the Work Breakdown Structure (WBS) is the technique used to aid project management. This structure provides a hierarchy decomposed into phases of the project, which will help the system developers identify the tasks and subtasks to be performed

3.5.3 Infrastructure

The software system's infrastructure and its requirements must be identified as the project cannot proceed without them. This infrastructure involves three key aspects: network requirements, server requirements, and software requirements.

Network Requirements

A high-bandwidth internet connection is essential for the system's development and operation to handle data traffic efficiently and ensure adequate upload and download speeds. Users and developers must be in a location with a strong network signal from the Wi-Fi modem. Additionally, to access the system, it is required to use the APC exclusive internet to gain access.

Server Requirements

A suitable server host is essential for the system's infrastructure. The SQL-based server used for the system software is provided by the School of Engineering at Asia Pacific College. It is located on the 8th floor of the APC building within the AIRHUB. The server host features an Intel i9 processor, 2 terabytes (2TB) of storage, an additional 2TB for backup, and 32 gigabytes (32GB) of RAM.

Software Requirements

The system's development will require several essential software tools. MySQL will be used for database storage and management. The front end will be developed using HTML (Hypertext Markup Language), CSS (Cascading Style Sheets), and JavaScript within IntelliJ IDEA. For backend development and API implementation, PHP will be utilized.

3.6 Managerial Process Plan

3.6.1 Work Breakdown Structure

Table 3.22 WBS

Data Gathering	Define Project Requirements
	Interview QA Personnel
	Interview Department Heads
	Interview Previous QA Director
	Interview Current QA Director
	Literature Study
Analysis and Documentation	Identify Current Processes
	Identify Use Case Scenarios
	Identify Pain Points
	Research and Data Documentation

Planning	Forming UI Hierarchy
	Wireframing
UI Design	Design Login and Sign-up Page
	Design Policy Repository
	Design Task Manager
	Design User Roles and Permissions
	Design Reports Dashboard
	Design Policy Submission, Verification, Approval, Upload, Document History
	Design Policy Folder and Department Organization
	Design Role Manager
System Design	Design Entity Relationship Diagram
	Design Data Flow Diagram
Development	Log In and Sign Up
	UI Design
	User Log In
	User Sign Up
	Forgot Password
	Email Authentication
	Policy Repository
	UI Design
	Policy Folder Organization
	Employee - Department Organization
	Submission of Feedback
	Policy Viewing and Download Audit
	Policy Submission
	UI Design
	Facility for Submission
	Integration with Notifications and Task Manager
	Policy Implementation Process Tracker

	UI Design
	View of submitted policies with implementation status
	Task Manager
	UI Design
	Policy Proposal Management (Reject, Upload, Feedback, Assign Verifier and Approver)
	Digital Signature for Review, Verification and Approval
	Integration with Feedback and Notifications Functions
	Role Management
	UI Design
	Manage User Roles
	Reports
	UI Design
	Reports Generation
Testing	Phase 1 - User Sign Up
	Phase 2 - User Log In
	Phase 3 - User redirected to Data Privacy Act
	Phase 4 - Search and Filter
	Phase 5 - Policy Submission
	Phase 6 - QA Personnel Policy Reviewal
	Phase 7 - Assigning Verifier and Approver
	Phase 8 - Assigning other people to Assign Roles
	Phase 9 - Assigned Verifier will Verify the proposed policy
	Phase 10 - Assigned Approver will approve the proposed policy
	Phase 11 - Quality Assurance Uploads the Policy
	Phase 12 - Notification Receiving
	Phase 14 - Feedback Submitting
	Phase 15 - Feedback Responding
	Phase 16 - Department Organization
	Phase 17 - Policy tracking

Review	System review with clients
	Revisions
Documentation	Chapter 4: Project Development
	Chapter 5: Testing and Evaluation
	Chapter 6: Summary and Recommendations

Table 3.22 presents the Work Breakdown Structure (WBS) for the system's development, organizing tasks into key modules: Data Gathering, Analysis and Documentation, Planning, UI Design, System Design, Development, Testing, Review, and Documentation. Most of the critical tasks fall under the Development phase, where core system functionalities are implemented.

3.6.2 Schedule Allocation

Table 3.23 *Term 1 Schedule*

		Term 1													
		Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9	Wk 10	Wk 11	Wk 12	Wk 13	
Graphic Design	UI Design														
	UI Designer 1					8 hrs	8 hrs	8 hrs	8 hrs	8 hrs	8 hrs	8 hrs	8 hrs		
	UI Designer 2					8 hrs	8 hrs	8 hrs	8 hrs	8 hrs	8 hrs	8 hrs	8 hrs		
	UI Designer 3					8 hrs	8 hrs	8 hrs	8 hrs	8 hrs	8 hrs	8 hrs	8 hrs		
	Project Management														
	Project Manager				30 hrs										
	Documentation														
	Documenter 1												4 hrs	5 hrs	

Table 3.23 presents the Term 1 work schedule followed by two more with each term having 13 weeks each. For the first term much of the work relegates to the Graphic Design of the website. The Graphic Design has three jobs, UI Design, Project Management and Documentation. There will be workers in Term 1, three UI Designers for the UI Design, one Project Manager for Project Management and one Documenter for the Documentation.

Table 3.24 *Term 2 Schedule*

		Term 2												
		Wk 14	Wk 15	Wk 16	Wk 17	Wk 18	Wk 19	Wk 20	Wk 21	Wk 22	Wk 23	Wk 24	Wk 25	Wk 26
Graphic Design	UI Design													
	UI/UX Designer 1			2 hrs	2 hrs	2 hrs					5 hrs	5 hrs		
	UI/UX Designer 2			2 hrs	2 hrs	2 hrs					5 hrs	5 hrs		
	UI/UX Designer 3			2 hrs	2 hrs	2 hrs					5 hrs	5 hrs		
	Project Management													
	Project Manager		4 hrs							3 hrs				
	Documentation													
	Documenter 1						2 hrs	2 hrs					4 hrs	2 hrs
Database	Database Design													
	Database Developer									10 hrs	10 hrs	15 hrs		
	Project Management													
	Project Manager								10 hrs					
	Testing													
	Software Tester 1												3 hrs	
	Documentation													
	Documenter 1													4 hrs

In Table 3.24 presents the term 2 schedule again with 13 weeks but it works on 2 modules. The first module is the Graphic Design much of the job done is more on ironing out the design. The design is not final since it may change if the client asked to or a more efficient design or system is discussed and agreed upon. The second module will be the Database where and there are four jobs which are the Database Design, Project Management, Testing and Documentation. Each job has one different worker.

Table 3.25 *Term 3 Schedule*

		Term 3												
		Wk 27	Wk 28	Wk 29	Wk 30	Wk 31	Wk 32	Wk 33	Wk 34	Wk 35	Wk 36	Wk 37	Wk 38	Wk 39
Sign-up & Sign-in	Front-end Development													
	Developer 1	12 hrs												
	Developer 2	12 hrs												
	Back-end Development													
	Developer 3	12 hrs												
	Project Management													
	Project Manager	2 hrs												
	Testing													
	Software Tester 2	2 hrs												
	Documentation													
	Documenter 2		4 hrs											

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Policy Submission	Front-end Development													
	Developer 1		12 hrs											
	Developer 2		12 hrs											
	Back-end Development													
	Developer 3		12 hrs											
	Project Management													
	Project Manager		2 hrs											
	Testing													
	Software Tester 1		2 hrs											
	Documentation													
	Documenter 1		4 hrs											
Task Manager Module	Front-end Development													
	Developer 1		12 hrs											
	Developer 2		12 hrs											
	Back-end Development													
	Developer 3		12 hrs	12 hrs										
	Project Management													

Verification/Approval Process	Project Manager		2 hrs										
	Testing												
	Software Tester 2			2 hrs									
	Documentation												
	Documenter 2		4 hrs										
	Front-end Development												
	Developer 1		12 hrs	12 hrs									
	Developer 2		12 hrs	12 hrs									
	Back-end Development												
	Developer 3			8 hrs									
	Project Management												
	Project Manager		2 hrs										
	Testing												
	Software Tester 1			2 hrs									
	Documentation												
	Documenter 1		4 hrs										

Policy Repository	Front-end Development														
	Developer 1			12 hrs	12 hrs										
	Developer 2			12 hrs	12 hrs										
	Back-end Development														
	Developer 3				12 hrs										
	Project Management														
	Project Manager			2 hrs											
	Testing														
	Software Tester 2				4 hrs										
	Documentation														
Documenter 2			6 hrs												
Organization of Policies and	Front-end Development														
	Developer 1				12 hrs										
	Developer 2				12 hrs										
	Back-end Development														
	Developer 3					12 hrs									
	Project Management														

	Project Manager				2 hrs								
	Testing												
	Software Tester 1				4 hrs								
	Documentation												
	Documenter 1				4 hrs								
Feedback Module	Front-end Development												
	Developer 1				12 hrs	12 hrs							
	Developer 2				12 hrs	12 hrs							
	Back-end Development												
	Developer 3					12 hrs							
	Project Management												
	Project Manager				3 hrs								
	Testing												
	Software Tester 2					2 hrs							
	Documentation												
	Documenter 2				5 hrs								

Process Tracker	Front-end Development													
	Developer 1					12 hrs								
	Developer 2					12 hrs								
	Back-end Development													
	Developer 3					12 hrs								
	Project Management													
	Project Manager					1 hrs								
	Testing													
	Software Tester 1					2 hrs								
	Documentation													
	Documenter 1					2 hrs								
Search and Filter Function	Front-end Development													
	Developer 1						12 hrs							
	Developer 2						12 hrs							
	Back-end Development													
	Developer 3						12 hrs							
	Project Management													

	Project Manager						1 hrs						
	Testing												
	Software Tester 2						2 hrs						
	Documentation												
Notification System	Documenter 2						2 hrs						
	Front-end Development												
	Developer 1						12 hrs						
	Developer 2						12 hrs						
	Back-end Development												
	Developer 3						12 hrs						
	Project Management												
	Project Manager						2 hrs						
	Testing												
	Software Tester 1						2 hrs						
	Documentation												
	Documenter 1						2 hrs						

Reports Module	Front-end Development													
	Developer 1							12 hrs						
	Developer 2							12 hrs						
	Back-end Development													
	Developer 3							12 hrs						
	Project Management													
	Project Manager							2 hrs						
	Testing													
	Software Tester 2							2 hrs						
	Documentation													
Documenter 2							4 hrs							
User Account Management	Front-end Development													
	Developer 1								12 hrs					
	Developer 2								12 hrs					
	Back-end Development													
	Developer 3									12 hrs				
	Project Management													

Project Manager										2 hrs			
Testing													
Software Tester 1										2 hrs			
Documentation													
Documenter 1										10 hrs			

In Table 3.25 presents the Term 3 Schedule where most of the work done here is the actual development of the system. There are 12 modules to be done which are the Sign-In & Sign-Up, Policy Submission, Task Manager Module, Verification/Approval Process, Policy Repository, Organization of Policies and Personnel, Process Tracker, Search and Filter Function, Reports Module, and lastly User Account Management. For each module there are five jobs which are Front-end development, Back-end Development, Project Management, Testing, and Documentation. For the workers there will be three Developers, two for the Front-end and one for the Back-end, there will also be one project manager, one software tester and two documenters.

3.6.3 Budget Allocation

Table 3.26 Average Salary per Job Hour

Manpower	Work Hours	Salary per Hour	Total Pay
UI/UX Designer 1	80	313	25040
UI/UX Designer 2	80	313	25040
UI/UX Designer 3	80	313	25040
Project Manager	66	231	15246
Documenter 1	45	102	4590
Documenter 2	25	102	2550
Database Developer	35	268	9380
Software Tester 1	17	239	4063
Software Tester 2	14	239	3346
Developer 1	180	382	68760
Developer 2	180	382	68760
Developer 3	152	382	58064
Total Hours:	954	Total Cost:	306879

Table 3.26 presents the expected salary of each job employee based on their work and hourly rates. Each job has different manpower categories, and these are the UI/UX Designers, Project Managers, Documenters, Database developers, Software Testers, and Developers. Each role has different kinds of jobs, hours of work, and salaries. The salaries of each manpower differ from one another and are scaled through their hourly rates. With a total of 954 work hours the total estimated cost comes around 306879 Pesos as per (Indeed, 2024).

3.6.4 Test Plan

3.6.4.1 System Test Plan

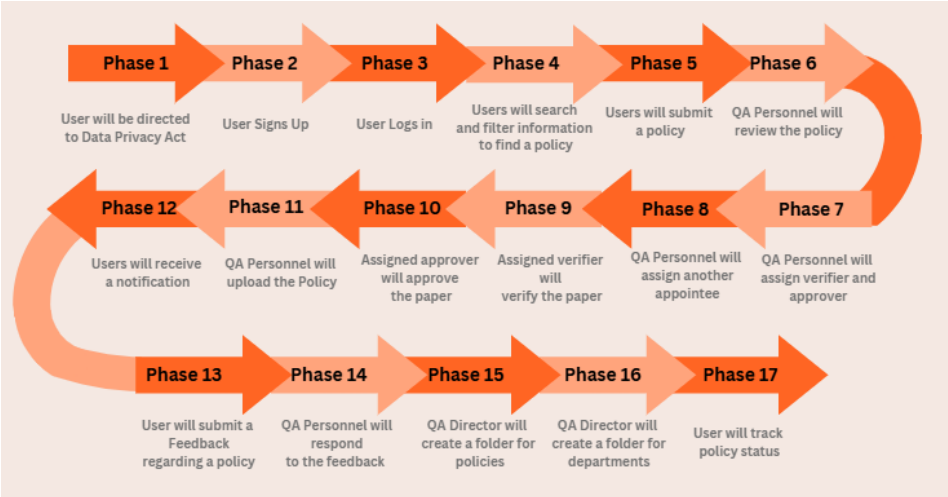


Figure 3.39 System Test Plan

Figure 3.39 illustrates the system testing process, highlighting key features and functionalities. The process begins with the Data Privacy Act page, which users must review and agree to before proceeding. After agreeing to the Data Privacy Act, users can register by entering their first name, last name, email, and password. Once registered, users log in to the system. After logging in, they can search and filter policies and submit policies for evaluation, which are reviewed by QA personnel. The QA personnel can assign a verifier and an approver with the appropriate expertise or delegate the task to another user. The verifier ensures that the policy meets institutional standards before it is approved. Users can track the status of their submitted policies. Once finalized, policies are uploaded to the system and made publicly available. Users are notified of any updates to stay informed throughout the process. Additionally, users can provide feedback to improve organizational quality. To maintain organization, the Quality Assurance Director creates folders for policies, categorized by type and department for easy access and retrieval.

3.6.4.2 Test Scenario

Table 3.27 Test Scenario

No.	Test Scenario
1	The user will be directed to the Data Privacy Act
2	User Signs in
3	User Logs in
4	Users will search and filter information to find a policy
5	Users will submit a policy
6	QA Personnel will review the policy
7	QA Personnel will assign a verifier and approver
8	QA Personnel will assign another appointee
9	The assigned verifier will verify the paper
10	The assigned approver will approve the paper
11	QA Personnel will upload the Policy
12	Users will receive a notification
13	User will submit Feedback regarding a policy
14	QA Personnel will respond to the feedback
15	QA Director will create a folder for policies
16	QA Director will create a folder for departments
17	The user will track the policy status

3.6.4.3 Test Case

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Table 3.28 Phase 1 – Test A- The user is directed to the Data Privacy Act

Test Case	Data Privacy Act of 2012	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The user is directed to the Data Privacy Act Page before signing up.	The user should be successfully redirected to the Data Privacy Act page.			
2	The user reviews the Data Privacy Act but does not click "I accept."	The user is unable to proceed without accepting the Data Privacy Act.			
3	The user attempts to skip the Data Privacy Act without accepting.	The system prevents skipping and displays			

		an error message or prompt.			
4	The user clicks "I accept" and submits, then proceeds to the Sign-up page.	The user is allowed to proceed to the Sign-up Page after accepting.			

As shown in Table 3.28, Before users can sign in to the system, they will first be directed to the Data Privacy Act Page. This page is very important because it shows the rules about how the system will handle and protect users' personal information. It's like a guide that tells users what the system can and cannot do with their data, ensuring that their privacy is respected. When users visit the Data Privacy Act Page, they must read and understand these rules before they can move forward. This step is important to make sure users know what they're agreeing to, so they are fully aware of how their information will be used. After reading the information on this page, there will be an option for users to either agree or disagree with the terms. If users agree with the Data Privacy Act, they can click a button that says something like "I Agree", and then they will be allowed to proceed to the Sign-In Page. If they don't agree with the terms, they won't be able to sign in or use the system. This process ensures that every user knows what the system will do with their data and gives them the choice to accept or decline those terms before moving forward.

Table 3.29 Phase 2 - User Signs Up

Test Case	User Signs Up	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The user fills in the First Name with text characters only (letters A-Z, a-z).	The system accepts the valid input.			
2	The user inputs a valid email (e.g., @staff.apc.edu.ph).	The system accepts the email and proceeds.			
3					

	The user inputs a invalid email (e.g., user@gmail.com).	The system rejects the input and prompts the user to enter a valid email.			
4	The user inputs an invalid email format (e.g., missing '@', missing domain).	The system rejects the input and prompts the user to enter a valid email.			
5	The user sets a password that meets the required criteria (e.g., at least 12 characters, and must include uppercase, lowercase, number, and special character).	The system accepts the password.			
6	The user enters a password with less than 12 characters.	The system rejects the password and prompts the user to meet the minimum character requirement.			
7	The user repeats information from an old account (e.g., using the same email as an existing account)	The system rejects duplicate accounts and prompts the user to choose a different email/username.			
8	The user inputs the correct verification code sent via email	The system accepts the code and proceeds to account activation			
9	The user inputs an incorrect verification code (e.g., entering an expired or random code).	The system rejects the code and displays an error message.			
10	The user completes the sign-up form correctly and clicks the "Next" button.	The system proceeds to the next step without errors.			
11	The user leaves required fields blank and clicks the "Confirm" button.	The system prevents the user from proceeding and highlights missing fields.			

Table 3.29 outlines the test plan for the user sign-up process, ensuring system accuracy, security, and usability. The process begins by validating name entries, accepting only alphabetic characters while rejecting special characters. For email input, the system accepts valid formats but rejects incorrect ones, such as those with missing '@' symbols or unapproved domains. Password validation ensures users meet security requirements, accepting passwords with at least 12 characters, including uppercase, lowercase, numbers, and special characters. If a password is too short, the system prompts the user to meet the criteria. The system also prevents duplicate accounts by rejecting repeated emails or usernames.

After registration, the system verifies the user's email through a verification code. A correct code allows progress, while an incorrect or expired code results in an error message. Finally, the system ensures that all required fields are completed before allowing users to proceed. If any field is left blank, the system highlights the missing information. This structured testing approach guarantees a secure and efficient sign-up process.

Table 3.30 Phase 3 – User Logs in

Test Case	User Logs in	Outcome	Validity		Remarks
			Successful	Not Successful	
1	User completes all log-in information (email and password), then clicks the log-in button.	User should be logged in successfully.			
2	User does not complete all required log-in information (missing email or password), then clicks the log-in button.	Error message for missing required fields.			
3	User inputs an email that is already registered in the system but with an incorrect password.	Error message for incorrect password.			
4	User inputs an email that is not registered in the system.	Error message for non-existent email.			
5	User enters the correct email but an incorrect password.	Error message for wrong password.			
6	User clicks 'show password' or 'hide password' to toggle visibility of the password field.	Password visibility toggles correctly.			

7	User clicks 'forgot password' link.	Redirected to password reset process.			
8	User inputs an email that is registered for resetting the password.	Password reset email should be sent.			
9	User inputs an email that is not registered for password reset.	Error message for unregistered email.			
10	Users input different passwords in 'password' and 'confirm password' fields.	Error message for mismatched passwords.			
11	User inputs the same password in both 'password' and 'confirm password' fields.	Password reset should be successful.			
12	User tries to log in using a different device or browser and enters valid credentials.	User should be logged in successfully.			
13	User inputs an email but does not provide a valid email format (e.g., missing '@').	Error message for invalid email format.			
14	User inputs an invalid password format (e.g., lacks required characters).	Error message for password format.			

Table 3.30 presents the test plan for the user login process, ensuring the system correctly handles various authentication scenarios before proceeding to the Data Privacy Act page. The process begins by verifying login credentials, allowing users with valid email and password combinations to access their accounts while rejecting incorrect or incomplete inputs. If a required field is missing, the system prompts the user to complete the information. An incorrect password or an unregistered email results in an error message. Users can toggle password visibility using the 'show/hide password' option and initiate a password reset if needed. The system sends a reset email for registered accounts, while unregistered emails receive an error message. During password reset, mismatched passwords trigger an error, while matching ones allow a successful reset. The system also ensures login functionality across different devices and browsers. Additionally, invalid email or password formats prompt respective error messages. This structured testing approach ensures secure and reliable login authentication.

Table 3.31 outlines the testing framework to evaluate the system's response when users are redirected to the **Home Page** within the **Policy Library Section, Role Manager, and Account Manager**. This testing framework is designed to cover a variety of potential user actions, such as searching for policies and applying filters to narrow down results. Each test case simulates a different scenario, ensuring that the system functions correctly across different situations. For example, when users search for a policy using text, the system should return relevant results. It also checks if the system can handle searches with special characters or numbers, ensuring that it can process various input types. In cases where the user inputs an invalid keyword, the system should either show no results or suggest similar policies. The framework also tests how the system responds when users apply filters based on policy categories, approval dates, or both. Additionally, the system should support searches even without keywords and should allow users to reset filters to view all policies. It also ensures that the system suggests relevant policies when no exact match is found. The testing framework ensures that all possible search and filtering scenarios are examined and validated before allowing users to proceed with policy submission, ensuring a smooth and efficient user experience.

Table 3.31 Phase 4 – The user will search and Filter Information to find a Policy

Test Case	User will search and filter information	Outcome	Validity		Remarks
			Successful	Not Successful	
1	User searches for an existing policy using only text characters (e.g., title, department name).	The system should return relevant policies based on the text search.			
2	User searches for an existing policy using special characters only (e.g., @, #, %, etc.).	The system should support searches with special characters and return matching results.			
3	User searches for an existing policy using numbers in the policy title or department name	The system should return policies that match the number or text combination in the title or department.			
4	User inputs an invalid keyword (e.g., a term not related to any policy).	The system should return a message indicating no results found or suggest similar policies.			
5	User applies a filter based on the policy category (e.g., "Privacy Policy," "Security Policy").	The system should filter and display policies relevant to the selected category.			

6	User applies a filter based on the policy approval date range (e.g., policies approved between 2020-2025).	The system should return policies within the specified date range.			
7	User applies a filter based on policy title and department name.	The system should display policies filtered by both the title and department name.			
8	User applies multiple filters (e.g., policy category and approval date range).	The system should return policies matching all selected filters.			
9	User applies filters without entering a search keyword.	The system should apply filters and return relevant policies even without a keyword.			
10	User resets all filters and search fields.	The system should clear all filters and search fields, showing all policies.			
11	User searches for a policy using an incorrect spelling or typo in the title.	The system should either show no results or suggest policies with similar titles or correct spelling.			
12	System suggests relevant policies when no exact match is found for the search term (e.g., related titles, departments).	The system should provide suggestions for policies related to the search term.			
13	User searches for a policy using only a department name.	The system should return all policies associated with the specified department.			
14	User applies a filter with no policies matching the selected criteria.	The system should return a message indicating no matching policies.			
15	User enters a keyword with both uppercase and lowercase letters (case sensitivity).	The system should perform a case-insensitive search, returning matching policies regardless of case.			
16	User searches for a policy using a title containing both numbers and text (e.g., "Policy 2022 - Data Protection").	The system should return policies that match the search term containing both numbers and text.			
17	User searches using a keyword, and the system returns all policies that contain that keyword.	The system should return all policies containing the keyword, whether it's in the title, department, or other fields.			

Table 3.32 presents a detailed testing plan for when a user is redirected to the **Home Page** in the **Policy Creation Section**. The test cases cover various scenarios that the system may encounter during the policy submission process. This ensures that the system functions properly

and efficiently before the user proceeds to review the proposed policy. For example, when the user clicks to download the policy template and chooses "Yes," the system should successfully download the template. If the user clicks "No," the download should not occur. When uploading a file, the system should accept a PDF file format, whether it's selected through the upload button or by dragging and dropping the file. If the user tries to upload an unsupported file format like DOCX or HTML, the system should reject the file. Additionally, if the user cancels the upload, the system should remove the file from the upload field. Once the user attaches a file and presses "Submit," a confirmation message will appear, asking the user to confirm the submission. If the user clicks "Yes," the policy will be submitted successfully. If the user clicks "Cancel" or fails to input a policy name or file, the system will prompt them to complete the required fields. This testing plan ensures that all possible user actions related to policy submission are tested, guaranteeing the system's functionality and helping prevent any errors before the user moves on to the next step.

Table 3.32 Phase 5 – The user will submit a policy

Test Case	User will submit a Policy	Outcome	Validity		Remarks
			Successful	Not Successful	
1	User clicks the download template and clicked yes	The template will be downloaded			
2	User clicks the download template and clicked no	The template will be not downloaded			
3	User clicks the upload file and put a PDF file format	The system should accept the file and put it in the upload file field			
4	User clicks the upload file and drag and drop a PDF File format	The system should accept the file and put it in the upload file field			
5	User clicks the upload file and put an unsupported format (eg., docx., ODT, HTML)	The system does not accept a file that is not in PDF Format			
6	User clicks the upload file and clicked 'X' button	The system will remove the upload file field in the screen			
7	User clicks the upload and attach a file, create a name and press 'Submit'	A message will ask confirm submission			

8	User clicks the upload and attach a file, create a name, press 'Submit' then a message asking for confirmation of submission the user will press 'Yes'	The system will be submitted			
9	The user click the submit button and entered a policy name then clicked cancel	The system will return to the policy submission page			
10	The user click the submit button but did not enter a policy name then clicked confirm	A message will ask to input a Policy Title			
11	The user click submit button without uploading any file	A message will ask to upload a file			

Table 3.33 provides a detailed testing plan for when a policy has been submitted and needs to be reviewed by QA Personnel. Each test case evaluates different scenarios to ensure the system works properly before moving on to assigning a verifier and an approver. For example, if the QA Personnel rejects a policy without providing feedback, the system will prompt them to enter feedback before proceeding. If feedback is given and the rejection is confirmed, the system will send the rejection and feedback to the author of the policy. If the QA Personnel sends the policy back for revision without submitting it, the system will return them to the policy viewing page, keeping the changes unsubmitted. Additionally, if the QA Personnel signs the document and clicks "submit," the policy moves forward. However, if they sign but do not submit, or if they don't sign at all and attempt to submit, the system will either remain in the review stage or prompt for required actions, such as signing. This testing plan ensures that the system responds accurately to all actions, helping to prevent errors before policies are finalized and approved.

Table 3.33 Phase 6 – The QA personnel will review the Policy

Test Case	QA personnel will review the policy	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA Personnel reject the policy without providing feedback	A message will appear asking to input feedback			
2	QA Personnel rejects the policy and provides feedback	A message will appear asking to confirm submission			
	QA Personnel rejects the policy, provides feedback and confirms the submission	The Rejection Feedback will be sent to the author of the proposed policy			
7	QA Personnel press the Accept Button	A text field will appear for additional suggestion			
	QA Personnel press the Accept Button then press Submit without writing in the Suggestions Text field	The System will ask to confirm the submission			
8	QA Personnel press the Accept Button writes a suggestion and press submit	The System will ask to confirm the submission			
9	QA Personnel press the Accept Button then press Submit without writing in the Suggestions Text field and confirm the submission	The system will record and submit the document			
10	QA Personnel press the Accept Button writes a suggestion and press submit and confirm the submission	The system will record and submit the document			

Table 3.34 presents a detailed testing plan for the stage after the QA Personnel has reviewed the policy. This plan outlines different scenarios to test how the system handles the assignment of a Verifier and Approver, ensuring smooth functionality. Each test case identifies potential system outcomes and specifies the expected behavior to ensure that any issues are resolved before moving on to the next stage. For instance, if the QA Personnel assigns a valid verifier and clicks submit but does not click confirm, the system will prompt the QA to assign a different person to approve the file. Similarly, if the QA Personnel assigns an approver but fails to click submit, the system will require them to complete the submission process. The system also checks for scenarios like when the QA assigns an unavailable verifier or approver, ensuring that

the assignment process is valid and the roles are properly filled. If the QA fails to assign either a verifier or approver, the system will flag this as an incomplete action and require the QA to make the necessary assignments. This thorough testing plan ensures that all steps in the process are followed correctly before policies are moved to the next stage.

Table 3.34 Phase 7 – The QA Personnel Assigns a Verifier and Approver

Test Case	The QA will assign Verifier and Approver	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA personnel assign a valid verifier and click submit and confirm	The system will ask the QA to Assign a different person to Approve the file			
2	QA personnel assigned a valid verifier and click submit but did not click confirm	The system will ask the QA again to Assign a different person to Approve the file			
	QA personnel assigned a valid verifier but did not click 'X'	The system will return the QA to the Quality			
3	QA personnel assign a valid approver and click submit and confirm	The system will submit the evaluators and return the QA to the Task Manager Page			
	QA assigns an unavailable verifier	The system will ask to reassign a different person as the verifier			
4	QA assigns an unavailable approver	The system will ask to reassign a different person as the approver			

Table 3.35 outlines a testing plan for situations where the QA Personnel has completed the review, but the verifier and approver are unavailable. The plan includes various test cases to ensure the system works as expected in these scenarios. For instance, when the QA Personnel assigns a valid person to handle the verifier or approver role, clicks submit, and confirms, the system should proceed as intended. If the QA skips either the submit or confirm step, the system will prompt the QA to complete the required action. Additionally, if the QA assigns an unavailable person to handle the verifier or approver role, the system should flag this and prevent further progress. This will ensure that the system only moves forward when a valid person is assigned to

each role. These tests are designed to catch potential issues, ensuring smooth functionality and that all necessary steps are followed before continuing with the process.

Table 3.35 Phase 8 – The QA Personnel Assigns Another Person to Assign Roles

Test Case	QA Personnel will assign another person to assign roles	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA personnel assigns a valid person to handle verifier selection and click submit and confirm	The system will notify the user of the task and return the QA to the Task Manager			
2	QA personnel assigns a valid person to handle verifier selection and clicked cancel	The QA personnel will be returned to the Assign Appointee window			
4	QA Personnel press select but didn't select a person	A error message will appear asking to choose a person			
5	QA personnel assigns a valid person to handle approver selection and clicked submit and confirm	The system will notify the user of the task and return the QA to the Task Manager			
6	QA personnel assigns a valid person to handle approver selection and clicked cancel	The QA personnel will be returned to the Assign Appointee window			
9	QA personnel assigns an unavailable person to assign a verifier	The system will ask the QA Personnel to choose another person as a verifier			
10	QA personnel assigns an unavailable person to assign approver	The system will ask the QA Personnel to choose another person as a approver			

Table 3.36 outlines a testing plan for scenarios where the assigned verifier is responsible for verifying the paper. The test cases in this plan examine different actions the verifier can take, ensuring that the system functions correctly before moving forward to the approval stage. In each test case, the verifier either provides feedback on the proposed policy, submits the feedback, and confirms the action or fails to complete the required steps. For example, if the verifier provides feedback and clicks submit but forgets to confirm, the system should prompt them to finalize the action. Similarly, if the verifier rejects the proposed policy but does not provide the necessary

feedback or submit the information, the system will highlight this as an issue. The test cases also account for actions such as signing the document and submitting it. If the verifier does not sign or submit the document, the system should not proceed until the necessary steps are completed. This testing ensures that all verification actions are handled properly, and no step is skipped before advancing to the next stage of the process.

Table 3.36 Phase 9 – The assigned Verifier will verify the Paper

Test Case	Assigned verifier will verify the paper	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The Verifier presses on the reject button	A text field would appear where the verifier would write the feedback			
2	Verifier makes feedback on why they reject the proposed policy and clicked submit and confirm	The feedback rejection is recorded and the system processes the feedback			
3	Verifier makes feedback on why they rejected the proposed policy and clicked submit but did not click confirm	Rejection feedback is successfully recorded, and the system processes the rejection.			
4	Verifier makes feedback about why they reject the proposed policy but did not click submit	Feedback is not recorded, and no action is taken.			
5	The verifier presses the Sign Document Button	The system will ask to confirm the Verification			
6	The verifier presses the Sign Document Button and confirms Verification	The document is successfully signed and submitted.			
7	The verifier presses the Sign Document Button and pressed cancel	The Verifier will be returned to the policy to be verified			

Table 3.37 presents a detailed testing plan for the scenario where the assigned approver is responsible for approving the paper. Each test case in this plan evaluates different situations the system might encounter to ensure all possible scenarios are accounted for before proceeding with the policy upload process. The test cases include various actions the approver might take, such as sending the policy back with feedback for revisions or rejection. The system should handle these actions appropriately, ensuring that the approver's feedback is submitted correctly when they click "submit" and confirm. If the approver sends back feedback but cancels the submission,

the system should not proceed and should alert the approver to complete the action. Additionally, the plan accounts for scenarios where the approver signs the document or does not, as well as cases where they fail to click submit after signing. The testing ensures that all required steps are completed and no action is skipped, preventing any delays in the approval process. This thorough testing guarantees the system's readiness before moving forward with the policy upload.

Table 3.37 Phase 10 – The assigned Approver will approve the Paper

Test Case	Assigned approver will approve the paper	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The approver presses on the reject button	A text field would appear where the verifier would write the feedback			
2	Approver makes feedback about why they reject the proposed policy and clicked submit and confirm	The rejection feedback is recorded, and the system processes the feedback			
3	Approver makes feedback about why they rejected the proposed policy and clicked submit but did not click confirm	Rejection feedback is successfully recorded, and the system processes the rejection.			
4	The approver makes feedback because they reject the proposed policy but did not click submit	Feedback is not recorded, and no action is taken.			
5	The approver presses the Sign Document Button	The system will ask to confirm the Verification			
6	The approver presses the Sign Document Button and confirms Verification	The document is successfully signed and submitted.			
7	The approver presses the Sign Document Button and pressed cancel	The Verifier will be returned to the policy to be verified			

Table 3.38 outlines a comprehensive testing plan for the stage after the assigned approver has approved the paper. Each test case in the plan addresses various possible scenarios that the system may encounter, ensuring all potential situations are thoroughly tested before moving on to uploading the policy and notifying the users. The testing plan covers multiple situations where

the QA personnel is required to fill in documentation information. In scenarios where the required fields are not completed or not checked, the system will display a message prompting the user to fill in the necessary information before submitting. If the QA personnel cancel the process, the system should remove the document information fields from the screen. The test cases also account for different actions, such as submitting the policy with incomplete information or canceling the action, ensuring that the system responds appropriately in each case. This testing ensures that the upload process is smooth, that all fields are completed properly, and that the system behaves as expected in case of incomplete submissions or cancellations.

Table 3.38 Phase 11 – The QA Personnel Will Upload the Policy

Test Case	The QA Personnel will upload the policy	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA personnel fill the documentation information completely and click submit	The system should put the uploaded policy in the policy repository			
2	QA personnel fill in the documentation information completely but did not click submit	The system should keep displaying the document information field until the user doesn't make any actions			
3	QA personnel checked the revision if minor/major but did not put description and clicked submit	The system should display a message that all fields should be filled in first before clicking submit			
4	QA personnel did not check the revision if minor/major but input description and clicked submit	The system should display a message that all fields should be filled in first before clicking submit			
5	QA personnel did not check whether the revision is minor/major and did not put a description but clicked submit	The system should display a message that all fields should be filled in first before clicking submit			
6	QA personnel fill the documentation information completely and click cancel	The system will remove the document information field to the screen			
7	QA personnel checked the revision if minor/major but did not put description but clicked cancel	The system will remove the document information field to the screen			
8	QA personnel did not check the revision if minor/major but input description and clicked cancel	The system will remove the document information field to the screen			

9	QA personnel did not check whether the revision is minor/major and did not put a description and clicked cancel	The system will remove the document information field to the screen			
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Table 3.39 outlines a testing plan to evaluate the notification system after the QA Personnel uploads the policy. It covers various scenarios to ensure that users receive notifications for new policies, feedback responses, policy modifications, and status updates on proposed policies. Each test case checks if users are correctly directed to the appropriate modules, such as the policy repository or task manager, when interacting with notifications. The plan also tests cases where users don't receive notifications to ensure the system is working properly in all situations. This ensures that the notification system functions as intended and provides users with the necessary information.

Table 3.39 Phase 12 – The users will receive a Notification

Test Case	The users will receive a Notification	Outcome	Validity		Remarks
			Successful	Not Successful	
1	User clicked the notification button	The system should show all the notifications			
2	User receives notification for the newly implemented policy and click it	The system should direct the user to the policy repository module			
3	Users receive notification for the response to their feedback	The system should direct the user to the task manager module			
4	Users receive notifications for the modified policies and click it	The system should direct the user to the policy repository module, specifically to the policy that has been modified			
5	Users receive notification for the policy status of their proposed policy and click it	The system should direct the user to the request tracker module			

Table 3.40 presents a testing plan for evaluating the user feedback process on a policy. The plan includes test cases that assess how the system handles different scenarios when users provide feedback. Each test case ensures that the system correctly processes the feedback, including notifying the QA Personnel and QA Director when comments are submitted. It also checks that if no comments are provided, the system won't accept the feedback and will remove the feedback field if the user cancels. Additionally, the test cases verify that users are correctly redirected to the policy repository module when they use the back button. This testing ensures smooth functionality for handling user feedback on policies.

Table 3.40 Phase 13 – The user will submit Feedback regarding a policy

Test Case	The user will submit a Feedback regarding a policy	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The user put comments on a policy then click submit	The system should accept the feedback and will notify the QA personnel and QA Director			
2	The user put comments to a policy then clicks 'X' button	The system will remove the report feedback field to the screen			
3	The user did not put any comments to a policy then click submit	The system should not accept blank feedback reports			
4	The user did not put any comments to a policy then clicked 'X' button	The system will remove the report feedback field to the screen			
5	The user clicks the back button	The system will direct the user to the main page of the policy repository module			

Table 3.41 outlines a testing plan for scenarios where the QA Personnel receives feedback on an existing policy. The test cases ensure the system's functionality in handling feedback notifications and responses. The first case checks if the QA Personnel receives the necessary notification when feedback is submitted. If no notification is received, the system has a flaw. The second case confirms that the QA Personnel correctly receives the feedback notification. The third and fourth cases test whether the QA Personnel responds to feedback, ensuring that the system supports both actions: responding and not responding. This comprehensive testing ensures smooth communication and proper handling of feedback within the system.

Table 3.41 Phase 14 – The QA Personnel will respond to the feedback

Test Case	The QA Personnel will respond to the feedback	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA Personnel press View Policy Button	The QA Personnel will be redirected to the Policy the feedback is based upon			
2	QA Personnel press View Report Feedback Button	The QA will be redirected to the Feedback of the Policy			
3	QA press the Hamburger Button	Reply Button, Request for Revision Button, and Download Change Request Form Button will show below the Hamburger Icon			
4	The QA presses the reply button	A Text field for the reply will show			
5	The QA presses the reply button inputs a reply and press submit	the reply is recorded and then will be submitted to the system			
6	The QA presses the reply button does not input a reply and press submit	An error message will show asking to input a reply			
7	The QA presses the Request for Revision Button	An overlay will show where there is a file Attachment, and Revision and Obsolete buttons will show			
8	The QA presses the Request for Revision Button, attach file and press Revision Button and press submit	The file attached is recorded for Revisions			
9	The QA presses the Request for Revision Button, attach file and press Obsolete Button and press submit	The file attached is recorded for Obsolete			

Table 3.42 presents a comprehensive testing plan for when the QA Director creates and manages folders for policies. It includes test cases that assess various scenarios to ensure the system functions properly in these situations. The plan covers the successful creation and display of folders when the name is assigned, as well as ensuring that the system requests a name if it's not provided and prevents submission until the name is entered. It also tests the deletion and renaming of folders, ensuring that these actions are reflected correctly in the system. Additionally, the plan verifies that PDF files can be uploaded into the folder, and that the system prompts the user when required fields are missing or when files are not uploaded. It also checks the handling of subfolders, confirming that the system responds appropriately when no name is assigned or when the action is canceled. Finally, the test cases address duplicate folder creation, ensuring that the system detects and prevents the creation of folders with the same name. These test cases ensure that all folder-related operations are handled efficiently, providing smooth functionality for the QA Director in managing policy files.

Table 3.42 Phase 15 – The QA Director will create a folder for policies

Test Case	The QA Director will create a folder for departments	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA Director successfully creates a folder for departments	The system should display the created folder			
2	The QA Director clicked the plus button and Assigns name for the department and clicked confirm	The system should display the created name for the folder			
3	The QA Director clicked the plus button but did not Assign a name for the department and clicked confirm	The system should display a message that the user should fill the field first before confirming			
4	The QA Director clicked the plus button but did not Assign name for the department and did not clicked confirm	The assigning of name feature will remain displayed in the users screen			
5	The QA Director clicked the plus button and Assigns name for the department and clicked cancel	The system will remove the assign name field in the screen			
6	The QA Director clicked the plus button but did not Assign name for the department and clicked cancel	The assigning of name feature will remain displayed in the users screen			

9	The QA Director clicks upload and uploads it in PDF Format	The system should accept the PDF file format			
10	The QA Director clicks assign role button, input the position, input the name, and assign role and click confirm	The role is successfully assigned to the specified person, and the system updates the user's position accordingly.			
11	The QA Director clicks assign role button, input the position, input the name, and assign role and click cancel	The system will remove the assign role field in the screen			
12	The QA Director clicks the assign role button, inputs the position, and inputs the name, but does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
13	The QA Director clicks the assign role button, inputs the position, does not input the name, but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
14	The QA Director clicks the assign role button, does not input the position, but input the name and assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
15	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
16	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
17	The QA Director clicks the assign role button, does not input the position but input the name and does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
18	The QA Director clicks the assign role button, inputs the position but did not input the name and does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
19	The QA Director clicks the assign role button, inputs the position, and inputs the name, but does not assign the role and clicks cancel	The system will remove the assign role field in the screen			
20	The QA Director clicks the assign role button, inputs the position, does not input the name, but assign the role and clicks cancel	The system will remove the assign role field in the screen			

21	The QA Director clicks the assign role button, does not input the position, but input the name and assign the role and clicks cancel	The system will remove the assign role field in the screen			
22	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks cancel	The system will remove the assign role field in the screen			
23	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks cancel	The system will remove the assign role field in the screen			
24	The QA Director clicks the assign role button, does not input the position but input the name and does not assign the role and clicks cancel	The system will remove the assign role field in the screen			
25	The QA Director clicks the assign role button, inputs the position but did not input the name and does not assign the role and clicks cancel	The system will remove the assign role field in the screen			
26	The QA director clicks the assign name button, put a name, and click confirm	The system should display the new name assigned to the folder			
27	The QA director clicks the assign name button, but did not put a name, and click confirm	The system should display a message that the user should fill the field first before confirming			
28	The QA director clicks the assign name button, put a name, and click cancel	The system will remove the assign name field in the screen			
26	The QA director clicks the assign name button, but did not put a name, and click cancel	The system will remove the assign name field in the screen			
27	QA Director attempts to create a duplicate folder with the same name and clicked cancel	The system will remove the assign name field in the screen			
28	The QA Director clicks the rename button, puts a new name, and clicked confirm	The system should display the new name assigned to the folder			
29	The QA Director clicks the rename button, did not put a new name, and clicked confirm	The system should display a message that the user should fill the field first before confirming			
30	The QA Director clicks the rename button, puts a new name, and clicked cancel	The system will remove the rename field in the screen			
31	The QA Director clicks the rename button, did not put a new name, and clicked cancel	The system will remove the rename field in the screen			

32	The QA Director clicks the delete button and clicks confirm	The system will remove the folder that the user deleted			
33	The QA Director clicks the delete button and clicks cancel	The system will remove the rename field in the screen			
34	QA Director attempts to create a duplicate folder with the same name and clicked confirm	The system should display a message that there's already a folder with the same name			
35	QA Director attempts to create a duplicate folder with the same name and clicked cancel	The system will remove the rename field in the screen			

Table 3.43 outlines a comprehensive testing plan for when the QA Director creates a folder for departments. It covers several potential scenarios to ensure the system functions as expected. Successful folder creation occurs when the QA Director assigns a name and confirms, with the system displaying the new folder. If the name is left blank, the system prompts the user to fill in the required information before proceeding. If the QA Director cancels any action, such as assigning a name or role, the system removes the corresponding field from the screen. Role assignment is also tested, where the system verifies that the position and name are input correctly; if any information is missing, a message appears indicating the user should fill in the required fields. In addition to creating folders, the QA Director can rename or delete them, with the system reflecting the changes by displaying the updated name or removing the folder entirely. The plan also checks for duplicate folder names, where the system will notify the user that the name already exists and prevent confirmation until a unique name is provided.

Table 3.43 Phase 16 – The QA Director will create a folder for departments

Test Case	The QA Director will create a folder for departments	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA Director successfully creates a folder for departments	The system should display the created folder			
2	The QA Director clicked the plus button and Assigns name for the department and clicked confirm	The system should display the created name for the folder			

3	The QA Director clicked the plus button but did not Assign a name for the department and clicked confirm	The system should display a message that the user should fill the field first before confirming			
4	The QA Director clicked the plus but did not Assign name for the department and did not click confirm	The assigning of name feature will remain displayed in the user's screen			
5	The QA Director clicked the plus button and Assigns name for the department and clicked cancel	The system will remove the assign name field in the screen			
6	The QA Director clicked the plus button but did not Assign name for the department and clicked cancel	The assigning of name feature will remain displayed in the user's screen			
9	The QA Director clicks upload and uploads it in PDF Format	The system should accept the PDF file format			
10	The QA Director clicks the assign role button, input the position, input the name, and assign role and click confirm	The role is successfully assigned to the specified person, and the system updates the user's position accordingly.			
11	The QA Director clicks the assign role button, input the position, input the name, and assign role and click cancel	The system will remove the assign role field in the screen			
12	The QA Director clicks the assign role button, inputs the position, and inputs the name, but does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
13	The QA Director clicks the assign role button, inputs the position, does not input the name, but assigns the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
14	The QA Director clicks the assign role button, does not input the position, but input the name and assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
15	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
16	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			

17	The QA Director clicks the assign role button, does not input the position but input the name and does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
18	The QA Director clicks the assign role button, inputs the position but does not input the name and does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming			
19	The QA Director clicks the assign role button, inputs the position, and inputs the name, but does not assign the role and clicks cancel	The system will remove the assign role field from the screen			
20	The QA Director clicks the assign role button, inputs the position, does not input the name, but assigns the role and clicks cancel	The system will remove the assigned role field from the screen			
21	The QA Director clicks the assign role button, does not input the position, but inputs the name and assigns the role and clicks cancel	The system will remove the assigned role field from the screen			
22	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks cancel	The system will remove the assigned role field from the screen			
23	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks cancel	The system will remove the assigned role field from the screen			
24	The QA Director clicks the assign role button, does not input the position but inputs the name, and does not assign the role and clicks cancel	The system will remove the assigned role field from the screen			
25	The QA Director clicks the assign role button, inputs the position but does not input the name, and does not assign the role and clicks cancel	The system will remove the assigned role field from the screen			
26	The QA director clicks the assign name button, puts a name, and clicks confirm	The system should display the new name assigned to the folder			
27	The QA director clicks the assign name button but does not put a name and clicks confirm	The system should display a message that the user should fill the field first before confirming			
28	The QA director clicks the assign name button, puts a name, and clicks cancel	The system will remove the assigned name field in the screen			
26	The QA director clicked the assign name but did not put a name and clicked cancel	The system will remove the assigned name field in the screen			

27	QA Director attempted to create a duplicate folder with the same name and clicked cancel	The system will remove the assigned name field in the screen			
28	The QA Director clicks the rename button, puts in a new name, and clicks confirm	The system should display the new name assigned to the folder			
29	The QA Director clicked the rename button, did not put a new name, and clicked confirm	The system should display a message that the user should fill the field first before confirming			
30	The QA Director clicks the rename button, puts in a new name, and clicks cancel	The system will remove the rename field in the screen			
31	The QA Director clicked the rename button, did not put a new name, and clicked cancel	The system will remove the rename field in the screen			
32	The QA Director clicks the delete button and clicks confirm	The system will remove the folder that the user deleted			
33	The QA Director clicks the delete button and clicks cancel	The system will remove the rename field in the screen			
34	QA Director attempts to create a duplicate folder with the same name and clicks confirm	The system should display a message that there's already a folder with the same name			
35	QA Director attempted to create a duplicate folder with the same name and clicked cancel	The system will remove the rename field in the screen			

Table 3.44 outlines a testing plan for when users track the status of their proposed policy, ensuring that different scenarios are evaluated. The first test case ensures that users can successfully track the policy status using the Request Tracker feature. The second test case examines the system's response to users using the scroll button, confirming that the system follows the user's scrolling pattern as expected. Lastly, the third test case tests the behavior when users click the back button, ensuring that the system correctly redirects them to the main page of the module. These cases are designed to ensure that the system functions smoothly during the tracking process.

Table 3.44 Phase 17 – The User will track policy status

Test Case	The User will track policy status	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The user successfully tracks policy status in the Request Tracker Feature	The user successfully tracks the policy as the policy goes through the Review, verification, and Approval process			
2	The user uses the scroll button	The system should follow the scrolling pattern of the user			
4	The users click the back button	The system should direct the user to the main page of the module			

CHAPTER 4

PROJECT DEVELOPMENT

4.1 Development Stages

MVP 1: User Access and Role-Based Management

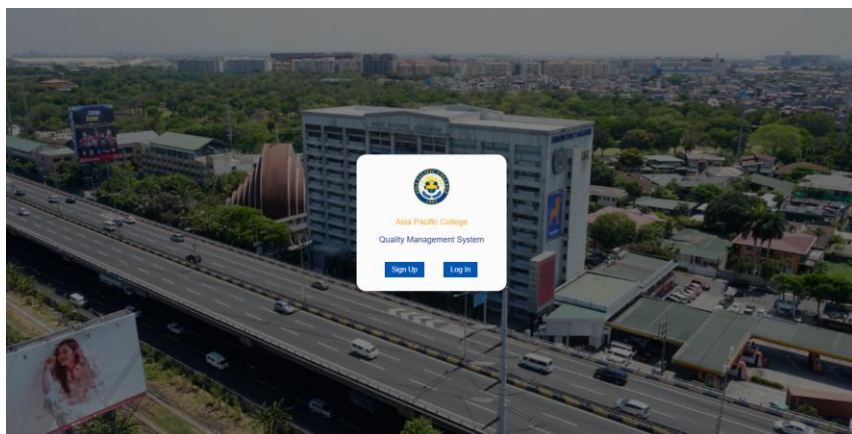


Figure 4.1

To begin the development of the system, we focused on building the front-facing screens that all users at the college encounter. As shown in **Figure 4.1**, the central access point provides two main options for entry: signing up for a new account or logging into an existing one. Building the sign-up process was more complex than a standard form.

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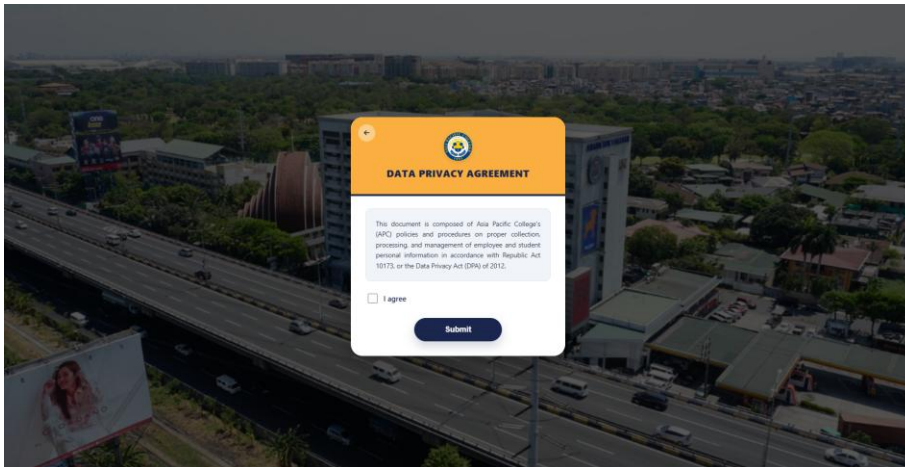


Figure 4.2

In line with project constraints and legal requirements, we integrated a mandatory Data Privacy Agreement screen, seen in **Figure 4.2**. A key development step ensures that users cannot proceed with account creation until they actively check the 'I agree' box and click submit. Once users navigate the privacy agreement, they are presented with the account registration form (**Figure 4.3**).

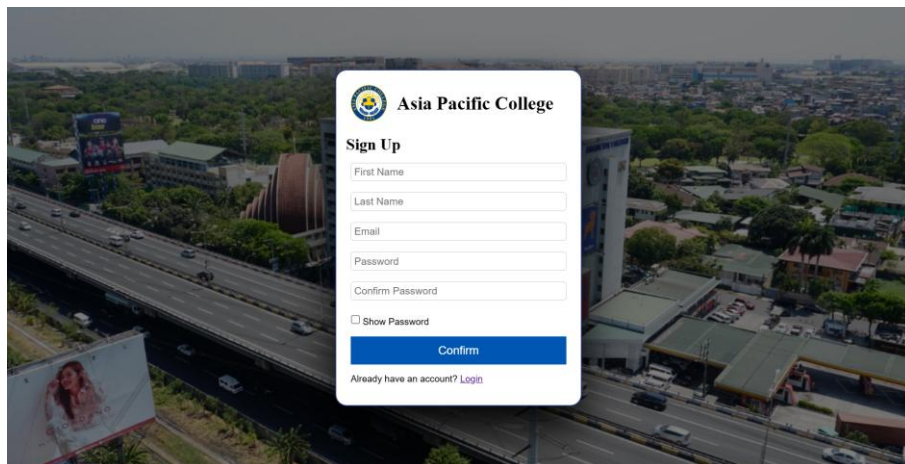


Figure 4.3

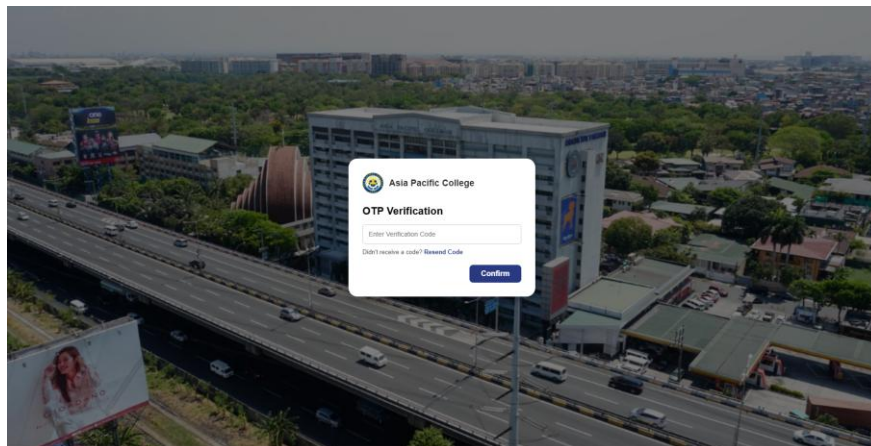


Figure 4.4

To complete the registration process and ensure only valid college emails are used, we implemented a verification feature. This tool, as seen in **Figure 4.4**, sends a code to the user's input email, which must be entered into the system to confirm and complete the account setup. Furthermore, standard system features, including the unified Log-In screen (**Figure 4.5**) and a self-service Forgot Password reset tool (**Figure 4.6**), were developed to provide a smooth and professional user experience from the moment of access.

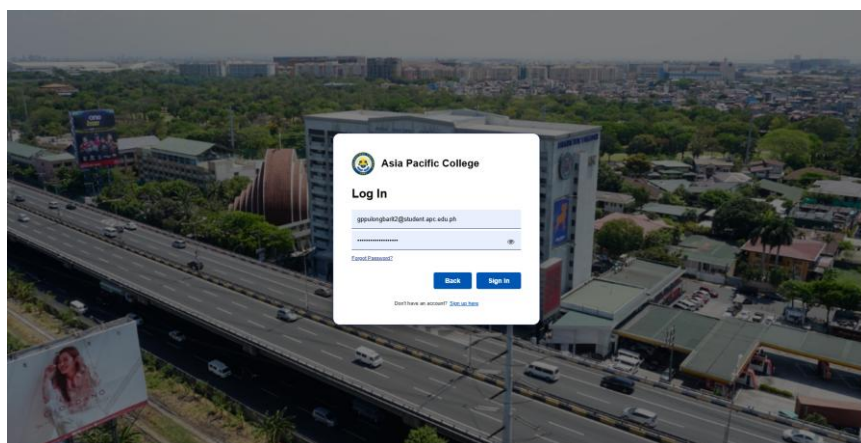


Figure 4.5

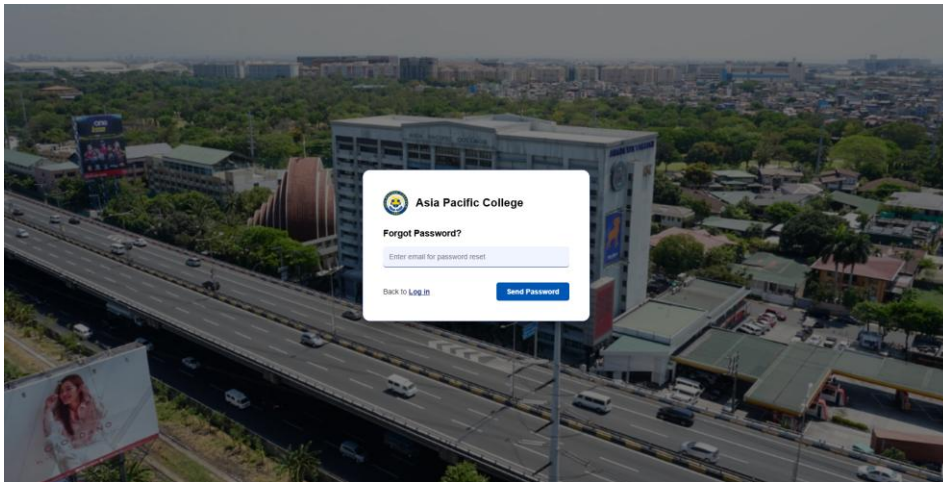


Figure 4.6

During this initial development stage, we encountered several challenges. One of the main hurdles was making sure the **One-Time PIN (OTP)** emails reached the user's inbox instead of getting lost or delayed. We also struggled with the logic behind the "Sign Up" and "Log In" buttons, specifically ensuring the system correctly checked the database to see if an account already existed before allowing a new registration. Additionally, it was challenging to make sure that once a person logged in, the system remembered their specific "role" so they wouldn't accidentally see pages or buttons they weren't supposed to access. These problems required us to double-check our code and simplify how the system talks to the database to ensure everything worked smoothly for the user.

MVP 2: Policy Management

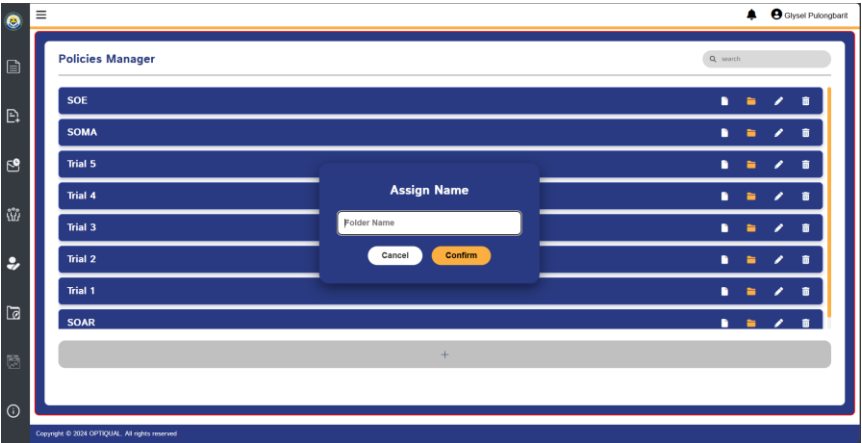


Figure 4.7

Following the establishment of user access, the second stage of development focused on the Policy Management module, which serves as the organizational core of the system. As shown in **Figure 4.7**, we developed a Policies Manager interface that allows authorized users to create a structured filing system by adding main folders for different departments or categories.

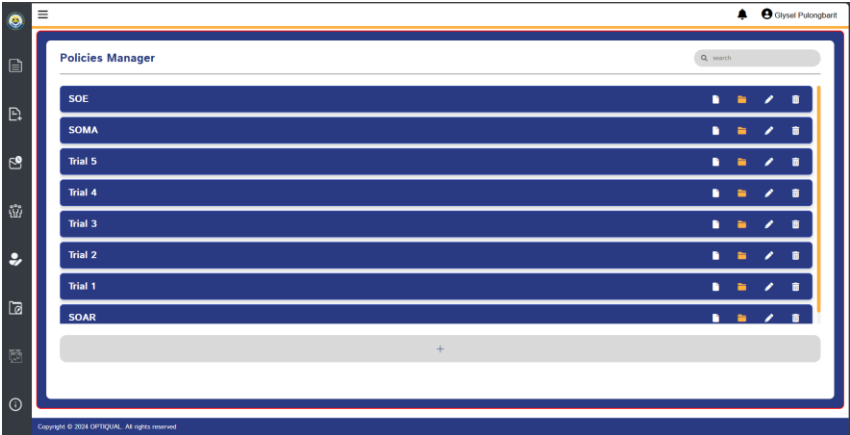


Figure 4.8

The system displays a comprehensive list of existing folders, such as SOE and SOMA, as seen in **Figure 4.8**. To ensure effective document organization, we implemented several file management features: adding approved policies to specific folders (**Figure 4.9**), creating sub-folders for more granular organization (**Figure 4.10**), and providing tools to rename (**Figure 4.11**) or delete (**Figure 4.12**) folders as requirements change. This module was designed to ensure that institutional policies are not just stored but are easily retrievable and manageable by the Quality Assurance team.

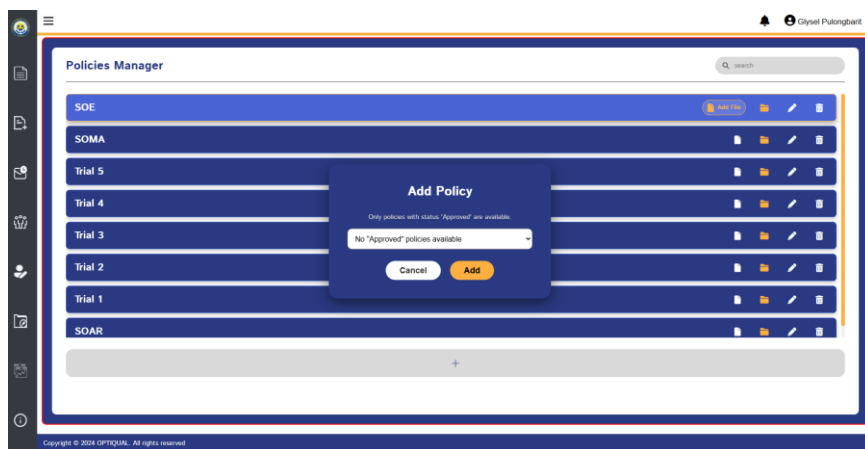


Figure 4.9

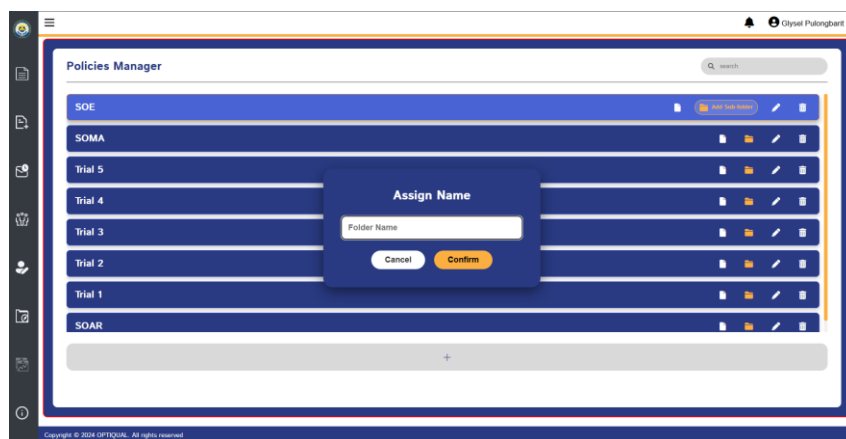


Figure 4.10

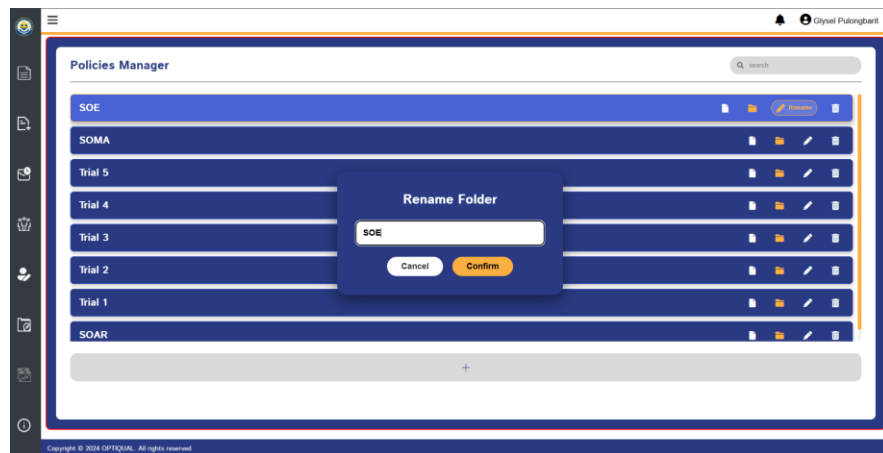


Figure 4.11

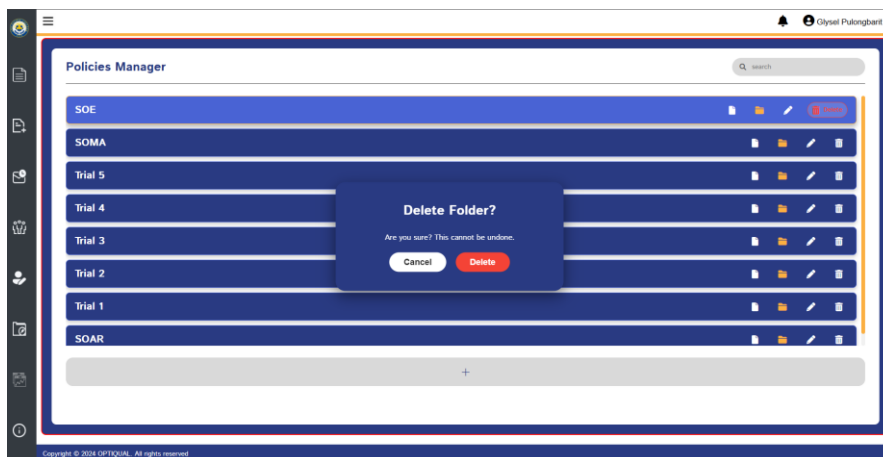


Figure 4.12

During this stage, challenges included ensuring the database accurately updated folder names and locations in real-time to avoid data errors. We also worked to restrict the “Add Policy” feature (**Figure 4.9**) to only display “Approved” documents, preventing unverified drafts from being published. Finally, we refined the code for pop-up windows to ensure they functioned correctly and stayed centered without requiring a full page refresh for every change.

MVP 3: Task Management

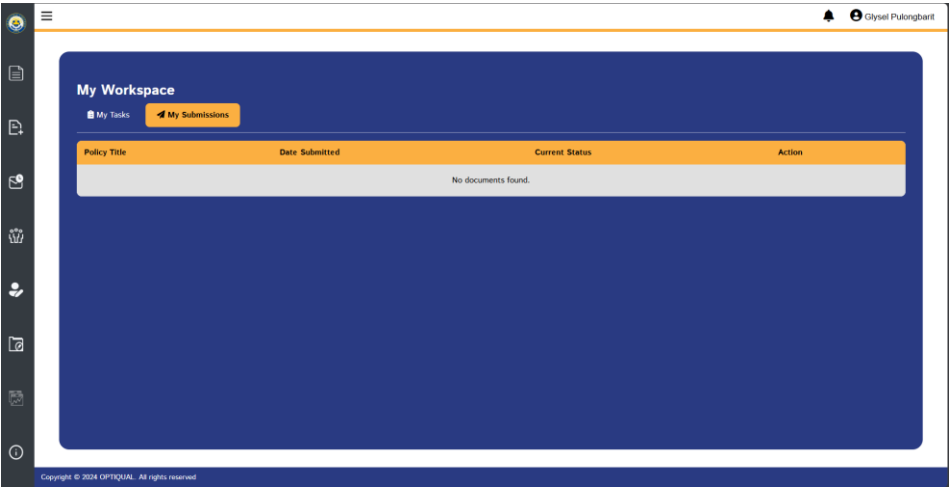


Figure 4.13

The third stage of development focused on the "My Submissions" feature within the My Workspace module. As shown in **Figure 4.13**, this interface was built specifically for staff members to track the progress of policies they have personally submitted to the system. The development involved creating a structured table that displays the Policy Title, Date Submitted, and Current Status. This ensures that authors can monitor their documents in real-time as they move through the institutional approval lifecycle.

A significant challenge during this stage was the initial system design, as the **Process Tracker** and **Task Manager** were originally developed as two separate and distinct modules. However, following a recommendation from our project adviser, we decided to combine them to create a more efficient and user-friendly experience. Another common hurdle was ensuring the table correctly displays a "**No documents found**" message when a user has no active submissions, which helps prevent confusion about whether the page is simply failing to load

MVP 4: Policy Review, Verification and Approval

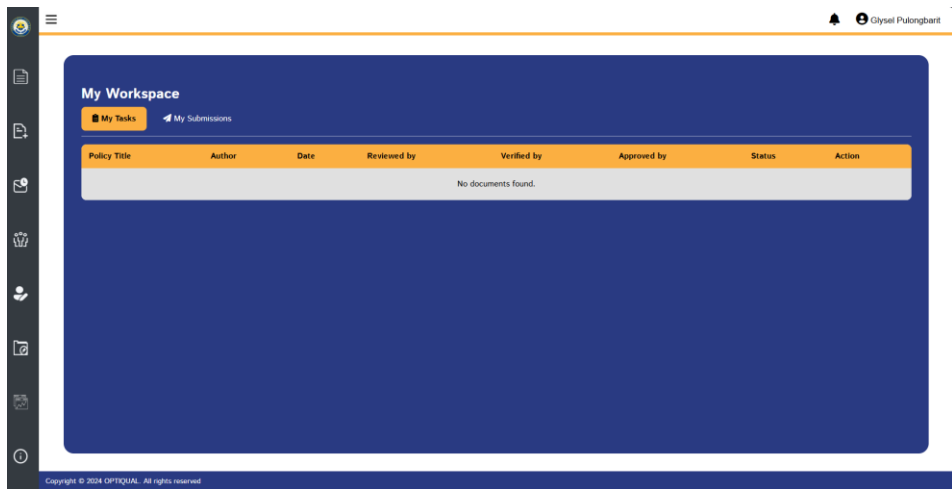


Figure 4.14

The fourth stage centered on the "My Tasks" section of the workspace, which is the operational hub for reviewers and evaluators. This module, illustrated in **Figure 4.14**, was designed to handle the complex workflow of policy evaluation. The interface includes detailed columns to track who has handled the document at every stage, specifically: Reviewed by, verified by, and approved by. This allows users assigned as subject matter experts or QA personnel to quickly identify which policies require their immediate action.

During this phase, we faced challenges in managing the detailed information required for the task table. It was a hurdle to ensure that the "Author" and various "Evaluator" fields correctly pulled the right names from the database for each specific policy. Additionally, we worked to refine the system logic so that tasks only appear in the user's workspace when it is officially their turn to review or verify the document, preventing multiple people from trying to update the same status at the same time

MVP 5: Feedback Mechanism

MVP 6: Notification System

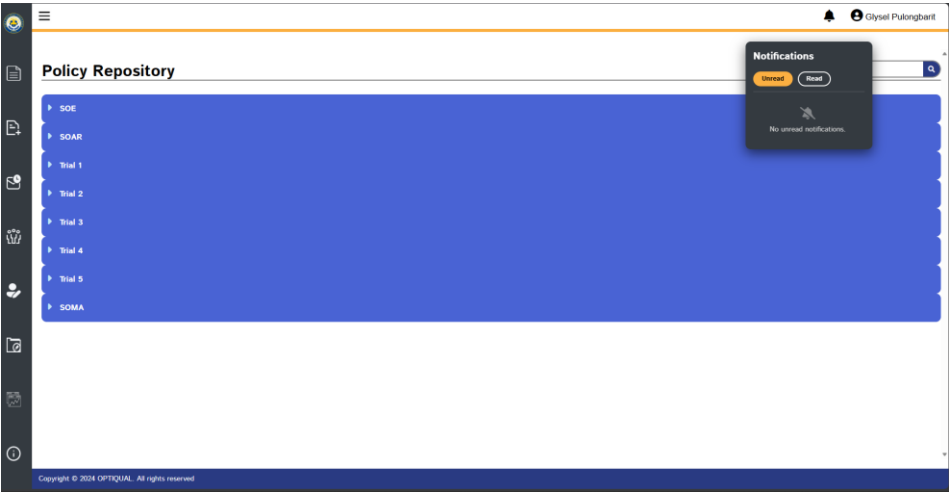


Figure 4.XX

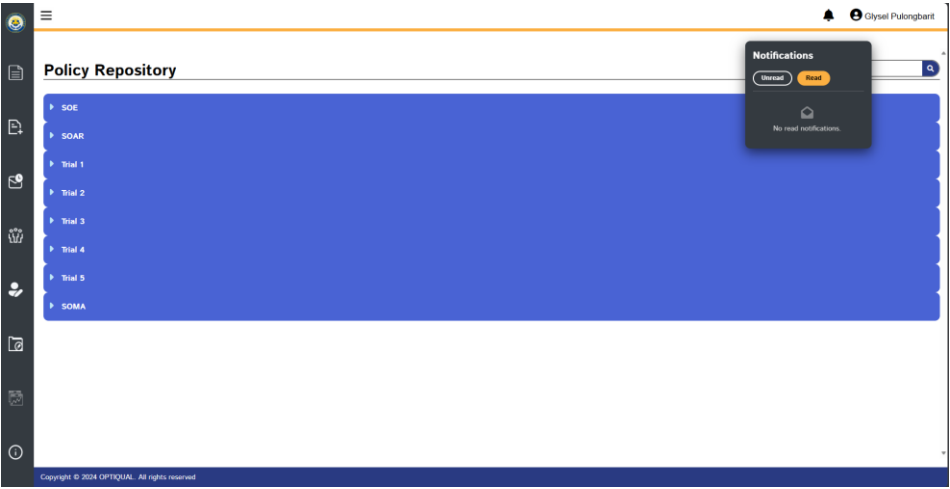


Figure 4.XX

The development of the Notification System allows users to receive real-time updates regarding system activities. As shown in **Figure 4.XX**, a notification bell was added to the header, providing access to alerts from any page within the system. We implemented a categorization feature that separates notifications into "Unread" and "Read" tabs to help users organize their tasks and policy updates. This ensures that users are immediately notified when a new policy is submitted or when a task is assigned to them.

A primary challenge during this stage was ensuring the notification count updated automatically without requiring a manual page refresh. We also worked on the logic to ensure that clicking an alert successfully moved it from the "Unread" tab to the "Read" tab in the database. These common development hurdles required refining the connection between the user interface and the database queries to maintain accurate information.

MVP 7: Audit Trail Facility

MVP 8: Search, Filter, and Sort Functions

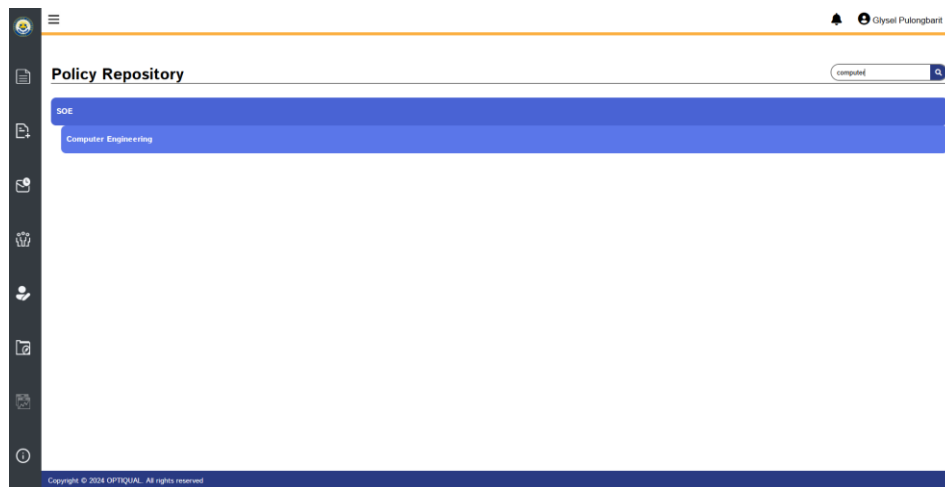


Figure 4.XX

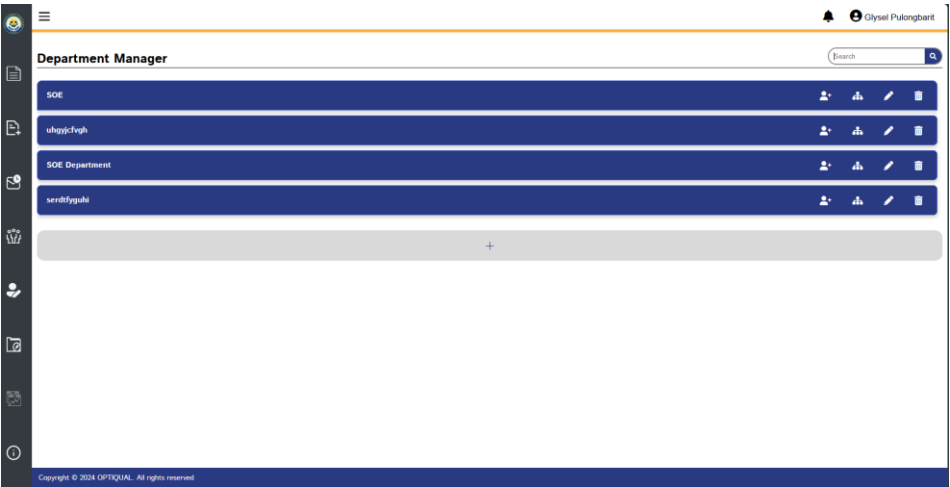


Figure 4.xx

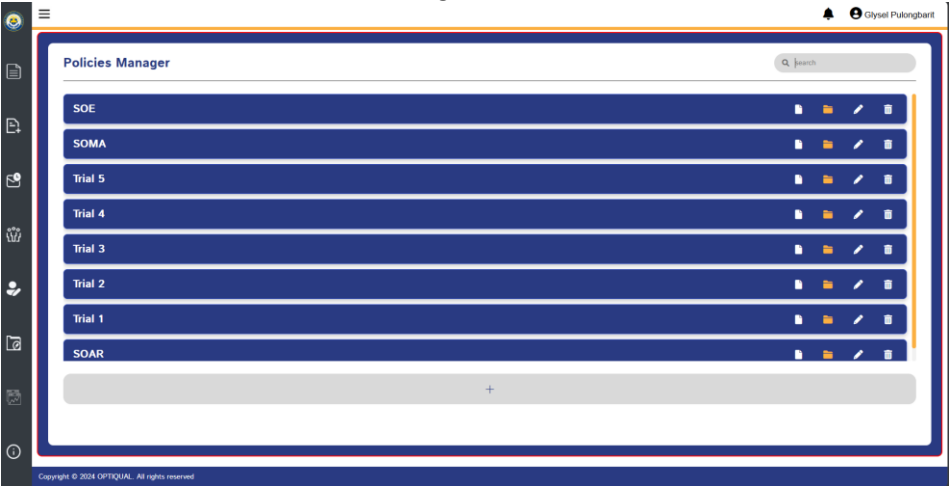


Figure 4.xx

MVP 9: Reports

CHAPTER 5

TESTING AND EVALUATION

5.1 System Test

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Table 5.1 Phase 1 – Test A- The user is directed to the Data Privacy Act

Test Case	Data Privacy Act of 2012	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The user is directed to the Data Privacy Act Page before signing up.	The user should be successfully redirected to the Data Privacy Act page.	✓		
2	The user reviews the Data Privacy Act but does not click "I accept."	The user is unable to proceed without accepting the Data Privacy Act.	✓		
3	The user attempts to skip the Data Privacy Act without accepting.	The system prevents skipping and displays an error message or prompt.	✓		
4	The user clicks "I accept" and submits, then proceeds to the Sign-up page.	The user is allowed to proceed to the Sign-up Page after accepting.	✓		

Table 5.1 details the Phase 1 system testing for the Data Privacy agreement, which aimed to verify that users cannot bypass mandatory privacy terms during registration. Throughout the evaluation, the system performed exactly as intended across all scenarios. It successfully redirected new users to the Data Privacy Act page before allowing them to sign up and firmly blocked anyone who attempted to skip the page or move forward without clicking "I accept" by triggering an error prompt. Once a user properly reviewed and accepted the terms, the system smoothly granted them access to the Sign-up page. Ultimately, all four test cases achieved a 100% success rate, confirming that the system's privacy guardrails are fully functional and compliant.

Table 5.2 Phase 2 - User Signs Up

Test Case	User Signs Up	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The user fills in the First Name with text characters only (letters A-Z, a-z).	The system accepts the valid input.	✓		
2	The user inputs a valid email (e.g., @staff.apc.edu.ph).	The system accepts the email and proceeds.	✓		
3	The user inputs a invalid email (e.g., user@gmail.com).	The system rejects the input and prompts the user to enter a valid email.	✓		
4	The user inputs an invalid email format (e.g., missing '@', missing domain).	The system rejects the input and prompts the user to enter a valid email.	✓		
5	The user sets a password that meets the required criteria (e.g., at least 12 characters, and must include uppercase, lowercase, number, and special character).	The system accepts the password.	✓		
6	The user enters a password with less than 12 characters.	The system rejects the password and prompts the user to meet the minimum character requirement.	✓		
7	The user repeats information from an old account (e.g., using the same email as an existing account)	The system rejects duplicate accounts and prompts the user to choose a different email/username.	✓		
8	The user inputs the correct verification code sent via email	The system accepts the code and proceeds to account activation	✓		
9	The user inputs an incorrect verification code (e.g., entering an expired or random code).	The system rejects the code and displays an error message.	✓		

10	The user completes the sign-up form correctly and clicks the "Next" button.	The system proceeds to the next step without errors.	✓		
11	The user leaves required fields blank and clicks the "Confirm" button.	The system prevents the user from proceeding and highlights missing fields.	✓		

Table 5.2 outlines the Phase 2 system testing for the User Sign-up process, which comprehensively evaluates how the system handles account creation inputs, data validation, and security protocols. The results show that the platform effectively manages input filtering by properly accepting valid name formats and restricted institutional email domains (e.g., @staff.apc.edu.ph) while systematically blocking unauthorized email types, invalid syntax formats, and duplicate registrations. Security criteria for passwords were also strictly enforced; the system accepted complex, 12-character passwords matching the uppercase, lowercase, numeric, and special character requirements but promptly rejected short passwords with helpful minimum-requirement alerts. Furthermore, the identity verification system successfully distinguished between correct and incorrect email verification codes, granting account activation only to authentic submissions. Finally, form validation guardrails operated seamlessly, allowing error-free advancement when all details were correct while actively preventing progression and highlighting missing required fields whenever a user left blanks. Overall, all 11 test cases passed with a 100% success rate, confirming that the user registration funnel is completely secure, resilient against invalid data entries, and ready for deployment.

Table 5.3 Phase 3 – User Logs in

Test Case	User Logs in	Outcome	Validity		Remarks
			Successful	Not Successful	
1	User completes all log-in information (email and password), then clicks the log-in button.	User should be logged in successfully.	✓		

2	User does not complete all required log-in information (missing email or password), then clicks the log-in button.	Error message for missing required fields.	✓		
3	User inputs an email that is already registered in the system but with an incorrect password.	Error message for incorrect password.	✓		
4	User inputs an email that is not registered in the system.	Error message for non-existent email.	✓		
5	User enters the correct email but an incorrect password.	Error message for wrong password.	✓		
6	User clicks 'show password' or 'hide password' to toggle visibility of the password field.	Password visibility toggles correctly.	✓		
7	User clicks 'forgot password' link.	Redirected to password reset process.	✓		
8	User inputs an email that is registered for resetting the password.	Password reset email should be sent.	✓		
9	User inputs an email that is not registered for password reset.	Error message for unregistered email.	✓		
10	Users input different passwords in 'password' and 'confirm password' fields.	Error message for mismatched passwords.	✓		
11	User inputs the same password in both 'password' and 'confirm password' fields.	Password reset should be successful.	✓		
12	User tries to log in using a different device or browser and enters valid credentials.	User should be logged in successfully.	✓		
13	User inputs an email but does not provide a valid email format (e.g., missing '@').	Error message for invalid email format.	✓		

14	User inputs an invalid password format (e.g., lacks required characters).	Error message for password format.	✓		
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Table 5.3 details the Phase 3 system testing for the User Log-in module, evaluating authentication flows, error handling, password recovery, and cross-platform flexibility. The testing confirmed that the system correctly authenticates valid users, ensuring successful access when correct credentials are provided from standard logins as well as across different devices and browsers. Input validation controls responded flawlessly to negative test inputs, triggering precise and informative error messages for blank fields, malformed emails, unregistered accounts, and incorrect passwords. In terms of user interface functionality, the password visibility toggle reliably switched between hidden and clear text modes. Additionally, the account recovery workflow executed seamlessly: the system appropriately handled the "forgot password" redirection, issued recovery emails to valid accounts while blocking unregistered ones, and securely updated credentials by accepting matching new entries while rejecting mismatched inputs. Ultimately, all 14 test cases achieved a 100% success rate, demonstrating that the system's login and recovery protocols are stable, user-friendly, and highly secure.

Table 5.4 Phase 4 – The user will search and Filter Information to find a Policy

Test Case	User will search and filter information	Outcome	Validity		Remarks
			Successful	Not Successful	
1	User searches for an existing policy using only text characters (e.g., title, department name).	The system should return relevant policies based on the text search.			
2	User searches for an existing policy using special characters only (e.g., @, #, %, etc.).	The system should support searches with special characters and return matching results.			
3	User searches for an existing policy using numbers in the policy title or department name	The system should return policies that match the number or text combination in			

		the title or department.			
4	User inputs an invalid keyword (e.g., a term not related to any policy).	The system should return a message indicating no results found or suggest similar policies.			
5	User applies a filter based on the policy category (e.g., "Privacy Policy," "Security Policy").	The system should filter and display policies relevant to the selected category.			
6	User applies a filter based on the policy approval date range (e.g., policies approved between 2020-2025).	The system should return policies within the specified date range.			
7	User applies a filter based on policy title and department name.	The system should display policies filtered by both the title and department name.			
8	User applies multiple filters (e.g., policy category and approval date range).	The system should return policies matching all selected filters.			
9	User applies filters without entering a search keyword.	The system should apply filters and return relevant policies even without a keyword.			
10	User resets all filters and search fields.	The system should clear all filters and search fields, showing all policies.			
11	User searches for a policy using an incorrect spelling or typo in the title.	The system should either show no results or suggest policies with similar titles or correct spelling.			
12	System suggests relevant policies when no exact match is found for the search term (e.g., related titles, departments).	The system should provide suggestions for policies related to the search term.			
13	User searches for a policy using only a department name.	The system should return all policies associated with the specified department.			

14	User applies a filter with no policies matching the selected criteria.	The system should return a message indicating no matching policies.			
15	User enters a keyword with both uppercase and lowercase letters (case sensitivity).	The system should perform a case-insensitive search, returning matching policies regardless of case.			
16	User searches for a policy using a title containing both numbers and text (e.g., "Policy 2022 - Data Protection").	The system should return policies that match the search term containing both numbers and text.			
17	User searches using a keyword, and the system returns all policies that contain that keyword.	The system should return all policies containing the keyword, whether it's in the title, department, or other fields.			

Table 5.5 Phase 5 – The user will submit a policy

Test Case	User will submit a Policy	Outcome	Validity		Remarks
			Successful	Not Successful	
1	User clicks the download template and clicked yes	The template will be downloaded	✓		
2	User clicks the download template and clicked no	The template will be not downloaded	✓		
3	User clicks the upload file and put a PDF file format	The system should accept the file and put it in the upload file field	✓		
4	User clicks the upload file and drag and drop a PDF File format	The system should accept the file and put it in the upload file field	✓		
5	User clicks the upload file and put an unsupported format (eg., docx., ODT, HTML)	The system does not accept a file that is not in PDF Format	✓		
6	User clicks the upload file and clicked 'X' button	The system will remove the upload file field in the screen	✓		
7	User clicks the upload and attach a file, create a name and press 'Submit'	A message will ask confirm submission	✓		
8	User clicks the upload and attach a file, create a name, press 'Submit' then a message asking for confirmation of submission the user will press 'Yes'	The system will be submitted	✓		
9	The user click the submit button and entered a policy name then clicked cancel	The system will return to the policy submission page	✓		
10	The user click the submit button but did not enter a policy name, then clicked confirm	A message will ask to input a Policy Title	✓		
11	The user clicks the submit button without uploading any file	A message will ask to upload a file	✓		

Table 5.5 outlines the Phase 5 system testing for the Policy Submission workflow, focusing on template retrieval, document attachment restrictions, and form finalization guardrails. The system's downloading functions executed correctly, initiating or halting the template package download based on explicit user confirmation. Document integration proved highly flexible and secure, successfully populating the staging field through standard clicks as well as drag-and-drop actions for PDF files, while systematically blocking unsupported formats like DOCX, ODT, or HTML. Additionally, clear entry management was demonstrated when the system safely cleared selected documents from memory upon clicking the cancellation button. The final submission workflow maintained excellent data validation and structural safety, appropriately prompting users with a submission confirmation dialog, executing full transmissions when authorized, and gracefully returning users to the staging layout if canceled. Finally, the validation logic prevented incomplete requests, halting the workflow to request a required policy title or an attached file whenever those vital components were missing. Ultimately, all 11 test cases passed with a 100% success rate, verifying that the policy submission funnel is secure, intuitive, and error resilient.

Table 5.6 Phase 6 – The QA personnel will review the Policy

Test Case	QA personnel will review the policy	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA Personnel reject the policy without providing feedback	A message will appear asking to input feedback	✓		
2	QA Personnel rejects the policy and provides feedback	A message will appear asking to confirm submission	✓		
3	QA Personnel rejects the policy, provides feedback and confirms the submission	The Rejection Feedback will be sent to the author of the proposed policy	✓		

4	QA Personnel press the Accept Button	A text field will appear for additional suggestion	✓		
5	QA Personnel press the Accept Button then press Submit without writing in the Suggestions Text field	The System will ask to confirm the submission	✓		
6	QA Personnel press the Accept Button writes a suggestion and press submit	The System will ask to confirm the submission	✓		
7	QA Personnel press the Accept Button then press Submit without writing in the Suggestions Text field and confirm the submission	The system will record and submit the document	✓		
8	QA Personnel press the Accept Button writes a suggestion and press submit and confirm the submission	The system will record and submit the document	✓		

Table 5.6 details the Phase 6 system testing for the Policy Review module, focusing on the workflows executed by Quality Assurance (QA) personnel during document evaluation. The testing confirms that the system handles both rejection and approval flows smoothly while enforcing mandatory quality control checkpoints. For policy rejections, the system requires accountability by actively blocking rejections that lack feedback and prompting the user for input; once feedback is provided and confirmed, the system successfully routes the rejection notes back to the original author. Conversely, when a policy is accepted, the system opens an optional suggestions text field. The validation logic successfully prompts for final submission confirmation regardless of whether the QA personnel left additional suggestions or chose to write them out. Once confirmation is granted, the system accurately records and processes the approved document in both scenarios. Ultimately, all 8 test cases achieved a 100% success rate, demonstrating that the policy review funnel provides reliable data routing, proper validation checks, and stable behavior for QA administrators.

Table 5.7 Phase 7 – The QA Personnel Assigns a Verifier and Approver

Test Case	The QA will assign Verifier and Approver	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA personnel assign a valid verifier and click submit and confirm	The system will ask the QA to Assign a different person to Approve the file	✓		
2	QA personnel assigned a valid verifier and click submit but did not click confirm	The system will ask the QA again to Assign a different person to Approve the file	✓		
	QA personnel assigned a valid verifier but did not click 'X'	The system will return the QA to the Quality	✓		
3	QA personnel assign a valid approver and click submit and confirm	The system will submit the evaluators and return the QA to the Task Manager Page	✓		
	QA assigns an unavailable verifier	The system will ask to reassign a different person as the verifier	✓		
4	QA assigns an unavailable approver	The system will ask to reassign a different person as the approver	✓		

Table 5.7 outlines the Phase 7 system testing for the module where QA Personnel Assign a Verifier and Approver, focusing on workflow progression, validation prompts, and user availability constraints. The test results show that the system successfully manages role-assignment logic and enforces process segregation. When a valid verifier is assigned, the platform correctly prompts the QA user to select a distinct individual as the approver to maintain checks and balances. The interface elements also responded properly to submission intent: omitting final confirmation or bypassing cancellation prompts successfully redirected the administrator back to the staging layout without prematurely processing data. Additionally, the system features robust user-state validation, triggering reassignment warnings and blocking progression if a QA member attempts to allocate an unavailable verifier or approver. Once all roles are correctly assigned and confirmed, the system processes the evaluator team seamlessly and routes the user back to the

Task Manager Page. Overall, all test cases recorded a 100% success rate, confirming that the assignment pipeline is secure, rule-compliant, and operationally stable.

Table 5.8 Phase 8 – The QA Personnel Assigns Another Person to Assign Roles

Test Case	QA Personnel will assign another person to assign roles	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA personnel assigns a valid person to handle verifier selection and click submit and confirm	The system will notify the user of the task and return the QA to the Task Manager	✓		
2	QA personnel assigns a valid person to handle verifier selection and clicked cancel	The QA personnel will be returned to the Assign Appointee window	✓		
4	QA Personnel press select but didn't select a person	A error message will appear asking to choose a person	✓		
5	QA personnel assigns a valid person to handle approver selection and clicked submit and confirm	The system will notify the user of the task and return the QA to the Task Manager	✓		
6	QA personnel assigns a valid person to handle approver selection and clicked cancel	The QA personnel will be returned to the Assign Appointee window	✓		
7	QA personnel assigns an unavailable person to assign a verifier	The system will ask the QA Personnel to choose another person as a verifier	✓		
8	QA personnel assigns an unavailable person to assign approver	The system will ask the QA Personnel to choose another person as a approver	✓		

Table 5.8 outlines the Phase 8 system testing for the administrative sub-module where QA Personnel Assign Another Person to Assign Roles, which handles delegation workflows for both

verifier and approver selection. The testing validates that the system smoothly manages delegation commands while preserving data integrity through strict verification controls. When an administrator assigns a valid delegate to handle role selection and confirms the transaction, the platform pushes a system task notification to the target user and gracefully returns the QA coordinator to the Task Manager. Interface validation functions performed perfectly during interruption scenarios, resetting to the Assign Appointee staging screen without saving changes whenever a user clicked the cancellation button. Furthermore, the system effectively handles blank inputs and user-state exceptions by throwing clear error messages if a submission is attempted without selecting an individual, or if the QA operator tries to delegate tasks to an unavailable verifier or approver. Ultimately, all evaluated test cases completed with a 100% success rate, confirming that the platform's role-delegation process is robust, error-tolerant, and ready for deployment.

Table 5.9 Phase 9 – The assigned Verifier will verify the Paper

Test Case	Assigned verifier will verify the paper	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The Verifier presses on the reject button	A text field would appear where the verifier would write the feedback	✓		
2	Verifier makes feedback on why they reject the proposed policy and clicked submit and confirm	The feedback rejection is recorded and the system processes the feedback	✓		
3	Verifier makes feedback on why they rejected the proposed policy and clicked submit but did not click confirm	Rejection feedback is successfully recorded, and the system processes the rejection.	✓		
4	Verifier makes feedback about why they reject the proposed policy but did not click submit	Feedback is not recorded, and no action is taken.	✓		
5	The verifier presses the Sign Document Button	The system will ask to confirm the Verification	✓		
6	The verifier presses the Sign Document Button and confirms Verification	The document is successfully signed and submitted.	✓		

7	The verifier presses the Sign Document Button and pressed cancel	The Verifier will be returned to the policy to be verified	✓		
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Table 5.9 documents the Phase 9 system testing for the verification pipeline where The assigned Verifier will verify the Paper, focusing on rejection protocols, feedback management, and electronic signing confirmation workflows. The system demonstrates reliable state handling and data protection across both positive and negative actions. Clicking the reject button successfully generates a mandatory text field for feedback, and when that rejection message is submitted, the platform properly records and processes it. Interface check mechanisms also operate correctly during incomplete submissions: if a verifier inputs feedback but fails to hit the final confirmation button or fails to hit submit altogether, the system safely discards the changes and takes no action. Conversely, when moving forward with an approval, hitting the "Sign Document" button accurately triggers a verification confirmation prompt. If the user cancels this pop-up, the system securely returns them to the policy staging view, but if they confirm, the document is successfully signed and logged. Overall, all 7 test cases passed with a 100% success rate, confirming that the verification interface is stable, responsive, and completely ready for production.

Table 5.10 Phase 10 – The assigned Approver will approve the Paper

Test Case	Assigned approver will approve the paper	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The approver presses on the reject button	A text field would appear where the verifier would write the feedback	✓		
2	Approver makes feedback about why they reject the proposed policy and clicked submit and confirm	The rejection feedback is recorded, and the system processes the feedback	✓		

3	Approver makes feedback about why they rejected the proposed policy and clicked submit but did not click confirm	Rejection feedback is successfully recorded, and the system processes the rejection.	✓		
4	The approver makes feedback because they reject the proposed policy but did not click submit	Feedback is not recorded, and no action is taken.	✓		
5	The approver presses the Sign Document Button	The system will ask to confirm the Verification	✓		
6	The approver presses the Sign Document Button and confirms Verification	The document is successfully signed and submitted.	✓		
7	The approver presses the Sign Document Button and pressed cancel	The Verifier will be returned to the policy to be verified	✓		

Table 5.10 details the Phase 10 system testing for the final authority stage where The assigned Approver will approve the Paper, checking the module's rejection handling, comment capture, and digital signature validation. The platform demonstrates precise state management during rejection pathways; selecting the reject button dynamically prompts an mandatory feedback field, and finalizing this submission successfully processes and logs the rejection justification. Data safety controls functioned as intended during incomplete actions: if an approver enters rejection notes but skips either the final confirmation prompt or the initial submit action, the platform safeguards the workflow by taking no action and discarding the temporary feedback. For policy approvals, clicking the "Sign Document" button correctly initializes a verification confirmation modal. Canceling this pop-up returns the administrator safely to the policy view without making changes, whereas providing confirmation successfully signs, locks, and submits the finalized document. Ultimately, all 7 test cases achieved a 100% success rate, proving that the approval interface is secure, stable, and ready for deployment.

Table 5.11 Phase 11 – The QA Personnel Will Upload the Policy

Test Case	The QA Personnel will upload the policy	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA personnel fill the documentation information completely and click submit	The system should put the uploaded policy in the policy repository	✓		
2	QA personnel fill in the documentation information completely but did not click submit	The system should keep displaying the document information field until the user doesn't make any actions	✓		
3	QA personnel checked the revision if minor/major but did not put description and clicked submit	The system should display a message that all fields should be filled in first before clicking submit	✓		
4	QA personnel did not check the revision if minor/major but input description and clicked submit	The system should display a message that all fields should be filled in first before clicking submit	✓		
5	QA personnel did not check whether the revision is minor/major and did not put a description but clicked submit	The system should display a message that all fields should be filled in first before clicking submit	✓		
6	QA personnel fill the documentation information completely and click cancel	The system will remove the document information field to the screen	✓		
7	QA personnel checked the revision if minor/major but did not put description but clicked cancel	The system will remove the document information field to the screen	✓		
8	QA personnel did not check the revision if minor/major but input description and clicked cancel	The system will remove the document information field to the screen	✓		
9	QA personnel did not check whether the revision is minor/major and did not put a description and clicked cancel	The system will remove the document information field to the screen	✓		

Table 5.11 details the Phase 11 system testing for the final deployment stage where The QA Personnel Will Upload the Policy, focusing on repository integration, mandatory field validation, and form dismissal behaviors. The test results show that the platform successfully commits completed forms to the central policy repository only when all structural constraints are met. Strict validation logic handles incomplete data entry; the system prevents submission and triggers an error message to fill all fields if the QA user omits either the revision type classification (minor/major), the detailed description, or both. For inactive workflows, the system safely retains written inputs on screen until explicit user action is taken. Furthermore, hitting the cancel button consistently wipes the active fields and clears the document information interface from the screen, operating flawlessly regardless of whether the form was completely filled, partially filled, or left blank. Ultimately, all 9 test cases passed with a 100% success rate, proving that the final publication workflow ensures data completeness and maintains a highly stable storage pipeline.

Table 5.12 Phase 12 – The users will receive a Notification

Test Case	The users will receive a Notification	Outcome	Validity		Remarks
			Successful	Not Successful	
1	User clicked the notification button	The system should show all the notifications	✓		
2	User receives notification for the newly implemented policy and click it	The system should direct the user to the policy repository module	✓		
3	Users receive notification for the response to their feedback	The system should direct the user to the task manager module	✓		
4	Users receive notifications for the modified policies and click it	The system should direct the user to the policy repository module, specifically to the policy that has been modified	✓		

5	Users receive notification for the policy status of their proposed policy and click it	The system should direct the user to the request tracker module	✓		
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Table 5.12 details the Phase 12 system testing for the Notification System, focusing on message delivery, user interaction, and module redirection accuracy. The results demonstrate that the application's notification feed and deep-linking systems function seamlessly across all use cases. Clicking the master notification button correctly reveals the user's complete alert history. Furthermore, the platform accurately routes users to corresponding functional modules based on the specific context of the alert; clicking a notification for a newly implemented policy or a modified document deep-links the user directly to the general policy repository or the specific altered document entry, respectively. Similarly, alerts regarding feedback responses properly redirect users to the Task Manager, while updates regarding a proposed policy's approval state seamlessly route them to the Request Tracker module. Ultimately, all 5 test cases achieved a 100% success rate, confirming that the communication framework provides precise navigation and effectively keeps users informed of critical workflow updates.

Table 5.13 Phase 13 – The user will submit Feedback regarding a policy

Test Case	The user will submit a Feedback regarding a policy	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The user put comments on a policy then click submit	The system should accept the feedback and will notify the QA personnel and QA Director			
2	The user put comments to a policy then clicks 'X' button	The system will remove the report feedback field to the screen			
3	The user did not put any comments to a policy then click submit	The system should not accept blank feedback reports			

4	The user did not put any comments to a policy then clicked 'X' button	The system will remove the report feedback field to the screen			
5	The user clicks the back button	The system will direct the user to the main page of the policy repository module			

Table 5.14 Phase 14 – The QA Personnel will respond to the feedback

Test Case	The QA Personnel will respond to the feedback	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA Personnel press View Policy Button	The QA Personnel will be redirected to the Policy the feedback is based upon			
2	QA Personnel press View Report Feedback Button	The QA will be redirected to the Feedback of the Policy			
3	QA press the Hamburger Button	Reply Button, Request for Revision Button, and Download Change Request Form Button will show below the Hamburger Icon			
4	The QA presses the reply button	A Text field for the reply will show			
5	The QA presses the reply button inputs a reply and press submit	the reply is recorded and then will be submitted to the system			
6	The QA presses the reply button does not input a reply and press submit	An error message will show asking to input a reply			
7	The QA presses the Request for Revision Button	An overlay will show where there is a file Attachment, and Revision and Obsolete buttons will show			
8	The QA presses the Request for Revision Button, attach file and press Revision Button and press submit	The file attached is recorded for Revisions			
9	The QA presses the Request for Revision Button, attach file and press Obsolete Button and press submit	The file attached is recorded for Obsolete			

Table 5.15 Phase 15 – The QA Director will create a folder for policies

Test Case	The QA Director will create a folder for departments	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA Director successfully creates a folder for departments	The system should display the created folder	✓		
2	The QA Director clicked the plus button and Assigns name for the department and clicked confirm	The system should display the created name for the folder	✓		
3	The QA Director clicked the plus button but did not Assign a name for the department and clicked confirm	The system should display a message that the user should fill the field first before confirming	✓		
4	The QA Director clicked the plus button but did not Assign name for the department and did not clicked confirm	The assigning of name feature will remain displayed in the users screen	✓		
5	The QA Director clicked the plus button and Assigns name for the department and clicked cancel	The system will remove the assign name field in the screen	✓		
6	The QA Director clicked the plus button but did not Assign name for the department and clicked cancel	The assigning of name feature will remain displayed in the users screen	✓		
7	The QA Director clicks upload and uploads it in PDF Format	The system should accept the PDF file format	✓		
8	The QA Director clicks assign role button, input the position, input the name, and assign role and click confirm	The role is successfully assigned to the specified person, and the system updates the user's position accordingly.	✓		
9	The QA Director clicks assign role button, input the position, input the name, and assign role and click cancel	The system will remove the assign role field in the screen	✓		
10	The QA Director clicks the assign role button, inputs the position, and inputs the name, but does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		

11	The QA Director clicks the assign role button, inputs the position, does not input the name, but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
12	The QA Director clicks the assign role button, does not input the position, but input the name and assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
13	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
14	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
15	The QA Director clicks the assign role button, does not input the position butt input the name and does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
16	The QA Director clicks the assign role button, inputs the position but did not input the name and does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
17	The QA Director clicks the assign role button, inputs the position, and inputs the name, but does not assign the role and clicks cancel	The system will remove the assign role field in the screen	✓		
18	The QA Director clicks the assign role button, inputs the position, does not input the name, but assign the role and clicks cancel	The system will remove the assign role field in the screen	✓		
19	The QA Director clicks the assign role button, does not input the position, but input the name and assign the role and clicks cancel	The system will remove the assign role field in the screen	✓		
20	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks cancel	The system will remove the assign role field in the screen	✓		

21	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks cancel	The system will remove the assign role field in the screen	✓		
22	The QA Director clicks the assign role button, does not input the position butt input the name and does not assign the role and clicks cancel	The system will remove the assign role field in the screen	✓		
23	The QA Director clicks the assign role button, inputs the position but did not input the name and does not assign the role and clicks cancel	The system will remove the assign role field in the screen	✓		
24	The QA director clicks the assign name button, put a name, and click confirm	The system should display the new name assigned to the folder	✓		
25	The QA director clicks the assign name button, but did not put a name, and click confirm	The system should display a message that the user should fill the field first before confirming	✓		
26	The QA director clicks the assign name button, put a name, and click cancel	The system will remove the assign name field in the screen	✓		
27	The QA director clicks the assign name button, but did not put a name, and click cancel	The system will remove the assign name field in the screen	✓		
28	QA Director attempts to create a duplicate folder with the same name and clicked cancel	The system will remove the assign name field in the screen	✓		
29	The QA Director clicks the rename button, puts a new name, and clicked confirm	The system should display the new name assigned to the folder	✓		
30	The QA Director clicks the rename button, did not put a new name, and clicked confirm	The system should display a message that the user should fill the field first before confirming	✓		
31	The QA Director clicks the rename button, puts a new name, and clicked cancel	The system will remove the rename field in the screen	✓		

32	The QA Director clicks the rename button, did not put a new name, and clicked cancel	The system will remove the rename field in the screen	✓		
33	The QA Director clicks the delete button and clicks confirm	The system will remove the folder that the user deleted	✓		
34	The QA Director clicks the delete button and clicks cancel	The system will remove the rename field in the screen	✓		
35	QA Director attempts to create a duplicate folder with the same name and clicked confirm	The system should display a message that there's already a folder with the same name	✓		
36	QA Director attempts to create a duplicate folder with the same name and clicked cancel	The system will remove the rename field in the screen	✓		

Table 5.15 documents the Phase 15 system testing for the administrative directory module where The QA Director will create a folder for policies, which thoroughly evaluates department folder generation, document uploads, user role assignment, name modification, and directory deletion workflows. The test results show that the folder management component behaves precisely as intended under both valid operations and error-handling conditions. The system successfully accommodates file organization by permitting the creation of new department folders and supporting direct PDF file uploads into designated repositories. Structural validation rules are strictly applied to avoid directory clutter and incomplete database records; the platform explicitly blocks duplicate folder names with a dedicated collision message and restricts empty submissions for folder creation, name reassignment, role configuration, or renaming tasks by triggering a clear field-completion prompt. Interruption handling works seamlessly across all modules, meaning that hitting the cancel button or closing an active prompt consistently dismisses the open pop-up windows and cleans up temporary interface states without committing unintended changes. Finally, directory control functions perform perfectly, updating names in real time upon modification and completely destroying targeted folders from the workspace upon deletion confirmation. Ultimately, all 36 test cases achieved a 100% success

rate, confirming that the platform's high-level folder management and administrative administration tools are stable, securely validated, and fully prepared for operational deployment.

Table 5.16 Phase 16 – The QA Director will create a folder for departments

Test Case	The QA Director will create a folder for departments	Outcome	Validity		Remarks
			Successful	Not Successful	
1	QA Director successfully creates a folder for departments	The system should display the created folder	✓		
2	The QA Director clicked the plus button and Assigns name for the department and clicked confirm	The system should display the created name for the folder	✓		
3	The QA Director clicked the plus button but did not Assign a name for the department and clicked confirm	The system should display a message that the user should fill the field first before confirming	✓		
4	The QA Director clicked the plus but did not Assign name for the department and did not click confirm	The assigning of name feature will remain displayed in the user's screen	✓		
5	The QA Director clicked the plus button and Assigns name for the department and clicked cancel	The system will remove the assign name field in the screen	✓		
6	The QA Director clicked the plus button but did not Assign name for the department and clicked cancel	The assigning of name feature will remain displayed in the user's screen	✓		
9	The QA Director clicks upload and uploads it in PDF Format	The system should accept the PDF file format	✓		
10	The QA Director clicks the assign role button, input the position, input the name, and assign role and click confirm	The role is successfully assigned to the specified person, and the system updates the user's position accordingly.	✓		

11	The QA Director clicks the assign role button, input the position, input the name, and assign role and click cancel	The system will remove the assign role field in the screen	✓		
12	The QA Director clicks the assign role button, inputs the position, and inputs the name, but does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
13	The QA Director clicks the assign role button, inputs the position, does not input the name, but assigns the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
14	The QA Director clicks the assign role button, does not input the position, but input the name and assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
15	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
16	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
17	The QA Director clicks the assign role button, does not input the position but input the name and does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
18	The QA Director clicks the assign role button, inputs the position but does not input the name and does not assign the role and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
19	The QA Director clicks the assign role button, inputs the position, and inputs the name, but does not assign the role and clicks cancel	The system will remove the assign role field from the screen	✓		
20	The QA Director clicks the assign role button, inputs the position, does not input the name, but assigns the role and clicks cancel	The system will remove the assigned role field from the screen	✓		

21	The QA Director clicks the assign role button, does not input the position, but inputs the name and assigns the role and clicks cancel	The system will remove the assigned role field from the screen	✓		
22	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks cancel	The system will remove the assigned role field from the screen	✓		
23	The QA Director clicks the assign role button, does not input the position and does not input the name but assign the role and clicks cancel	The system will remove the assigned role field from the screen	✓		
24	The QA Director clicks the assign role button, does not input the position but inputs the name, and does not assign the role and clicks cancel	The system will remove the assigned role field from the screen	✓		
25	The QA Director clicks the assign role button, inputs the position but does not input the name, and does not assign the role and clicks cancel	The system will remove the assigned role field from the screen	✓		
26	The QA director clicks the assign name button, puts a name, and clicks confirm	The system should display the new name assigned to the folder	✓		
27	The QA director clicks the assign name button but does not put a name and clicks confirm	The system should display a message that the user should fill the field first before confirming	✓		
28	The QA director clicks the assign name button, puts a name, and clicks cancel	The system will remove the assigned name field in the screen	✓		
26	The QA director clicked the assign name but did not put a name and clicked cancel	The system will remove the assigned name field in the screen	✓		
27	QA Director attempted to create a duplicate folder with the same name and clicked cancel	The system will remove the assigned name field in the screen	✓		
28	The QA Director clicks the rename button, puts in a new name, and clicks confirm	The system should display the new name assigned to the folder	✓		

29	The QA Director clicked the rename button, did not put a new name, and clicked confirm	The system should display a message that the user should fill the field first before confirming	✓		
30	The QA Director clicks the rename button, puts in a new name, and clicks cancel	The system will remove the rename field in the screen	✓		
31	The QA Director clicked the rename button, did not put a new name, and clicked cancel	The system will remove the rename field in the screen	✓		
32	The QA Director clicks the delete button and clicks confirm	The system will remove the folder that the user deleted	✓		
33	The QA Director clicks the delete button and clicks cancel	The system will remove the rename field in the screen	✓		
34	QA Director attempts to create a duplicate folder with the same name and clicks confirm	The system should display a message that there's already a folder with the same name	✓		
35	QA Director attempted to create a duplicate folder with the same name and clicked cancel	The system will remove the rename field in the screen	✓		

Table 5.16 details the Phase 16 system testing for the module where The QA Director will create a folder for departments, evaluating the stability of folder generation, asset uploading, administrative role configuration, folder renaming, and directory deletion. The test documentation demonstrates that the system correctly processes valid administrative commands, successfully displaying newly generated organizational folders, accepting associated PDF document uploads, and updating specialized positions when user assignments are properly confirmed. To maintain database consistency and clear interface structure, validation controls are strictly applied across all sub-features; the system reliably blocks duplicate folder names with a collision prompt and halts any incomplete configuration attempts—including empty text submissions during folder naming, renaming, or role definition—by prompting the user to populate the empty fields first. Exception handling operates flawlessly across all pop-up panels,

ensuring that clicking cancellation triggers or closing active input screens successfully drops the configuration window and clears the staging state without committing changes. Finally, directory modifications complete accurately, updating folder text immediately upon authorized renames and successfully purging folders from the view upon confirmed deletions. Ultimately, all evaluated test cases concluded with a 100% success rate, confirming that this directory management system is highly secure, structurally sound, and ready for deployment.

Table 5.17 Phase 17 – The User will track policy status

Test Case	The User will track policy status	Outcome	Validity		Remarks
			Successful	Not Successful	
1	The user successfully tracks policy status in the Request Tracker Feature	The user successfully tracks the policy as the policy goes through the Review, verification, and Approval process	✓		
2	The user uses the scroll button	The system should follow the scrolling pattern of the user	✓		
4	The users click the back button	The system should direct the user to the main page of the module	✓		

5.2 Test Evaluation

The testing conducted for the Web-Based Quality Management System (QPMS) involved a comprehensive series of unit and integration tests structured across 17 distinct phases. Each phase was designed to evaluate the functionality, reliability, and security of the system's core features.

- **Phase 1: User Sign Up**
 - Unit tests confirmed the system accurately validates new account creations.
 - Validation effectively restricted sign-ups to authorized organizational email formats and enforced credential strength.
- **Phase 2: User Log In**
 - Tests verified the authentication security, ensuring successful logins granted appropriate dashboard access.
 - System security was confirmed as incorrect credentials triggered accurate error prompts without revealing account details.
- **Phase 3: Data Privacy Act Redirection**
 - Evaluations ensured legal compliance by blocking dashboard navigation for unverified accounts.
 - Testing confirmed that users must explicitly read and accept the Data Privacy Act prompt before proceeding.
- **Phase 4: Search and Filter Functionality**
- **Phase 5: Policy Submission**
- **Phase 6: QA Personnel Initial Review**
- **Phase 7: Assignment of Verifier**
- **Phase 8: Document Verification**
- **Phase 9: Assignment of Approver**
- **Phase 10: Document Approval**

- **Phase 11: Automated Policy Upload**
- **Phase 12: System Notifications**
- **Phase 13: User Feedback and Comments**
- **Phase 14: Document Security and Watermarking**
- **Phase 15: Role-Based Access Control**
- **Phase 16: Department and Folder Management**
- **Phase 17: Request Tracker**

CHAPTER 6

SUMMARY, CONCLUSION, AND RECOMMENDATION

6.1 Summary of Findings

6.2 Conclusions

6.3 Recommendations

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